

BAC – DTB BHS

ALS-19057-BH-SPC-0192

DTB LLC 4 LANE STRATEGY (STAGE SP5)

FDS

Ind	Description	Written by	Date
Rev. C	Update to 4 Lane Strategy	J Duley	04/12/2025



DTB LLC Functional Design Specification



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APPENDECIES

1. INTRODUCTION

1.1. Purpose

The purpose of this document is to present the Functional Design Specification for the Baggage Handling System (BHS) at the Brisbane Airport Domestic Terminal, known as the Separable Portion 5 (SP5) stage. In this stage four CBS lines will be utilised for BHS operation. This specification provides an overview of the design principles and a comprehensive description of the system's functionality.

The following diagram represents the functional layout and CCP splits for the SP5 stage. Please refer to Overall System FCP Splitting Layout drawing for details

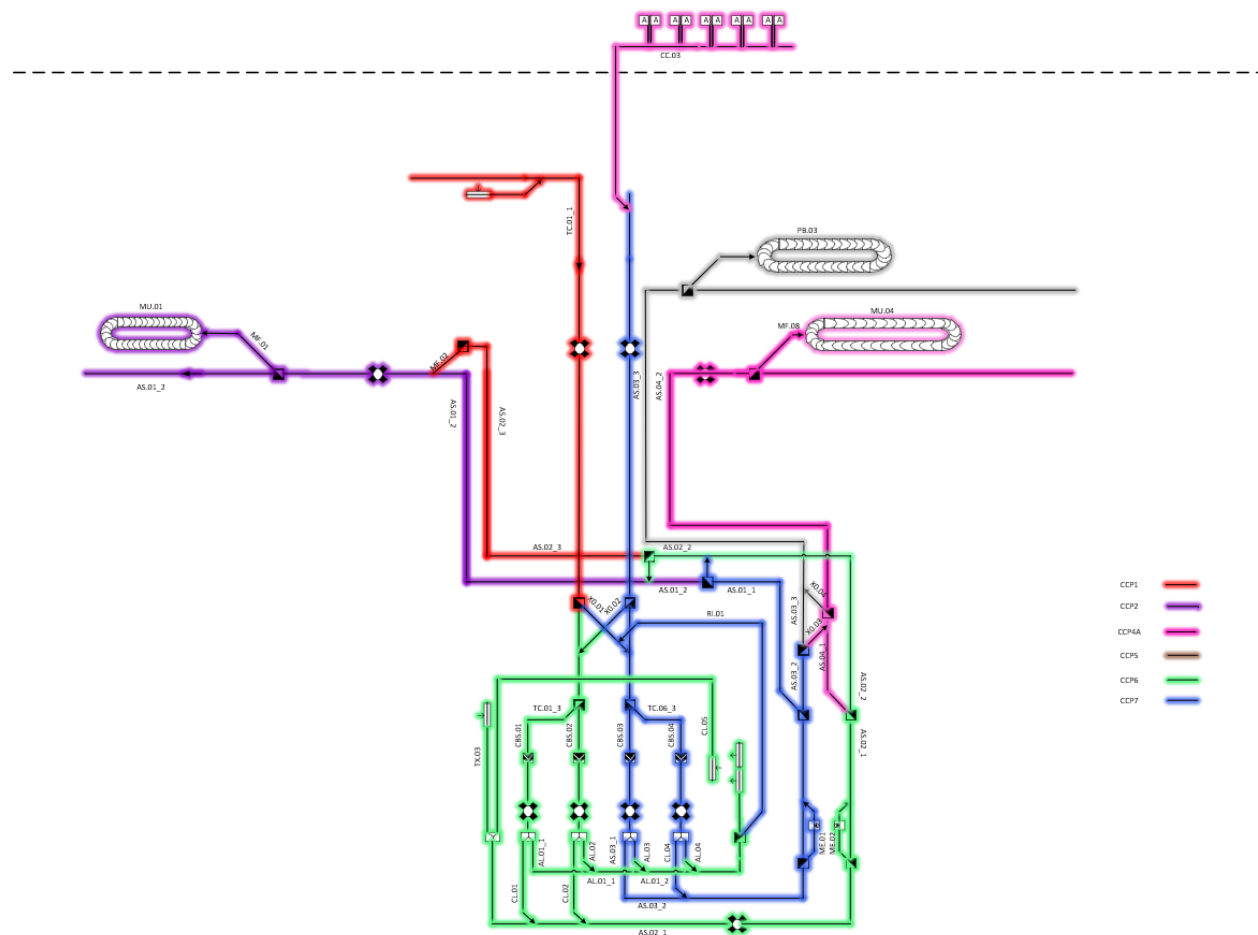


Figure 1: 4 Lane Strategy SP5 Stage Functional Layout and CCP Splits

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1.3. Document References

This specification complies with:

Document No.	Date	Rev	Title
BAC-19057-BH-SPC-0039		2	DTB SUP BHS – Design and Construct – Project Requirements – Volume 1 – General
BAC-19057-BH-SPC-0040		2	DTB SUP BHS - Design and Construct - Project Requirements - Volume 2 - Product
ALS-19057-BH-SPC-0025		1	Simulation FDS_DTB
ALS-19057-BH-SPC-0186-A-Brisbane_Airport (BNE) DTB HLC Phase 2 FDS		B	Brisbane Airport (BNE) DTB HLC Phase 2 Functional Design Specification
ALS-19057-BH-SPC-0191		1.2	Interface Description - Airport Luggage Identification System - Standard Interface Specification (EIP)
RAP-23061-BH-SPC-4001		8	RTT 110 BHS Protocol Specification – Airport- TSA
ALS-19057-BH-SPC-0281		A	DTB Network Topology
ALS-19057-BH-MAN-0007		4.5	Logistics Analytics User Manual
PCL-BHS INTERFACE v007– (BNE DTB Central check-in precinct)		007	Amadeus ABD FDS
ALS-19057-BH-SPC-0188		2.2	PCL-UL Transaction Messages
ALS-19057-BH-SPC-0074		A	DTB2 Auto Routing FDS
19057-ALS-00-M1-DR-BH-070107		B	PDP Splitting Layout Overall
19057-ALS-00-M1-DR-BH-070108		B	CCP Splitting Layout Overview
19057-ALS-00-M1-DR-BH-070034		B	Overall System FCP Splitting Layout

19057-ALS-00-M1-DR-BH-070035		B	Sub-System CCP1 FCP Split Layout
19057-ALS-00-M1-DR-BH-070036		B	Sub-System CCP2 FCP Split Layout
19057-ALS-00-M1-DR-BH-070038		B	Sub-System CCP4AT FCP Split Layout
19057-ALS-00-M1-DR-BH-070040		B	Sub-System CCP6 FCP Split Layout
19057-ALS-00-M1-DR-BH-070041		B	Sub-System CCP7 FCP Split Layout
19057-ALS-00-M1-DR-BH-070056		B	Sub-System CCP4AT E-Stop Zone Layout
19057-ALS-00-M1-DR-BH-070066		B	Sub-System CCP1 E-Stop Zone Layout
19057-ALS-00-M1-DR-BH-070067		B	Sub-System CCP2 E-Stop Zone Layout
19057-ALS-00-M1-DR-BH-070072		B	Sub-System CCP6 E-Stop Zone Layout
19057-ALS-00-M1-DR-BH-070242		A	Sub-System CCP1 Jam Reset Layout
19057-ALS-00-M1-DR-BH-070243		A	Sub-System CCP2 Jam Reset Layout
19057-ALS-00-M1-DR-BH-070245		A	Sub-System CCP4A Jam Reset Layout
19057-ALS-00-M1-DR-BH-070248		A	Sub-System CCP6 Jam Reset Layout
19057-ALS-00-M1-DR-BH-070249		A	Sub-System CCP7 Jam Reset Layout

Table 1: Document References

1.4. List of Abbreviations

Abbreviation	Meaning
ABD	Automated Bag Drop
ABF	Australian Border Force
AS-i	Actuator Sensor Interface
ATR	Automatic Tag Reader
BAC	Brisbane Airport Corporation
BHS	Baggage Handling System
BMS	Building Management System
BNE	Brisbane International Airport
BNP	BNP Associates, Inc.
BPH	Bags Per Hour
BPM	Bags Per Minute or Baggage Processed Message, context dependent
BSM	Baggage Source Message
CBS	Checked Baggage Screening
CBSS	Checked Baggage Screening System
CCP	Central Control Panel
CIP	Common Industrial Protocol
CPU	Central Processing Unit
DCS	Departure Control System
DIO	Distributed Input / Output
DTB	Domestic Terminal Building
ECM	Energy Conservation Mode
EDS	Explosive Detection System
EIP	EtherNet/IP
EPROM	Erasable Programmable Read Only Memory
ETD	Explosive Trace Detection
E-Stop	Emergency Stop
FAT	Factory Acceptance Test
FCP	Field Control Panel
FIFO	First In First Out

Abbreviation	Meaning
GPO	General Power Outlet
HLC	High Level Control
HMI	Human Machine Interface
HOA	Hand/Off/Auto
HSD	High Speed Diverter
HVAC	Heating, Ventilating and Air Conditioning
I/O	Input/output
IATA	International Air Transport Association
ID	Identification
IEC	International Electrotechnical Commission
IP	Internet Protocol
ISO	International Organization for Standardization
ITB	International Terminal Building
LAN	Local Area Network
LCD	Liquid Crystal Display
LED	Light Emitting Diode
LLC	Low Level Control
LPN	License Plate Number
MEC	Manual Encoding Console
MES	Manual Encode Station
MPM	Metres per Minute
MPS	Metres per Second
m/s	Metres per Second
MTBF	Mean Time Between Failures
O&M	Operations and Maintenance
OOG	Out of Gauge (AKA Oddsize and Oversize)
OSR	On-Screen Resolution
PAX	Passenger
PDP	Power Distribution Panel
PEC	Photo Electric Cell

Abbreviation	Meaning
PLC	Programmable Logic Controller
PM	Project Manager
PR	Project Requirements
QF	Qantas
REP	Resilient Ethernet Protocol
RF	Radio Frequency
RFI	Request for Information
RFID	Radio Frequency Identification
SAC	Sort Allocation Controller
SATP	Site Acceptance Testing Period
SCADA	Supervisory Control and Data Acquisition
SIT	Systems Integration Testing
Sim-FDS	Simulation Functional Design Specification
SMI	Security Manual Inspection
SVS	Secondary Viewing Station
UAT	User Acceptance Test
ULD	Unit Load Device
UPS	Uninterruptable Power Supply
VFD	Variable Frequency Drive
VMU	Vertical Merge Unit
VSU	Vertical Sort Unit
WAN	Wide Area Network

Table 2: Glossary

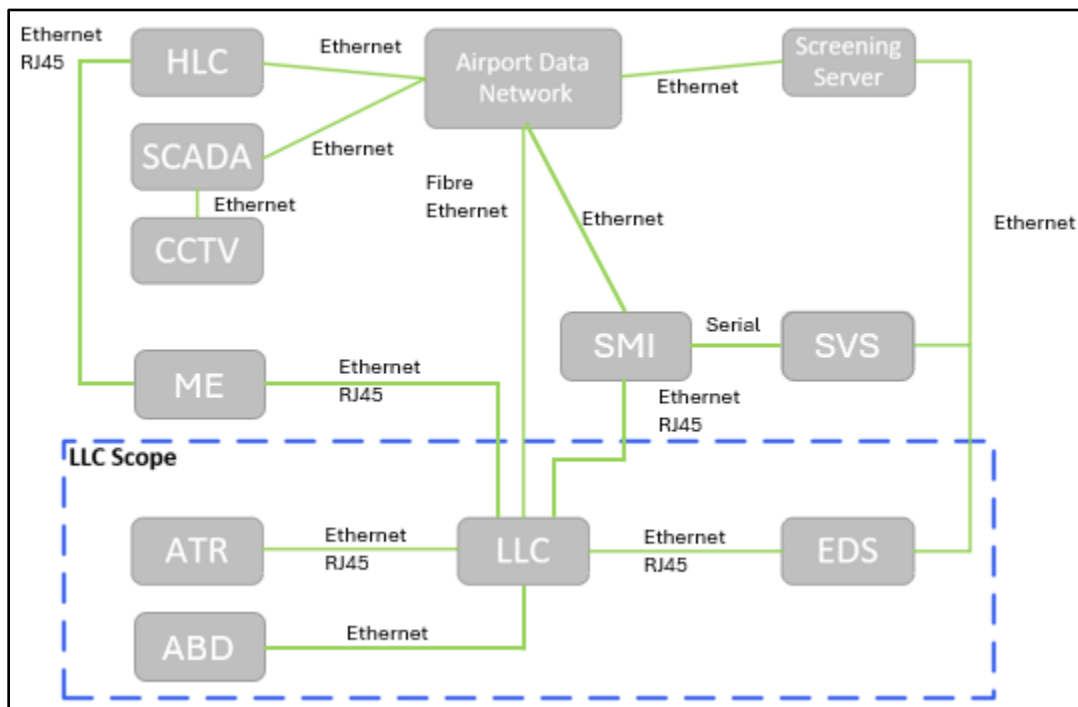
2. CONTROL SYSTEM INTERFACE OVERVIEW

2.1. System Topology

Function Layout

The diagram below demonstrates the LLC scope within the relevant network topology, providing a clear representation of its place in the overall system.

2.1.1.



2.1.2.

Figure 2: Network Function Layout

Network Redundancy

Please refer to the DTB Network Topology for the network layout for each CCP & FCP.

2.1.2.1. Distribution Layer

Network configuration for the distribution layer incorporates several key design elements to ensure efficient and reliable communication throughout the system. These elements are as follows:

- The distribution layer utilises EtherChannel connection, a highly robust architecture that facilitates rapid convergence and minimises downtime in the event of network failures.
- At the core of the distribution layer are the two stacked distribution switches.
- Each PLC subsystem is equipped with an access switch that links to the distribution switches via an aggregation connection.

EtherChannel connections provide fault-tolerance and high-speed links between switches.

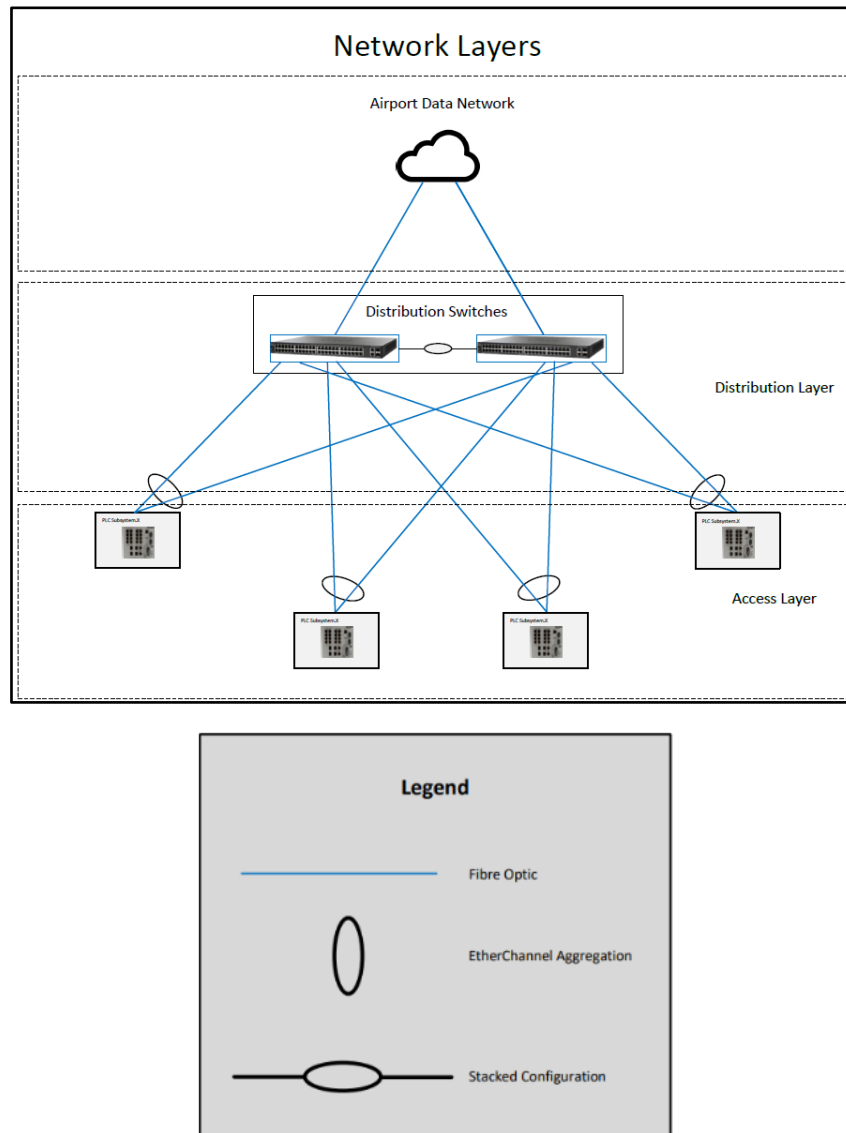


Figure 3: EtherChannel Configuration Example

2.1.2.2. Access Layer

At this level, access switches connect directly to the PLC OT Network, serving as a node for data uplink and facilitating communication between different system components. Additionally, the access switch acts as a Device Level Ring (DLR) supervisor, forming an OT network. Each FCP switch is configured to a DLR node.

DLR topology offers substantial benefits to the network's overall resilience, providing swift recovery and fault tolerance in situations where network failures or device issues occur. When faced with a failure, the DLR identifies and isolates the affected segment, and reconfigures the network autonomously to ensure uninterrupted communication between the functioning components.

Additionally, each PLC has a secondary connection to the access switch of the neighbouring subsystem. This redundancy ensures that in the unlikely event of a failure with the PLC's primary access switch, the local OT network's I/O will continue to function. This ensures that the subsystem remains operational, and external communications can still be routed via the secondary IT connection.

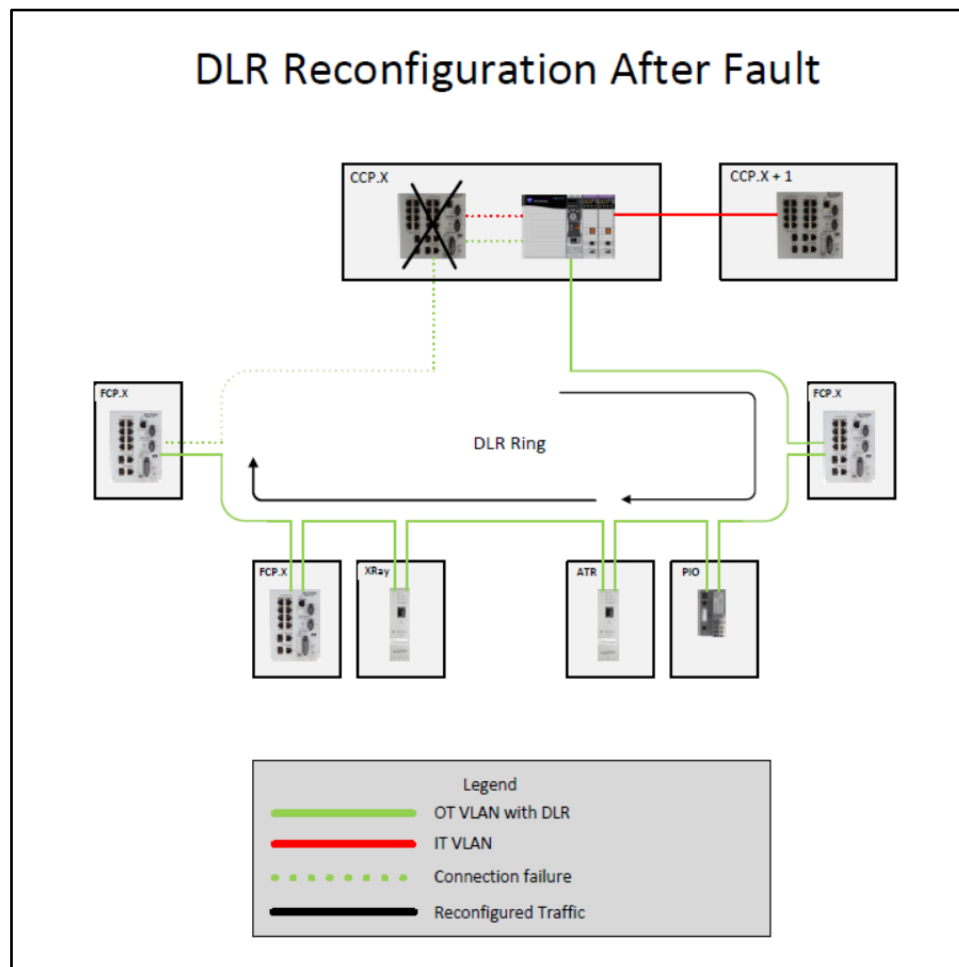


Figure 4: DLR Reconfiguration After Fault

2.2. LLC Hardware

PLC

2.2.1.1. Outbound

DTB outbound main control system utilises multiple Rockwell 1756-L83E controllers. This model offers high-performance capabilities and advanced processing power. This model provides fast scan times, supports enhanced communication, and has a large memory capacity. Each controller is equipped with two 1756-EN4TR dual-port network adapters dedicated to IT and OT networks. Additionally, the controller supports CompactFlash cards for non-volatile memory storage.

2.2.1.2. Firmware Revision

All controllers utilised in the system will be equipped with Rockwell's firmware V33, ensuring compatibility across all devices.

2.2.1.3. Programming

The PLC software uses ladder logic language for much of the PLC software, ensuring a well-organised and easily understandable programming approach. For more complex algorithms, structured text is employed, aiming to provide concise and readable syntax while maintaining efficient system operation. The software includes detailed comments in English, which offer clear explanations and insights into the code's functionality.

2.2.1.4. Variable Naming Conventions

The variable naming convention utilised in the PLC programme is designed to be organised and logical, facilitating easy understanding and maintenance of the code. Variable names are chosen to be mnemonic, effectively conveying their intended use to casual observers and enhancing readability.

2.2.1.5. Fieldbus

All new outbound subsystems utilize the Ethernet/IP (EIP) protocol for both local and remote I/O communication. This architecture is supported by Stratix 5200 (**1783-CMS20DP**) industrial-grade switches, providing network connectivity to Rockwell 1756-EN4TR adapters, SEW MOVIGEAR® drives, Explosive Detection Systems (EDS), Automatic Tag Readers (ATRs), and Bihl+Wiedemann BWU3683 Ethernet/IP AS-i gateways.

The AS-i gateways serve a dual function—handling standard I/O communications while also acting as AS-i safety monitors. Each gateway supports two AS-i buses, which manage the safety input signals from field devices such as Emergency Stop (E-Stop) buttons.

2.2.1.5.1. Fieldbus Health Monitoring

The PLC continuously monitors the health status of all fieldbus-connected devices on both the Ethernet/IP and AS-i networks. For detailed information on SCADA representation please refer to the Airport (BNE) DTB HLC Phase 2 Functional Design Specification section 5.2.

2.2.1.5.2. Ethernet/IP Devices

The PLC uses the GSV (Get System Value) instruction to retrieve each module's Connection Status. This includes:

- **Connected:** Communication is established and operating normally.
- **Faulted:** The device is in an error state or communication has been lost.

2.2.1.5.3. AS-i Gateways

General bus health indicators and fault signals are read into the PLC over Ethernet/IP using the diagnostics port of the Gateway. However, detailed diagnostic information (e.g., node-specific faults, configuration issues) are accessed via the gateway's built-in web server. This web interface provides a comprehensive view of the AS-i network and is made accessible directly from SCADA via a hyperlink.

2.2.1.6. Controller Redundancy

PLCs for any subsystem whose failure would affect more than 25% of the capacity of a specific function within the BHS are set up in a redundant configuration. In case of a failure incapacitating the primary PLC, operation team is notified via the SCADA system. Operator can then manually switch over to the secondary PLC to resume operations via the steps below.

While the primary PLC is connected to the IT network through the access switch in its local panel, the secondary PLC is linked to the access switch of the adjacent subsystem, ensuring switch redundancy. In the event of a switch failure, the primary PLC that it serves will automatically fail over to the secondary PLC. A SCADA alarm is generated in the event of a switch failure.

From the Security area onwards, tracking data is constantly synchronised between the primary and secondary PLCs, ensuring the tracking state of any bag within the system is retained after a failover.

When a Primary PLC fails, operators are required to follow the procedures below:

- Gain access to the faulted PLC.
- Disconnect the power from the primary PLC.
- Toggle a key switch to activate the standby, secondary PLC.
- Clear any faults.
- Switch over to the secondary PLC via SCADA.
- Startup the system.
- Contact Alstef support.

Please refer to the Airport (BNE) DTB HLC Phase 2 Functional Design Specification , section X for further information on the PLC redundancy indication and control from the SCADA.

2.2.1.7. PLC Assignment

2.2.1.7.1.1. Overview

For the current stage of works in the CBS, the new outbound system is split into 5 PLC groups. The CCP4A panel will contain the PLC for CCP5, this will only control CCP4A-FCP5. Please refer to the CCP Splitting Layout Overview document for further details.

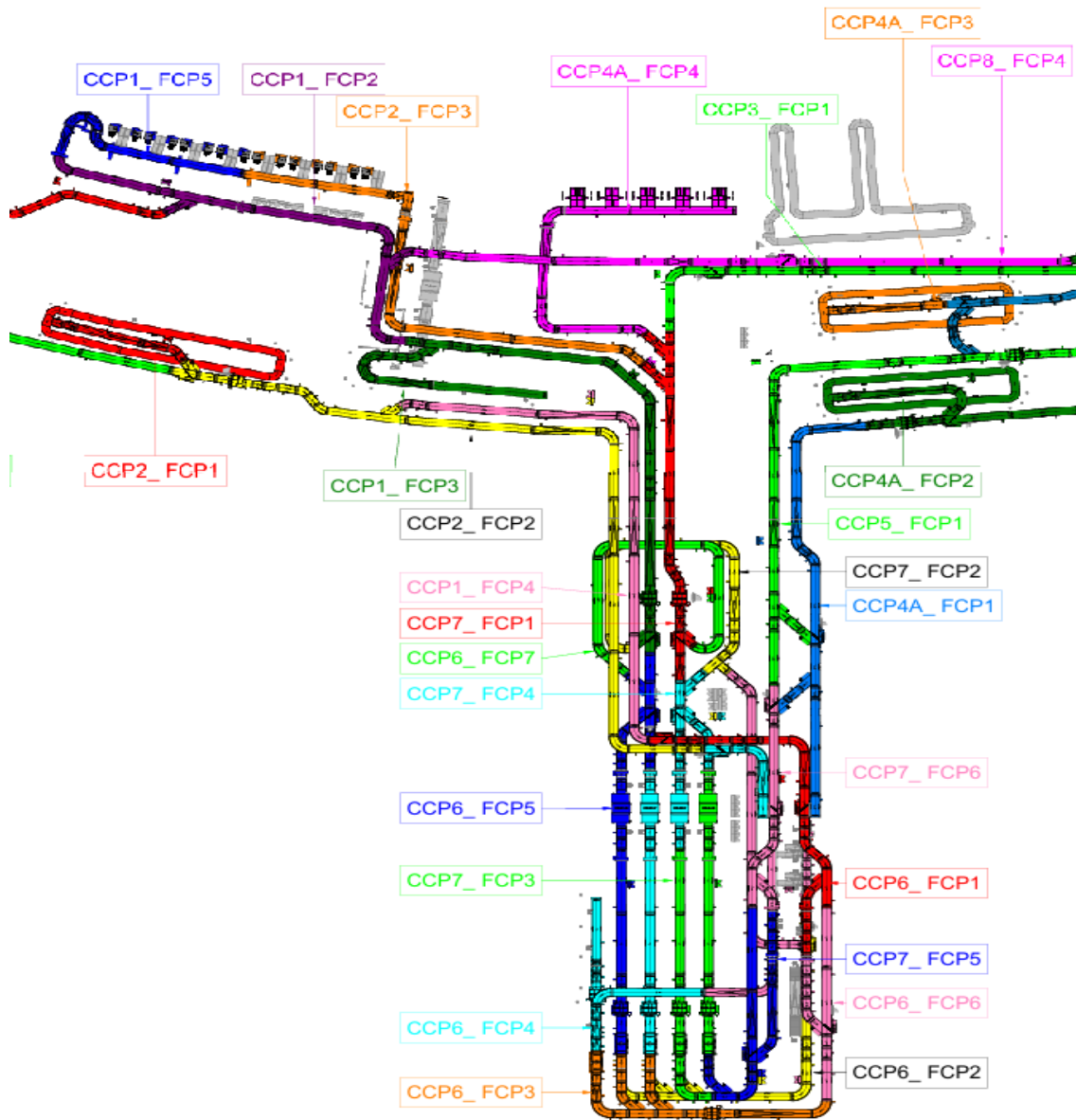


Figure 5 4 Lane Strategy CBS PLC Assignment

2.2.1.7.1.2. PLC Split Scheme

The entire BHS is controlled by a series of PLCs. The boundaries between the PLC's are positioned such that failure of one PLC will not reduce system redundancy built into the mechanical layout.

Network Switch

2.2.2.1. CCP

2.2.2. A Stratix 5200 (1783-CMS20DP) industrial managed switch serves as the access switch for each PLC control system. Up to four of these switches form an REP ring along with distribution switches. It supports integrated DLR with up to three-ring support. Each switch is connected to the BAC network using a fibre network connection.

2.2.2.2. FCP

One level below the CCP, each FCP is equipped with a Stratix 5200 (1783-CMS10DP) industrial managed switch. This switch serves as a Device Level Ring (DLR) node, providing a resilient network architecture for field I/Os.

2.2.2.3. VLAN Configuration

Each PLC is equipped with two network adapters, 1756-EN4TR, dedicated to IT and OT VLANs respectively. The IT VLAN is used for SCADA, HLC, and programming access, while the OT VLAN accommodates all I/Os, ATRs, and EDS machines. This separation of IT and OT VLANs ensures a structured network architecture, providing enhanced security and organisation for improved system performance and manageability.

2.2.3. Safety Monitor

2.2.3. In addition to its communication capabilities, the Bihl+Wiedemann BWU3683 EIP AS-i gateway also serves as a safety monitor, ensuring that the system's safety components function effectively and reliably. The gateway is responsible for processing and monitoring safety-related inputs, such as those from Emergency Stop devices, and initiating appropriate responses to power supply connection.

Check-in

2.2.4.1. Scale

Check-in scales serve the purpose of allowing operators to verify if a bag is overweight according to airline limits. Since these airline limits are much lower than the BHS equipment capacity, the BHS does not need to acquire bag weight information, and LLC connectivity to the scale unit is not within the scope of the BHS project. For the ABD scales, these serve as a contained part of the ABD unit, only the ABD unit can stop the ABD conveyors due to an overweight bag.

2.2.4.2. ABD

Alstef collaborates with the ABD supplier in the Central check-in area to ensure that the Baggage Handling System (BHS) provides the necessary interfaces and system capabilities to fully utilise the designed functionality of the ABDs.

- **Central Check-in Area:**

The area is equipped with Amadeus ABD units, which will be fully supported and interfaced by the BHS (Please refer to section 3.2 for details).

The BHS PLCs will integrate with the ABD controllers to enable the exchange of data and control signals required for system operation. Detailed interface design and communication requirements are described in the ABD FDS and/or the applicable interface specification

Motor Controller

2.2.5. The new BHS utilises motors integrated with SEW Movigear decentralised Variable Frequency Drive (VFD) solution. Each motor functions as an EtherNet/IP (EIP) device on the Operational Technology (OT) network.

ATR

2.2.6.

ATRs are directly connected to access switches using a star topology and configured to the OT VLAN. Each ATR for the new system has a redundant controller configuration, both primary and secondary controllers have identical IP addresses and are connected to the access switch but only one control is active at any given time. Failover is initiated from the ATR end, and LLC reports the event to SCADA. The communication between the ATR and the PLC for digital I/O and messages is facilitated via Ethernet/IP (EIP), ensuring a reliable and efficient data exchange between these critical system components. See Section 10 ATR for

2.2.7 details.

MES

The LLC system is responsible for controlling the conveyors of the Manual Encode Station (MES) route. However, the ME station and its associated Human-Machine Interface (HMI) and input devices are part of the HLC system. Please refer to the Airport (BNE) DTB HLC Phase 2 Functional Design Specification, section 4.4 for further details .

CBSS

EDS machine communicates with the PLC through the OT Ethernet device level ring using a DLR interface module (1783-ETAP). The communication protocol will be EtherNet/IP using Cyclic I/O. This communication will be using a Type 2 Comms Interface fitted with an AnyBus Gateway AB7831. This network is separate from the CBS machine's internal network, which is not within the scope of the LLC.

The EDS Control System features dedicated communication modules fitted within its interface chassis/assembly. Please refer to the RTT BHS Protocol Specification - Airport -TSA, section 4 for protocol details.

In addition to the physical data interface for item data, transport management, and machine control, EDS is connected to the BHS via a set of dry contact wired discrete signalling circuits. These circuits are:

- Two EDS E-Stop circuits (front & rare)
- EDS E-Stop Remote Reset
- EDS Interface E-Stop Feedback
- BHS Interface E-Stop Feedback

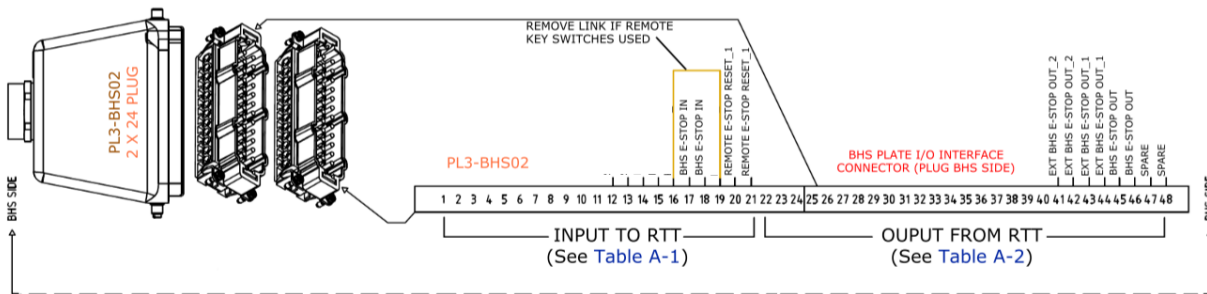


Figure 6 EDS RTT110 Interface connector – 48 Pin

Table A-1: BHS INPUT SIGNALS	
BHS E-STOP IN BHS E-STOP IN	Pins #17 and #18 are free of voltage E-Stop signal from the BHS system to the RTT®110.
REMOTE RESET_1_1 REMOTE RESET_1_2	Pins #20 and #21 are the free of voltage input signal for resetting the machine safety circuits from remote.
Table A-2: BHS OUTPUT SIGNALS	
EXT BHS E-STOP OUT_2_1 EXT BHS E-STOP OUT_2_2	Pins #41 and #42 are the free of voltage E-stop signal from the RTT®110 to the BHS system. This is normally a closed contact of an E-stop button. It is an open circuit while the E-stop button is pressed and closes when the E-stop button is released and therefore is independent on the RTT®110 power condition.
EXT BHS E-STOP OUT_1_1 EXT BHS E-STOP OUT_1_2	Pins #43 and #44 are the free of voltage E-stop signal from the RTT®110 to the BHS system. This is normally a closed contact of an E-stop button. It is an open circuit while the E-stop button is pressed and closes when the E-stop button is released and therefore is independent on the RTT®110 power condition.
BHS E-STOP OUT BHS E-STOP OUT	Pins #45 and #46 are the free of voltage E-stop signal from the RTT®110 to the BHS system. This is normally a closed relay contact of a safety relay in the machine and it remains open (E-stop condition) during machine power down.

Figure 7 Rapiscan RTT110 IO for the 48 Pin Connector

BMS

Each PLC is fitted with a dedicated hardwired input for fire alarm signals. This input is wired to a relay in the fire relay panel, located next to the PDP and routed via the PDP to the PLCs within the same fire zone.

2.2.9.

2.2.10.

Regardless of how the signals are distributed, each PLC retains its own independent hardwired input for receiving fire signals.

UPS
The BHS control system is supplied with power from the UPS power supply network, which is provided by Alstef. This network is intended to supply various components, including:

- PLCs
- AS-i Bus
- 24VDC power supplies
- Control circuits
- EIP remote I/O
- LLC Network switches
- ATRs

2.2.11.

SCADA

SCADA graphics use colours to indicate the various operational states for different devices. Please refer to the Brisbane Airport (BNE) DTB HLC Phase 2 Functional Design Specification, section 5.2 for the colour assignments that are specific to items of equipment (e.g. conveyor, diverter, ATR, etc).

2.2.12.

HLC

The PLCs connect to multiple HLC modules, such as sortation and system map, via Ethernet. The messaging utilises the Common Industrial Protocol (CIP) over a standard Ethernet physical layer.

2.2.13.

For detailed information on the message interface, please refer to the PCL-UL Transaction Messages for further details, including the message map.

Power Meter

Power meters are installed in the PDPs, CCPs, and FCPs to monitor the energy consumption of the BHS system. The BHS PLCs acquire data from the power meters via the EtherNet IP protocol and send it to the SCADA system for monitoring and reporting purposes.

2.2.14.

For additional information, please refer to the Brisbane Airport (BNE) DTB HLC Phase 2 Functional Design Specification, Section 5.

Voltage Monitor

Voltage monitors are installed in the CCPs to monitor the health status of the three-phase supply voltage. The monitor's relay output wired back to the CCP PLC input as fault feedback. If a voltage fault is detected

(unhealthy condition), a critical fault is generated and all devices controlled by the CCPS will be stopped. An alarm will generator on the SCADA system.

Surge Arrester

2.2.15. Surge arresters are installed in the CCPs to protect equipment from transient overvoltages caused by events such as lightning strikes, switching operations or fault conditions. When a surge is detected, the device activates a relay that is wired to the CCP PLC input. This fault condition generates an alarm on the SCADA system.

3. CHECK-IN AREA

3.1. General

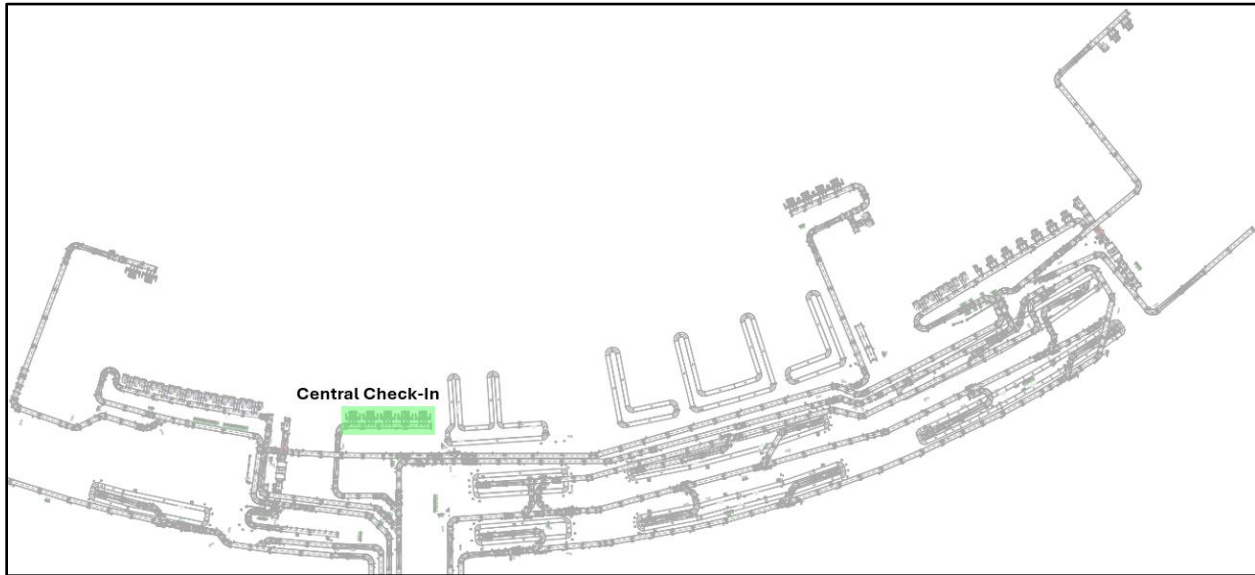


Figure 8: *Check-In Location*

3.2. ABD

3.2.1.

General

Alstef collaborates with the ABD supplier to ensure that the BHS provides adequate interface and system capabilities to effectively utilise the designed functionality of the ABDs.

The functionality of the ABD will be covered by the vendor, please refer to Amadeus ABD FDS **Signal Interface**

The signals exchange between the ABDs and the BHS PLC is carried out via EthernetIP remote IO.

The ABD sends the following signals to the PLC :

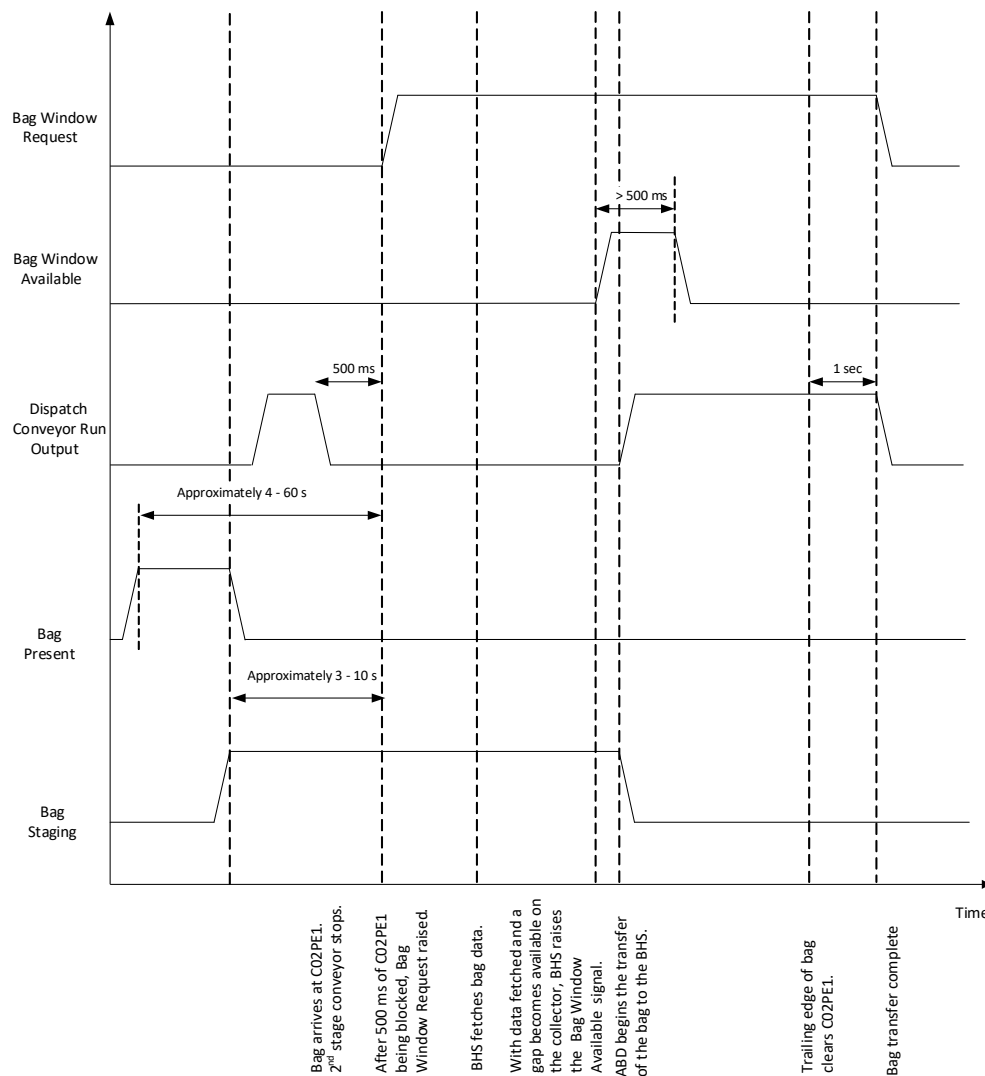
- Bag Window Request – ABD is requesting to induct a bag to the collector conveyor.
- ABD E-stop Fault – ABD E-Stop is activated.
- ABD Operational/Healthy – ABD is operational and healthy.
- ABD Security Intrusion – Security intrusion passing through the ABD light curtain.
- ABD Fault – Inform the BHS the ABD is in fault (unspecified fault from the ABD).
- ABD PE Jam Fault – ABD has a jam on the merge conveyor.
- Bag Present – Bag is present at the ABD (BHS to use this for ECM on the collector conveyors).
- Bag Staging – Bag data has been sent from HLC to ABD.

The PLC sends the following signals to the ABD:

- Bag Window Available – ABD has been given an allocated window that it can induct the bag into.
- BHS Operational/Healthy – BHS is operational and healthy.
- ABD Security Intrusion feedback – Mirror of the Security Intrusion signal the ABD sends.
- Please refer Amadeus FDS, section 5.1 for further details.

The Diagram below illustrates the interface signals timing and handshaking.

ABD to BHS Bag Transfer Signal Handshaking



3.2.2.

Safety Interface (add reference to security door)

Pressing an ABD Emergency Stop will trip the BHS collector conveyor safety zone. While the BHS zone is affected, only the specific ABD where the button was pressed will enter an E-stop state. This triggers a red

beacon and local sounder per Section 16.4. Conversely, any BHS Emergency Stop (including the FCP E-stop) within the collector conveyor zone will trip all connected ABD safety zones.

The safety interface between the ABD and BHS systems are done via AS-I safety relays, described below:

- Input relay taking the feedback **from** dual channel volt-free contacts from the Amadeus ABD – for the status of the ABD emergency stop pushbutton.
- Output relay providing the feedback of dual channel volt-free contacts **to** the Amadeus ABD – for the status of the BHS emergency stop pushbuttons in the shared safety zone.

The Amadeus ABDs do not have a built-in emergency stop reset; therefore a keyed emergency stop reset will be used to reset the ABDs. This will be installed front of house near the ABD check-in area.

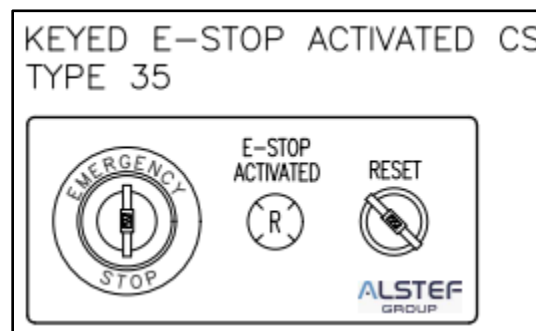


Figure 9 Keyed Emergency Stop with keyed Reset for the ABDs

3.2.3.

Low Level Signal Interface

The ABD communicates with the BHS PLC over Ethernet/IP. The ABD must be supplied with Ethernet/IP I/O modules.

The signals exchanged between the BHS system and ABD are described in Section 5.1 of the Amadeus ABD FDS .

3.3. Induction Window Allocation

In a check-in island with a long collector and multiple ABD check-ins, it is crucial to have an algorithm that ensures each check-in point has an equal opportunity to induct bags into the collector, regardless of their position along the collector (upstream or downstream).

The purpose of induction windowing is to ensure that bags injected from multiple ABDs are correctly spaced and sequenced onto a shared collector conveyor without collision, congestion, or tracking errors. This is critical for maintaining efficient throughput and avoiding merge jams.

For the 4-lane strategy, all window allocation is opportunistic. The ABDs will wait for a window if none is present on the takeaway conveyors of CC0301 and CC0302.

Functional Description

- 3.3. • Each ABD zone operates under a collector window allocation algorithm managed by the PLC.
- A “window” is a reserved, position-based space on the collector conveyor allocated to an individual ABD.
- The windowing logic prevents overlapping injections from multiple ABDs and ensures safe, FIFO-aligned spacing.

3.3.2. Key Functions

3.3.2.1. Window Reservation

- When an ABD requests to send a bag, the controller checks the availability of an open window.
- If available, the bag is released and tracked within the allocated window.
- If no window is available, the bag will wait at the ABD until the next safe window opens. The ABD can still accept another bag on the scale conveyor while there is a bag waiting on the merge conveyor.

3.3.2.2. Priority Handling

- Priority can be given based on either ABD location (e.g., furthest first) or timestamp of the request.

3.3.2.3. Spacing Enforcement

- Spacing between bags is enforced using PPI-based tracking on the collector conveyors to ensure the downstream BHS can reliably detect and track bags.

3.3.2.4. Collision Avoidance

- Bags will not be released if:
 - Another bag is still occupying the window.
 - The conveyor is not in the correct direction of flow – Only applies to reversible collector conveyors.
 - A fault condition is active downstream (e.g., Jam, Estop, Technical Fault).

3.3.2.5. Direction Change Handling (if applicable)

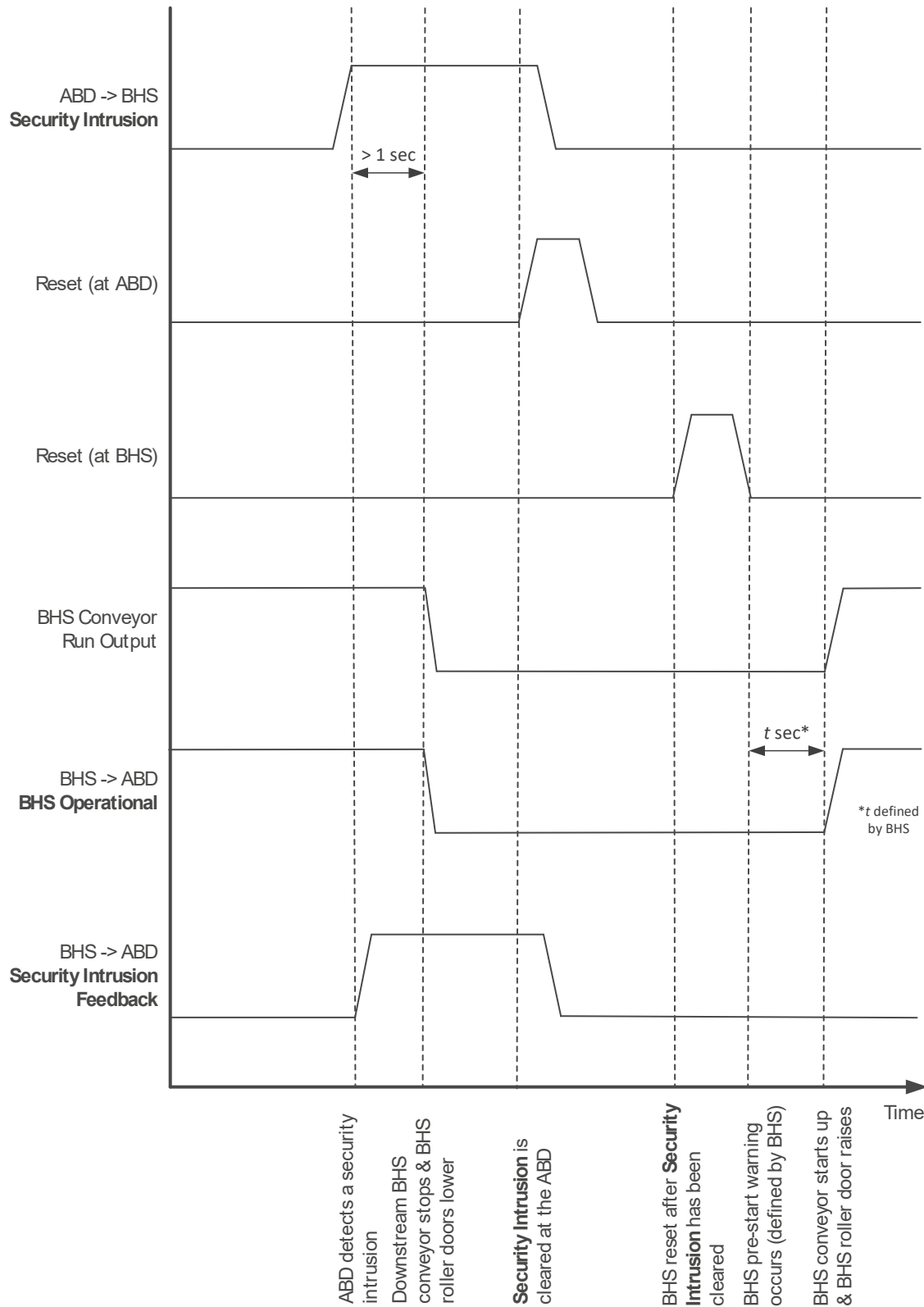
- If collector direction changes, existing window reservations are cancelled or reallocated.
- Bags already in motion will be allowed to exit before new ones are inducted.

3.3.2.6. ABD Security Intrusion Detection (update based on ABD FDS)

Each ABD unit is designed to detect unexpected objects entering the ABD induction tunnel, this is deemed a security intrusion to the BHS. As part of the interface between the ABDs and BHS, each ABD can send a security intrusion signal if a security intrusion is detected. The BHS must send back a security intrusion feedback signal to the ABD.

The BHS will respond to a security intrusion:

- Immediately stop the collector conveyors in the check-in area.
- Close the security door at the check-in area.
- Security intrusion shall be investigated and cleared by Operations team.
- Once the intrusion is cleared, the intrusion fault must be reset locally at the affected ABD by authorised personal. The Security Intrusion feedback signal will turn off to inform both the PLC and SCADA.
- The restart key switch located in the check-in area (the same one as for an ABD Estop activation reset) must be switched on and off to restart the collector conveyors and open the security door.



4. CONVEYOR

4.1. General

The Conveyor system is the core of a BHS. Each conveyor is equipped with at least one Photo Electric Cell (PEC) or photo eye near the downstream tip and a pulse wheel encoder if bag tracking is implemented. These components work together to ensure accurate and efficient transportation of bags throughout the system. PECs detect the presence of bags, while pulse wheel encoders can accurately register belt movement.

4.2. Conveyor Operation Status

Each conveyor has the following operation status:

- Disabled/Inactive/Shut down.
- Pre-Start Warning.
- Running.
- Indexing – Accumulating.
- Timed out/energy conservation mode (ECM).
- Cascade/Metering stopped.
- Isolated Status.
- Faulted.
- Fault interlocked – affected by nearby fault condition.
- Emergency Stopped.
- Running by manual override.
- Stopped by manual override.

4.3. CCP Subsystem

Subsystem Start

This section defines the startup logic for each subsystem within the Baggage Handling System (BHS). Startup can be triggered manually or automatically and is subject to safety interlocks and operational conditions.

4.3.1.

4.3.1.1. Startup Preconditions

Subsystem startup is only permitted when all the following conditions are met:

- A valid signal from the building fire alarm system must be present.

If an E-Stop or technical fault occurs, the conveyor or conveyors impacted will not startup and will remain stopped. This will not stop the reset of the unimpacted conveyors from subsystem starting.

4.3.1.2. Startup Triggers

Subsystem startup can be initiated by any of the following:

- Start pushbutton on the Central Control Panel (CCP).
- Start pushbutton on the Field Control Panel (FCP).
- SCADA-issued subsystem startup request.
- Bag approaching from an upstream conveyor or diverter (triggered by the last two photoeyes of the upstream subsystem).
- Start pushbutton on a local control station located at the transfer lines and at check-in.

4.3.1.3. Startup Sequence Behaviour

The start-up sequence will be as below:

- A 5-second audible and visual warning is activated using amber beacons and sounders positioned along the conveyor route and at the CCP & FCP(s). Please refer to [Section 16 Sounder and Beacon Indication](#) for in-depth information.
- The subsystem conveyor startup follows a downstream-to-upstream sequence:
 - The most downstream conveyors start first. These conveyors are identified within the control system as “can Start First” conveyors. the following diagram illustrates location of these conveyors. For full list of conveyor names please refer to the appendices section 23.9.
 - Each upstream conveyor starts after a 0.5-second delay, ensuring smooth and coordinated flow.
- If other parts of the system are already running:
 - Audible and visual alarms are activated only for the subsystem being started.

Subsystem Stop

4.3.2.1. Flush and Stop

4.3.2. This section defines the behaviour of a Flush and Stop sequence, which safely clears bags from a subsystem before shutdown. The logic ensures no bags remain stranded in the subsystem and that conveyors stop in a controlled upstream-to-downstream manner.

4.3.2.1.1. Flush and Stop Preconditions

Flush and Stop Preconditions are as follows:

- Subsystem must be in a healthy and safe state.
- No Emergency Stop is active.
- No Fire System Inhibit is active.

4.3.2.1.2. Flush and Stop Triggers

Flush and Stop can be initiated by:

- Stop pushbutton at the Central Control Panel (CCP).
- Stop pushbutton at the Field Control Panel (FCP).
- SCADA-issued Stop request.
- SCADA-configurable automatic shutdown timer expiration (inactivity-based).

The subsystem shutdown timer defines the maximum allowed time the subsystem can remain idle (i.e., with no detected baggage). If this time elapses without any PEC activation, the flush sequence begins. The timer resets to zero if:

- Any Photo-Electric Cell (PEC) in the subsystem detects a bag.
- A bag is detected approaching from an upstream subsystem.

4.3.2.1.3. The presence of any active Automatic Bag Drops (ABDs) will prevent automatic initiation of the Flush and Stop sequence. Flush and Stop Sequence Behaviour

- The first (most upstream) conveyor in the subsystem begins the Flush and Stop process
- A conveyor starts its Flush Shutdown Timer, which is SCADA-configurable:
 - By default, this is set to 150% of the time it takes the conveyor to run its full length.
 - If the conveyor's Photo-Electric Cell (PEC) detects a bag during this period, the timer resets to zero and any conveyors stopped restart.
- Once a conveyor shuts down, the next immediate downstream conveyor begins its Flush and Stop process.

- If a conveyor is already in Energy Conservation Mode (ECM) and all upstream conveyors have stopped, the conveyor will shut down immediately without reactivating.

4.3.2.2. Immediate Stop

This section describes the Immediate Stop function, which allows the operations team to halt a subsystem without completing the Flush and Stop procedure. Upon activation, all conveyor operations within the subsystem stop simultaneously, and the subsystem transitions to an inactive state.

4.3.2.2.1. Immediate Stop Triggers

- Long press (5 seconds) of the Stop pushbutton on the Central Control Panel (CCP).
- SCADA-issued Immediate Stop request by pressing the Lock Out pushbutton on the CCP Popup.

Pre-Start Warning

4.3.3. This section describes the Pre-Start Warning process, which is required when a conveyor recovers from a fault or override operation, or the system is started after a shutdown. The warning sequence ensures that field personnel are alerted before any conveyor resumes operation.

- A 5-second pre-start warning is issued before the conveyor resumes operation.
- During this period, both audible and visual alarms are activated in the field:
 - **Visual:** Amber beacon lights
 - **Audible:** Beacon sounders
- The audible alarm during this process cannot be muted.
- The pre-start warning sequence is required after fault recovery or override reset.

For further information on beacon and sounder indications, please refer to [Section 16 Beacon And sounder Indication](#).

4.4. Cascade Stop/Start

This section outlines the logic governing cascade-based control between interconnected conveyors. Cascade logic ensures conveyors only transfer bags when the next downstream conveyor can receive them, preventing system congestion and allowing for safe bag handling.

Cascade Stop Conditions

Cause	Effect (System Response)
4.4.1. A bag reaches the cascade PEC of a conveyor, and the downstream conveyor is stopped (and not in ECM) for any reason.	The upstream conveyor stops and enters a cascade-stopped state.
Cascade stop occurs during a transition between two conveyors.	Both conveyors involved stop in the transition simultaneously when downstream is unavailable.
	Both conveyors involved in the transition start simultaneously when downstream becomes available.
Conveyor is in cascade-stopped state, and PEC is cleared but downstream is still not running.	Conveyor remains stopped (cascade stop is latched and cannot be cleared by unblocking PEC alone).

Table 3 Cascade Stop Conditions

4.4.2. Cascade Restart Condition

Cause	Effect (System Response)
Downstream conveyor restarts after being stopped	The upstream (cascade-stopped) conveyor restarts after a cascade start delay (0.5 seconds).

Table 4 Cascade Restart Condition

Cascade Start Delay

A default delay of 0.5 seconds shall be applied to ensure smooth transition, avoid bag collisions, and allow the downstream conveyor to stabilise before receiving new bags. The delay is specific to each conveyor and can be adjusted during commissioning via PLC programming software.

4.5. Energy Save Conservation Mode

Energy Conservation Mode (ECM) is implemented to reduce energy consumption and mechanical wear by stopping conveyors when no bags are detected for a configured period of time. See Section 22.10 [Energy Conservation Mode](#) for details on Make-Up Loop ECM.

During a fire alarm, conveyors adjacent to fire/security roller doors will attempt to clear any bags (even if in ECM).

ECM Activation

Cause	Effect (System Response)
4.5.1. Conveyor is running and no bag is detected for the configured timeout period.	Conveyor stops and enters Energy Conservation Mode (ECM).
ECM timeout duration.	Default is set to the time required for the conveyor to run one full rotation of the belt; configurable via SCADA.
ECM enable/disable switch toggled via SCADA (subsystem-wide setting).	ECM is either activated or deactivated across the affected subsystem.

Table 5 ECM Activation Causes

The ECM timeout is configurable in SCADA from 10 seconds to 30 minutes. The delay is specific to each conveyor.

4.5.2. ECM Wake-up Condition (Automatic Operation)

A conveyor in ECM can automatically restart under the following conditions:

- The conveyor's own PEC is toggled (i.e., detects a bag).
- An approaching bag is detected at the upstream PEC.
- A plough diverter or VSU sends a wake-up signal indicating a bag is about to be diverted.

4.6. Metering

Metering is the function of introducing a minimum gap between consecutive bags. This is typically required to satisfy the operating constraints of downstream equipment such as ATRs, EDS machines, and high-speed ploughs.

The required spacing is determined by the system's maximum flow rate, average bag length, and conveyor speed. While any conveyor can be configured to perform metering, the function is usually assigned to queueing conveyors.

There are two methods used to implement metering functionality:

- Passive Metering (gap creation through speed differential).
- Active Metering (gap control via PLC logic and PE sensors feedback).

Please refer to appendices, section 23.4 for a list of metering conveyors (Active Metering).

Passive Metering

4.6.1.

Passive metering is a gapping technique used to introduce separation between consecutive bags to maintain consistent product flow. It typically consists of a series of two or three conveyors, where each downstream conveyor operates at a higher speed than the upstream one. As bags transfer from a slower upstream conveyor to a faster downstream conveyor, a natural gap is created due to the speed differential.

This technique does not provide precise or repeatable spacing and is therefore not considered a true metering function. Its primary purpose is to reduce the likelihood of bags overlapping or bunching together. Passive metering is mechanical in nature and does not rely on sensors or PLC-based control logic.

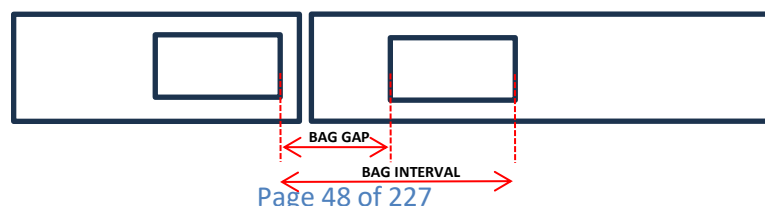
4.6.2.

Active Metering

Active metering is controlled by PLC logic and is typically implemented on the last conveyor in the metering group. This conveyor acts as the metering conveyor.

When the leading edge of a bag activates the metering photoelectric cell (PEC), the conveyor stops. It will only restart when both of the following conditions are met:

- Bag Interval is greater than the pre-set minimum*.
- Bag Gap is greater than the pre-set minimum*.



← Bag Flow Direction

*Metering bag interval & bag gap distance can **only** be configured within the PLC program.

4.7. Indexing

Indexing is the function added to a conveyor to enable it to accumulate bags when the downstream conveyors are unable to receive bags. It may also be initiated to absorb bag flow in a situation when the line feeds on to another and a defined flow-rate threshold on the receiving line has been exceeded. It leaves indexing mode when the downstream line returns to full flow.

Cause	Effect (System Response)
Downstream conveyors are unable to receive bags or receiving line has exceeded defined flow-rate threshold.	Bag activates the cascade stop sensor of an upstream conveyor.
	Indexing conveyor* (or multiple connecting conveyor) runs and continues to run for a configurable amount of distance (default is 500mm).
	Once the head end PEC of the indexing conveyor is activated then the conveyor goes into a cascade stop.
	Once the downstream conveyor or receiving line is cleared, the system resumes operation, and the conveyors leave indexing mode.

Table 6 Indexing Cause & Effect

*All GT conveyors that are level (not inclined or declined), excluding power curves and metering belts, can have indexing enabled.

*Laterals that are used as make-up devices are set to indexing mode by default.

Please refer to appendices, section 23.5 for a list of Indexing conveyors.

4.8.1. 4.8. Manual Override

General

Conveyor functions can be overridden using the SCADA system or individual SEW local maintenance selector switches. In addition, the cascade PEC and pulse encoder can be overridden via the SCADA.

Due to security considerations, SCADA override is not available by default to regular default user/guest operators. The hierarchy and role definitions for SCADA access levels are formally defined in the HLC/SCADA FDS document. Note that SCADA override is requires higher user privileges for components like conveyors, PECs, and pulse wheels within the CBS security tracking zone. This restriction is in place due to the potential for compromised tracking accuracy and risks associated with undermining the CBS fail-safe mechanisms.

SCADA Conveyor Override

Once SCADA override is enabled on a conveyor, there are two available options: "Force Run" and "Force Stop". In "Force Run" state, the conveyor operates continuously and disregards any faults other than emergency stops and isolate fault. On the other hand, in "Force Stop" state, the conveyor remains idle.

4.8.2 When a conveyor that has been running in automatic control is switched to manual override, it defaults to the "Force Stop" state. If it is switched to the "Force Run" state, a 5-second pre-start warning is activated before the conveyor starts if there is no active Estop or Isolate state. Using the override function, individual conveyors can be started and stopped regardless of whether the system is running or not.

When in the manual override state, there is no automatic control of the conveyor, which means there will be no cascade stopping or energy conservation mode. The conveyor will indicate to the upstream conveyor that it is ready to receive when the "Force Run" state is selected, and the conveyor run feedback is positive.

Turning off the SCADA override will start a 5-second pre-start warning before the conveyor runs in automatic control again.

4.8.3. Cascade PEC Override

When a PEC sensor is overridden from the SCADA system, the signal from the device is effectively ignored, and it ceases to have any effect on the operation of the conveyor. This practice is generally used as a temporary measure during a PEC hardware malfunction.

When a cascade PE is overridden, the associated conveyor is effectively connected (logically) to the downstream conveyor. Both conveyors operate (start, stop, cascade stop) as if they were one conveyor responding to the cascade PEC of the downstream conveyor.

Tracking still functions in less critical areas when the cascade PEC has been overridden. The tracking model will be updated at the next functioning PEC downstream.

4.8.4 Overriding a Photo-eye that has caused a Jam condition, will allow the jam condition to be reset as per [4.9.3.6 Fault Reset](#).

Pulse Encoder Override

When a pulse encoder is overridden through the SCADA system, the control system relies on "simulated" pulses (created within the PLC program) to maintain operation. The speed obtained from the simulated pulse stream may be less accurate than that from the physical pulse encoder, potentially impairing tracking performance if the override mode is used for a prolonged duration.

SEW Local Maintenance Selector

Conveyors equipped with IE5 class drives each feature a SEW Local Mode Pendant. This device allows for individualised conveyor maintenance control, allowing a conveyor to be run locally by the SEW VFD.

4.8.5.

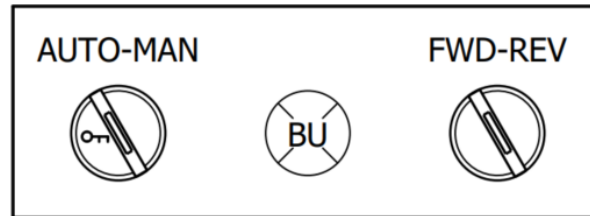


Figure 10 SEW Manual Selector Station

Control	Device	Colour
Auto/Manual	Two-Position Key Switch	Black
FWD/REV	Two-Position Selector Switch	Black
Local Mode Indicator	LED Indicator	Blue

Table 7 SEW Local Mode Pendant Station

By default, the key switch is set to the "Auto" position, allowing the PLC to control the conveyor in its normal operational mode. Switching the key to the "Man" position activates local mode, which takes over conveyor control through the "FWD/REV" selector switch. In local mode the following applies:

- The conveyor's forward direction and speed matches the Auto mode configuration.
- Reverse function can be disabled if it could harm the conveyor.
- The blue LED indicator illuminates when the drive is in local mode.

In Local mode, the conveyor is not affected by PLC software faults and latches such as jam and encoder fault, but can still be stopped by E-Stop, isolators, or any condition causing a power loss. Switching to Local mode does not clear existing software faults; upon returning to Auto mode, any pre-existing faults must be cleared and reset for normal operation to resume.

When the key switch is in the Auto position, the selector switch does not influence motor operation. These SEW Local mode Pendants are installed on IE5 class drives on standard conveyors. However, conveyors in VSU & VMU modules, horizontal diverter paddle conveyors, make-up carousels and actuators for switching diverter devices **do not** have the local mode pendants. These each feature a separate local control mechanism for maintenance purposes.

Transitions between Auto and Local modes, restoration of power after disconnection, or any interruption that causes the conveyor to stop while in local mode will require the Auto/Manual selector switch to be toggled out of and back into the 'Local Mode' position.

4.9. Fault and Recovery

General

The BHS control algorithm identifies abnormal operating conditions or events, signalling them as faults. Common faults include:

- 4.9.1. Bag jam
- Bag oversize (over-height/over-length)
 - Motor isolator
 - Motor feedback (run) faults
 - Encoder fault
 - Emergency Stop
 - Fire Alarm

When a fault occurs, the control algorithm may stop some or all controlled devices in the subsystem. Fault conditions are typically indicated by beacons, sounders, fault reset, and CCP/FCP indicators. Fault alarms also appear on the SCADA system, and the subsystem will not start while a fault condition is present.

The reset function, accessible through the reset button at the CCP or FCP, can reset all faults in the subsystem if the fault situation has been resolved (except for emergency stop faults). Motor feedback faults can only be reset at the CCP for the specific motor. Motor isolator faults are auto reset when the motor isolator is switched to the ON position.

Fault conditions (other than motor feedback and motor isolator faults) associated with field equipment like ploughs, VSUs/VMUs, and doors can be reset using one or more Fault Reset buttons located near the equipment that triggered the fault. An illuminated Fault Reset button indicates that the current fault can be reset with that button. Illuminated jam reset pushbuttons are also located throughout the BHS for resetting local jam faults after bag jams have been cleared.

If the fault has been resolved, pressing an illuminated Fault Reset button once will deactivate beacons and sounders, clear the alarm state, and initiate a subsystem start (including pre-start warning). If the fault has not been cleared, the first press will deactivate (mute) any sounders and leave the button illuminated. After the fault is cleared, pressing the Fault Reset button again will deactivate the beacons & sounders, clear the alarm state, the illuminated fault reset button will extinguish, and initiate a subsystem start with a pre-start warning.

Motor Isolator

4.9.2.1. Overview

Each conveyor motor is equipped with an electrical isolation switch to disconnect it from its power source.

4.9.2. This allows each motor to be completely de-energised during maintenance, repair, and other activities where unexpected motor operation would be hazardous. The control system monitors a separate contact within each motor isolator to determine the isolator's on/off status.

4.9.2.2. Cause

A Motor Isolator Fault occurs when a motor's isolator is switched off.

4.9.2.3. Effects

When a motor isolator is turned off, the following actions occur:

- The motor is stopped by the control system.
- Local blue beacons and sounders activate. Please refer to Section [16 Beacon And sounder Indication](#) for further information.
- The affected conveyor turns blue on the SCADA.
- The "Isolator Fault" alarm message appears in the SCADA alarm banner, specifying which conveyor is at fault.

4.9.2.4. Alarm Muting

While the motor isolator is off, pressing any of the Jam Reset buttons in the same emergency stop zone as the isolated conveyor will cause the sounders in the emergency stop zone, that the isolated conveyor is a part of, to be muted/silenced. The CCP and FCP sounders will also turn off/mute when any motor fault alarm is muted.

Pressing the 'Alarm Silence' button on a CCP or FCP will also mute the activated sounders.

4.9.2.5. Fault Reset

An isolator fault is cleared by returning the switch to the ON position; no separate reset action is required. If the conveyor is set to restart when the isolator is switched on, a pre-start warning will be issued before operation resumes. Any fault conditions that existed prior to the isolator fault will not be cleared by restoring the isolator to ON. In such cases, one or more additional reset procedures may be required before the affected conveyor can resume operation..

Jam

4.9.3.1. Overview

4.9.3. The control system employs all PE sensors to detect baggage jam events on transport conveyors, power curves, merge conveyors and diverters. A jam fault occurs after a PE, with jam detection functionality, remains blocked for the 'Jam Fault Time'. The jam timer retains its value when the conveyor stops. Only the allocated Jam Reset button to the affected conveyor is allowed to reset the jam fault.

4.9.3.2. Cause

A jam fault is triggered by all cascade PECs and other PEs designated for jam fault, when the PEC/PE remains blocked for more than a preset time while the conveyor is running, as follows:

- For conveyors running between 0.5m/s and 0.7m/s, the preset time is 4 seconds.
- For conveyors operating above 0.7 m/s, up to 1.0 m/s, the preset time is 3 seconds.
- For conveyors operating above 1.0 m/s, the preset time is 2.5 meters divided by the conveyor speed.

4.9.3.3. Effects

When a jam fault occurs, the following actions take place:

- The sounder at the CCP and FCP turns on.
- The amber beacon at the CCP and FCP turns on. Please refer to Section [16 Beacon and Sounder Indication](#) for further information.
- Sounders within the Jam Reset zone of the jammed conveyor turn on.
- Amber beacons within the Jam Reset zone of the jammed conveyor turn on.
- The allocated jam fault reset button is illuminated (solid).
- The conveyor with the affected PEC, stops.
- If within 5m of the PEC that caused the jam fault, the conveyor immediately downstream also stops.
- If the conveyor immediately downstream is a power curve, and is within 5m of the PEC that caused the jam fault, both the power curve and the conveyor immediately downstream of the power curve stop.
- The graphic for the jammed conveyor on the SCADA changes to the appropriate 'jam fault' colour.
- The SCADA graphics for beacons in the jam reset zone turn amber.
- A 'Jam Fault' alarm is displayed in the SCADA alarm banner, identifying the conveyor that is in the 'Jam' state.

4.9.3.4. Alarm Muting

While the jam still exists (PEC is blocked), pressing any of the Jam Reset buttons in the same Jam Reset zone as the jammed conveyor will cause the sounders in the whole Jam Reset zone to turn off. The CCP and FCP sounder will also turn off when any jam fault alarm is muted.

4.9.3.5. Fault Clearance

After the jam has been cleared, but before the jam fault is reset, the following occurs:

- The amber beacons begin to flash.
- The allocated Jam Reset button changes from being solidly illuminated to flashing.

This indicates that the fault is now able to be reset.

4.9.3.6. Fault Reset

After the jam has been cleared by O&M personnel, the allocated Jam Reset button (i.e. the one that is flashing) will reset the jam fault when pressed. Once the jam fault has been reset, the following actions occur:

- The sounders in the affected jam reset zone turn off (if not already muted).
- The amber beacons in the affected jam reset zone turn off.
- The allocated Jam Reset button changes from flashing to off.
- The sounders at the CCP and FCP turn off (if not already muted).
- The amber beacons at the CCP and FCP turn off.
- The 5-second pre-start warning occurs, and then the conveyors restart.

Notes:

- Pressing any of the Jam Reset buttons other than the Allocated illuminated button **will not** reset the jam fault.
- The relevant jam fault beacons & sounders are inhibited if the affected conveyor is emergency stopped while a jam fault is active. Once the emergency stop has been reset, the jam fault is again indicated by the relevant amber beacons and sounders, unless the jam has been cleared while the affected conveyor was emergency stopped, in which case resetting the emergency stop also resets the jam fault.
- All PECs are wired in a normally closed configuration, meaning that if they detect an object in their field of view, the circuit opens. Therefore, misalignment or malfunction of the PECs can result in a jam fault in the system.

For further details on the jams reset stations layout and zones, please refer to the following documents:

- Sub-System CCP1 Jam Reset Layout
- Sub-System CCP2 Jam Reset Layout
- Sub-System CCP4A Jam Reset Layout
- Sub-System CCP6 Jam Reset Layout
- Sub-System CCP7 Jam Reset Layout

Emergency Stop

The general Emergency Stop (E-Stop) and reset functionality for the BHS control system is detailed in Section 13.3 Activation and Reset.

When a conveyor is emergency stopped:

4.9.4.

- The conveyor stops immediately.
- Red visual beacons and sounders within the affected E-Stop zone are activated.
- Red visual beacons and sounders at the affected CCP & FCP are activated.
- The affected area and the triggering E-Stop station are displayed on the SCADA system.
- A SCADA alarm for the triggered E-Stop is displayed in the SCADA alarm banner.
- Upstream conveyors will cascade-stop bags at the last PEC position, preventing further bag accumulation.

After the emergency stop condition is cleared (i.e. the associated E-Stop pushbutton(s) are released), and the local Estop Restart button has been pressed, See Section 13.3 Activation and Reset, the affected conveyors will:

1. Undergo a 5-second pre-start warning
2. Resume operation automatically.

Priority and Fault Interaction:

- An Emergency Stop alarm takes precedence over all other fault types (e.g. Jam Fault).
- While an E-Stop alarm is active, indications of other faults (e.g., on SCADA graphics, beacons, or sounders) within the same zone are inhibited.
- Resetting the Emergency Stop does not automatically clear any other faults that may have been active prior to or during the E-Stop condition. These faults must be cleared via their respective reset mechanisms.

Encoder Fault

Pulse encoders are rotary sensors that generate a stream of electrical pulses proportional to the conveyor's speed. These encoders are installed on conveyors that require precise speed or position monitoring — such as tracked conveyors within the BHS.

4.9.5 The control system continuously compares the encoder-derived speed against a configured reference value. If the measured speed deviates beyond a defined threshold — for example, due to:

- Disconnection of the pulse encoder.
- Damaged or misaligned encoder.
- Incorrect pulse frequency.
- Torn or worn belt.
- Warped roller.
- Broken Encoder Bracket.
- Mechanical vibrations.

A Pulse Encoder Fault is triggered.

When a pulse encoder fault occurs:

- The affected conveyor is immediately stopped.
- Blue warning beacons and audible sounders near the conveyor at the relevant CCP and FCP are activated.
- The allocated jam fault reset button is illuminated (solid).

Reset and Override Behaviour:

- The fault can be cleared by pressing the allocated illuminated Jam Reset pushbutton.
- If the encoder remains in a faulty condition and continues generating invalid or no pulse data, the fault will re-trigger automatically.
- On Sortation/non-CBS conveyors only, a SCADA-based Encoder Override function can be enabled as a temporary bypass, allowing the conveyor to resume operation.
 - Use of this override must be closely monitored and logged.
 - This function is not available on CBS-tracked conveyors, where accurate position tracking is mandatory.
- A permanent resolution requires repair or replacement of the faulty encoder or mechanical fault to restore proper system performance and tracking integrity.

Motor Feedback Fault

4.9.6.1. Overview

The BHS control system continuously monitors motor the run feedback during operation. The motor run feedback is a signal provided by a Variable Frequency Drive (VFD), indicating its current operational status. This information is used to ensure that a motor is running as intended.

When the PLC instructs a motor to run, it expects positive run feedback from the VFD, confirming the physical movement. If valid feedback is not received within a short delay, the PLC cuts off the run request and triggers a motor feedback fault.

4.9.6.2. Cause

A motor feedback fault is triggered after a short time delay if the motor run input is not received from the motor's VSD (via Ethernet) or from the motor's run contactor if a DOL motor (via digital PLC input), at any time when the control system is commanding the motor to run.

4.9.6.3. Effects

When a motor feedback fault occurs, the following actions take place:

- The affected conveyor is stopped.
- Sounders in the affected conveyor's emergency stop zone turn on.
- Blue warning beacons in the affected conveyor's emergency stop zone and at the CCP and FCP are illuminated (solid). Please refer to [16 Beacon And sounder Indication](#) for further information.
- The sounder at the relevant CCP and FCP turns on.
- The blue 'Technical Fault Reset' button at the relevant CCP and FCP is illuminated (solid).
- The SCADA system displays the associated alarm, identifying the conveyor which has the fault.

4.9.6.4. Alarm Muting

Pressing any 'Jam Reset' button in the same emergency stop zone as the affected conveyor while the fault is still active will cause the alarm to be muted. Sounders in the same emergency stop zone, and at the CCP and FCP, as the affected conveyor will turn off. The blue beacons will remain illuminated.

4.9.6.5. Fault Clearance

The motor feedback fault must be cleared by correcting the underlying cause of the loss of the run feedback signal.

4.9.6.6. Fault Reset

A motor feedback fault can only be reset by pressing the CCP or FCP 'Technical Fault Reset' button after the fault has been cleared.

Note:

- The motor feedback fault state can only be reset by pressing the illuminated CCP or FCP Technical Fault Reset button. It cannot be reset using field reset buttons. This ensures proper acknowledgment and resolution of the fault before resuming operation.
- The relevant technical fault beacons & sounders are inhibited if the affected conveyor is emergency stopped while a motor feedback fault is active. Once the emergency stop has been reset, the motor feedback fault is again indicated by the relevant blue beacons and sounders.

Motor VFD Not Ready Fault

4.9.7.1. Overview

The Motor VFD ready status bit is received by the PLC via the Ethernet IP Interface and monitored for the motor control and alarming.

4.9.7.2. Cause

A motor VFD Not Ready Fault is triggered if the VFD ready status bit is off.

4.9.7.3. Effects

When a motor VFD Not Ready fault occurs, the following actions take place:

- The affected conveyor is stopped.
- Sounders in the affected conveyor's emergency stop zone turn on.
- Blue warning beacons in the affected conveyor's emergency stop zone and at the CCP and FCP are illuminated (solid). Please refer to [16 Beacon And sounder Indication](#) for further information.
- The sounder at the relevant CCP and FCP turns on.
- The blue 'Technical Fault Reset' button at the relevant CCP and FCP is illuminated (solid).
- The SCADA system displays the associated alarm.

4.9.7.4. Alarm Muting

Pressing any 'Jam Reset' button in the same emergency stop zone as the affected conveyor while the fault is still active will cause the alarm to be muted. Sounders in the same emergency stop zone, and at the CCP and FCP, as the affected conveyor will turn off. The blue beacons will remain illuminated.

4.9.7.5. Fault Clearance

The motor VFD must be inspected to determine the reason the "Ready" status is not available. Possible causes include a VFD fault condition or an incorrect operation setting.

4.9.7.6. Fault Reset

The VFD Not Ready fault can be reset by pressing the CCP or FCP 'Technical Fault Reset' button after the ready status bit is activated.

Oversize Detection

4.9.8.1. Over-Height Detection

4.9.8. An over-height fault is signalled when the height of an item on a conveyor exceeds the limit set for the BHS.

- Over height PECs are located prior to security doors at check-in collector conveyors.
- Over height PECs are located after transfer line loading conveyor(s).
- Over height swingarms, with a proximity switch, are located prior to EDS Machines.

When an over-height fault is triggered:

- The conveyor under the over-height PEC (or over-height swingarm) stops immediately.
- The next downstream conveyor stops immediately.
- The oversize fault reset button on the allocated reset station is illuminated solid in white.
- The white oversize beacon and sounder for that oversize activate.
- The fault is indicated on the SCADA System (both visually and in the SCADA alarm banner).
- The bag needs to be removed from the photo eye PEC or swingarm before the fault can be reset.

4.9.8.2. Over-Length Detection

Bag length is measured before the EDS machines, at transfer input loading conveyors and at check-in collector conveyors.

Bag length is used by the control system to determine if the bag can be conveyed by the BHS system and to ensure the bags can be screened by the EDS machine.

Bags longer than 1.2 meters are not permitted onto the BHS and are to be removed from the system and taken to the Out of Gauge sub-systems. Usually, an item that appears to the BHS to be longer than 1.2 metres is because of two or more bags overlapping.

When an over-length fault is triggered:

- The conveyor, with over-length detection turned on, stops immediately.
- The next downstream conveyor stops immediately.
- The oversize fault reset button on the nearest reset station is illuminated solid in white.
- The white beacon and sounders for that area activate.
- The fault is indicated on the SCADA System (both visually and in the SCADA alarm banner).
- The bag needs to be removed from over length PEC before the fault is reset.

4.9.8.3. Oversize Fault Recovery

Adjacent to each oversize detection location, a dedicated oversize reset station is available, either integrated into the existing control station panel or configured as an independent control station. When an oversize fault is detected, the white indicator illuminates. Upon removal of the oversize bag from the PEC, the indicator flashes at a frequency of 1 Hz to signal that the fault is now resettable. Pressing the "Oversize Reset" pushbutton clears the fault, initiates a pre-start warning, and allows the system to resume normal operation. The Oversize bag will then to be manually transfer to an OOG line to be processed.



Figure 11 Oversize Reset Control Station

Controls	Device	Colour
Oversize Reset	Illuminated Push Button	White

Table 8 Oversize Reset Control Station

Fire Alarm

In the event of a fire alarm signal being triggered, the system shall respond as outlined below:

4.9.9.1. Conveyor Shutdown Logic

4.9.9.1. All conveyors shall initiate an immediate shutdown, except for conveyors located directly upstream and downstream of a roller door within the designated fire zone. These conveyors are permitted to continue operation temporarily to allow for bag clearance. Please refer to Section 7.3 Normal Operating Modes for details on the roller door.

4.9.9.2. Actuator Motion Completion

Actuators associated with Ploughs and Vertical Sort Units (VSUs) shall attempt to complete their current motion cycle prior to stopping. This is necessary to avoid leaving the actuators in an indeterminate state, which could cause faults during recovery or restart.

4.9.9.3. SCADA Alarm

The fire alarm status shall be displayed on the SCADA alarm banner. The operator will be alerted to the fire condition until the relay signal is restored. Please refer to the Airport (BNE) DTB HLC Phase 2 Functional Design Specification, section 5.3 for details.

4.9.9.4. Fire Alarm Reset Behaviour

Once the fire relay signal is determined to be in a healthy state, the fire alarm on SCADA shall automatically reset. Despite this reset, all devices affected by the alarm shall remain in a stopped (shutdown) state until a start command has been registered. Please refer to 4.3.1 Subsystem Start for details on Subsystem Start.

5. MERGE

5.1. General

A merge point is created to combine bags from one transport line into another.

Bags are merged onto a straight section of conveyor on the receiving or primary line. Generally, the merging line joins the primary line at 45° or 90° via a merge conveyor.

Devices and I/O required at a merge and owned by the merge conveyor are:

- Merge PEC, mounted on merge conveyor at 90 degrees (for 45° layout only) to priority line takeaway conveyor. The speed setting of the merge conveyor and the PEC angle provide accurate placement of a bag into a bag window on the takeaway conveyor.
- Update PEC on the primary conveyor, upstream of the merge point, if windowing is required.
- Bag jam PEC on the primary conveyor, after the merge point.

5.2. 45 Degree Merge

The 45° merge conveyor is designed to ensure smooth and efficient merging of bags onto the primary conveyor.

- Merge conveyor belt is positioned at least 50 mm above the primary conveyor to allow bags to drop down smoothly.
- Runs at a fixed speed that is $\sqrt{2}$ times faster than the primary conveyor to match longitudinal speeds during merging.
- In some cases, the merge conveyor speed can be increased to provide more momentum, helping bags land more centrally on the primary conveyor.
- A PEC is tilted to align the holding position of merging bags with their geometric projection onto the primary conveyor.
- This setup ensures proper alignment regardless of how close the bag is to either side guard.

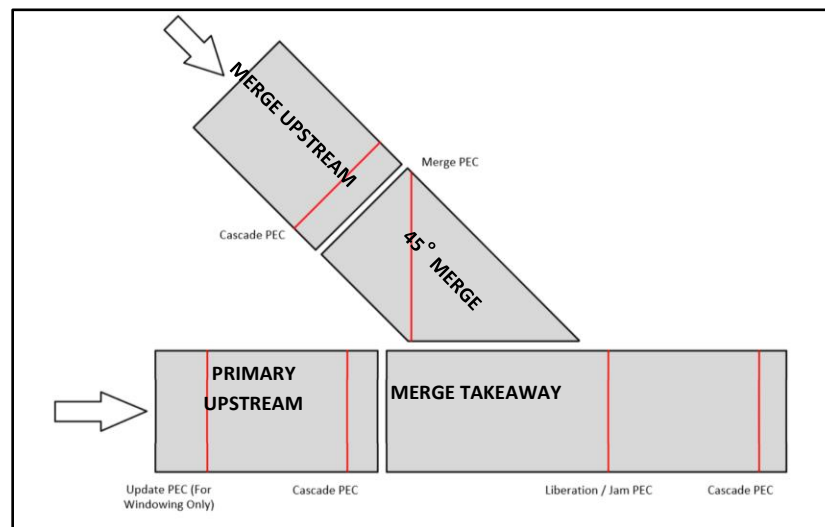


Figure 12: Typical 45 Degree Merge Layout

5.3. 90 Degree Merge

The 90° merge conveyor is designed to facilitate smooth bag transfer onto the primary conveyor with flexible speed control.

- Conveyor belt is positioned at least 50 mm above the primary conveyor to allow bags to drop down onto it.
- Unlike the 45° merge, the 90° merge conveyor can operate at the same speed as the primary conveyor
- The belt speed is more flexible and can be adjusted as needed for optimal merging.
- In certain cases, the speed is increased to provide more momentum, ensuring a smoother transition onto the primary conveyor.

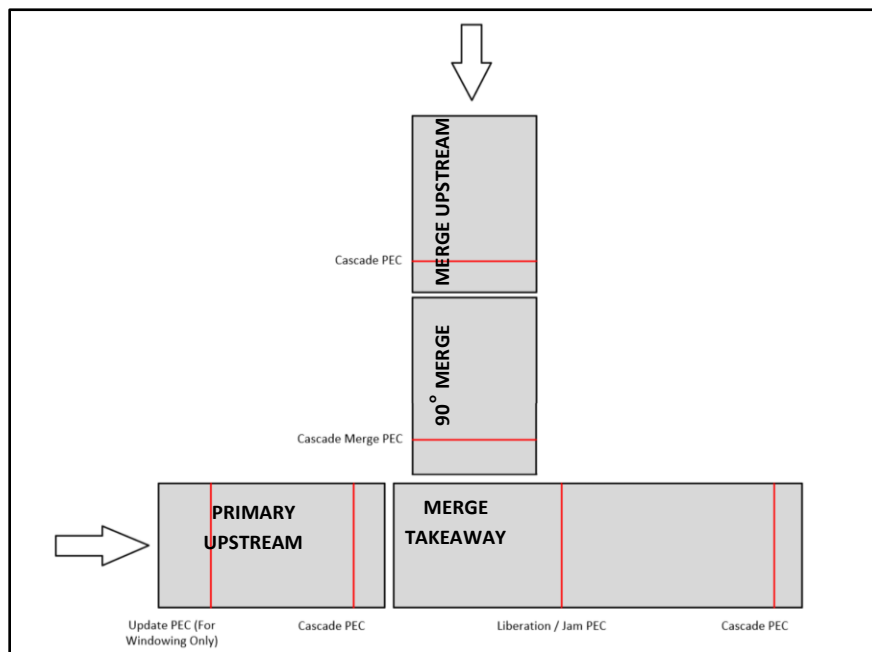


Figure 13: Typical 90 Degree Merge Layout

5.4. Priority Configuration

One on One Merge

Priority alternates between each merging line to control bag entry.

- The first bag to reach the cascade PEC on its path is granted clearance to proceed.

5.4.1. **Once the released bag reaches the liberation or jam PEC, the held bag on the alternate line is allowed to move forward.**

- A pulse counter is used to maintain the required gaps between output bags for smooth operation.

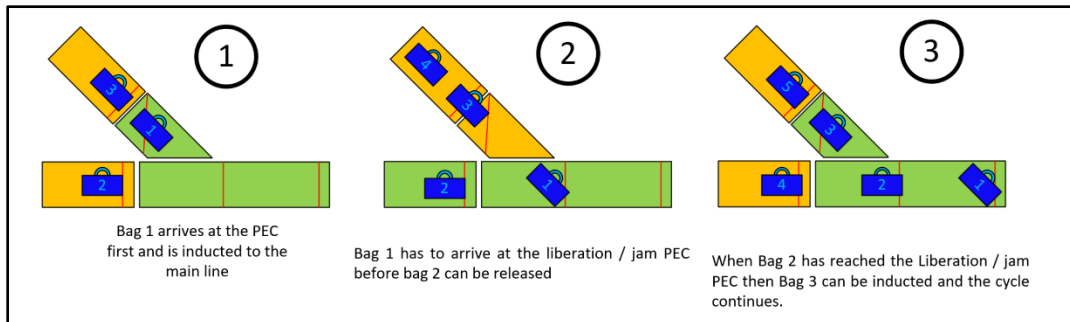


Figure 14: One on One Merge

5.4.2. Primary Priority Merge

A bag on the merge line scans the primary conveyor for a bag window (adequate space)

- When a suitable gap is found, the bag is injected onto the primary conveyor.
- Until a gap is detected, the bag on the merge conveyor remains in a holding position.
- A configurable flush timer is used to flush bags from the merge conveyor if the timer expires.

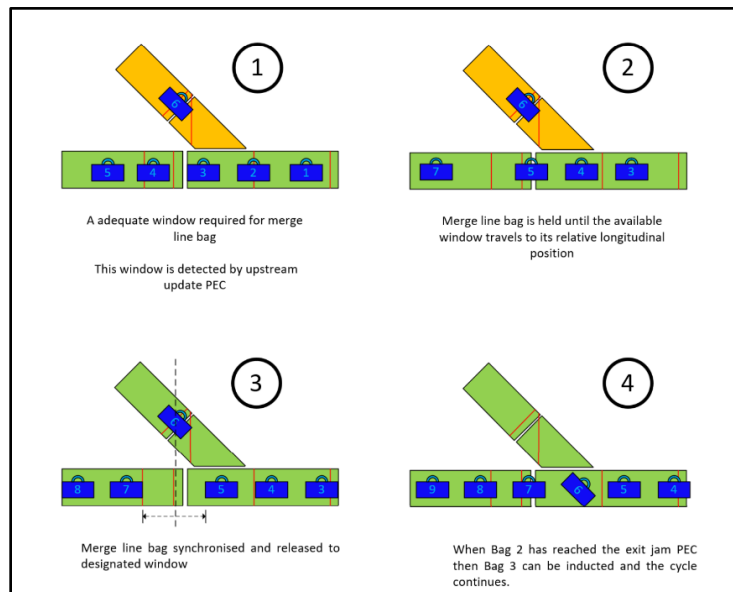


Figure 15: Primary Priority Merge

Primary Priority Merge Timeout and Purge

In a primary priority setup, merge line access is regulated through timed requests to maintain balance with the primary line. The table below describes the cause-and-effect sequence of this process.

5.4.

Cause	Effect	SCADA Configurable	Notes
Bag on merge conveyor exceeds maximum wait time without bag window	Merge line sends a priority request	✓ Yes	Timeout value is configurable
A preconfigured number of bags (typically 3) queue on merge conveyor	Merge line sends a priority request	✓ Yes	Queue length threshold is configurable
Priority granted to merge line	Primary line is stopped upstream of merge takeaway conveyor	N/A	Merge line gains control
Merge line gains control	Merge line initiates bag flush	✓ Yes	Flush window duration is configurable
Merge flush window expires	Merge line relinquishes control; primary conveyor resumes with priority	✓ Yes	Ensures primary regains control after purge
Priority granted to merge line	A cooldown timer begins during which further merge priority requests are blocked	✓ Yes	Prevents merge line from continuously interrupting the primary line
Any priority transition or SCADA condition change	Events are logged and optionally alarmed on SCADA	✓ Yes	For diagnostics, operator awareness, and auditing

Table 9 Primary Priority Merge Cause & Effect

Merge Priority Merge

A bag on the main line scans the primary conveyor for a bag window (adequate space).

- Once a suitable gap is found, the bag is injected onto the primary conveyor.
- Until a bag gap is detected, the bag remains on hold on the main conveyor.

5.4.4.

Merge Priority Merge Timeout and Purge

In a merge priority setup, the main line can request temporary priority to clear congestion and maintain flow balance with the merge line. The table below describes the cause-and-effect sequence of this process.

5.4.5.

Cause	Effect	SCADA Configurable	Notes
Bag on main conveyor exceeds maximum wait time without bag window	Main line sends a priority request	✓ Yes	Timeout value is configurable
A preconfigured number of bags (typically 3) queue on main conveyor	Main line sends a priority request	✓ Yes	Queue length threshold is configurable
Priority granted to main line	Merge conveyor is held	✗ No	Main line gains control
Main line initiates bag flush	Main line starts flushing bags	✓ Yes (duration only)	Flush duration is configurable, initiation is system-driven
Flush window expires/main line clears	Merge line regains priority; system resumes normal merge logic	✓ Yes (duration only)	Flush timeout is SCADA-configurable
Cooldown timer starts before next priority request	A cooldown timer begins before main line can request priority again; Prevents repeated priority grants	✓ Yes	Prevents main line from continuously interrupting the merge line; Cooldown is operator-tuneable via SCADA
Any priority transition or SCADA condition change	Events are logged and optionally alarmed on SCADA	✓ Yes	For diagnostics, operator awareness, and auditing

Table 10 Merge Priority Cause & Effect

5.5. Merge Configuration

Please refer to appendices section 23.6 for merge configuration.

6. TRACKING

6.1. General

Baggage tracking is the process of monitoring the location of each bag as it travels along the conveyor system. This allows status information to be maintained for each bag, which can then be used to determine the appropriate processing actions.

The following features facilitate baggage tracking:

- An encoder on each conveyor to give incremental position feedback.
- Variable Frequency drives (VFD) to give precise control of the belt speed, especially during starting and stopping.
- A synchronised model within the PLC memory that tracks movement of baggage along the conveyor.
- Synchronisation of modelled bag position with actual bag-position at photo-eyes on tracked conveyor sections.
- A database (or lookup table) in the PLC to store the information for each bag being tracked.

The PLC control algorithm accurately tracks bag positions using encoders and sensors to maintain smooth and reliable conveyor flow.

- Incremental position encoders are used to maintain an in-memory model of bag positions on the conveyor system.
- The encoder resolution ensures tracking accuracy within approximately 50 mm of belt travel.
- Each time a bag passes a PEC, its position in the tracking model is resynchronized with its real-world location.
- A strap filter is activated at the leading edge of a bag passing a PEC to reduce false sensor triggers.
- Bags are spaced on the belt to ensure any slip within the allowable tolerance does not affect adjacent bags as illustrated below.
- The bag length check function is implemented within the CBS area, from the EDS exit to the CBS VSU. Bag length is measured in this zone, and if a change in length is detected while the bag is travelling through the check area, the bag status is set to Force Loss. Please refer to appendices section 23.7 for list of the bag length check conveyors.

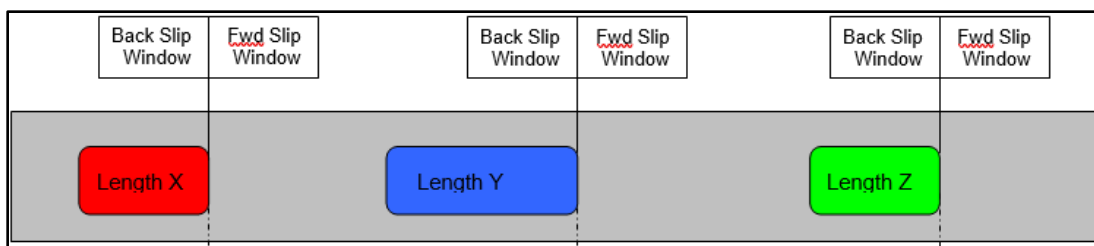


Figure 16 Tracking Example

6.2. Slip Window

Bags can move while being transported (e.g., slip on inclines/declines or during starting and stopping) thus tolerance is provided to account for minor bag position shifts during conveyor transport. This is referred to as the slip window.

- The model position at each PEC is determined by counting encoder pulses between successive tracking PECs.
- The histogram (below) of pulse-count deviations shows the range of possible bag positions, with most variance arising from conveyor-to-conveyor transitions.

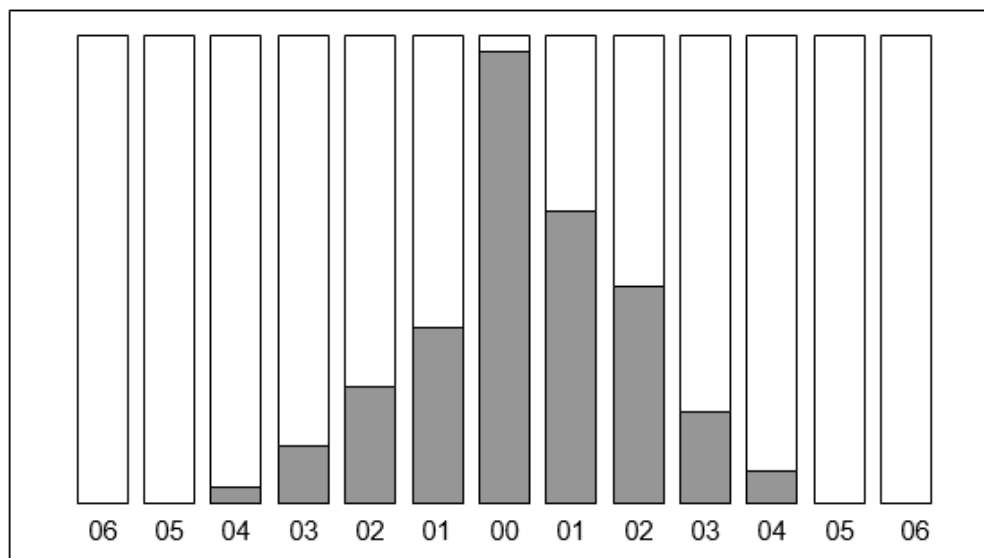


Figure 17 Encoder Pulse-Count Deviation

Conveyor Type	Belt Travel per Encoder Pulse
General Transport Conveyor	56 mm
Metering Conveyor	53 mm

Table 11 Belt Travel Per Encoder Pulse

Each histogram bar represents a pulse count, with encoders providing 5 pulses per revolution.

For example:

- With a slip window of ± 150 mm, bags within bars -03 to 03 (including bar 00) are well tracked, and their positions are updated in the tracking model.
- Bags outside bars -05 to 05 (including bar 00) of the histogram are considered lost.
- Bag transitions across conveyor gaps follow a normally distributed arrival profile, regardless of physical movement.

The tracking is usually divided into two areas, CBS Tracking and Sortation/General Transport Tracking. Both areas have different slip windows.

Tracking area	Slip Window	Uncleared Bags Present	Cleared Bags Present	Notes
CBS Tracking	±250 mm*	✓ Yes	✓ Yes	Tighter slip window
General Transport / Sortation Tracking	±500 mm*	✗ No	✓ Yes	Wider slip window

Table 12 Tracking Area Slip Windows

*To test the validity of the slip window, the bag must be shown to have moved outside the predicted tolerance.

6.3. Bag Status

If a bag's position difference exceeds the slip window limits, it is classified as missing or unknown.

Bag Status	Cause	Effect
Missing Bag	Tracking PEC sensor does not detect a bag presence within the slip window tolerance	Bag is declared missing
Unknown Bag	PEC sensor detects an unexpected contact	New LLC ID is assigned to the bag.
	New LLC ID is assigned	Downstream conveyors start tracking new LLC ID
	Bag has no CBS status	Bag is considered uncleared

Table 13 Bag Status Cause & Effect

6.4. Mistrack Scenarios

When a bag is mistracked, it loses its reference and information within the control system. This is because the PLC can no longer guarantee the identity of the bag. Mistracking in a BHS typically occurs due to unexpected physical interactions between bags and the conveyor system. Some common mistracking scenarios include:

- Bag Jams/PEC malfunction.
- Bag slip backward - acceleration, caught on side guard, incline tumble backward.
- Bag slip forward – deceleration, decline tumble forward.
- Drastic orientation changes on power curve.
- Bag with wheels facing downwards, creating counterforce against the conveyor grip.
- Parts broken off from a bag.
- Inappropriate bag profile – soft, round, or satchel object not put in tubs.
- Unsuccessful divert.

- Rolled by EDS curtains.

6.5. Fault Handling

Fault	Cause	Effect	Tracking Information Retained*	Immediate Fault Fix Required	Notes
Conveyor Stoppage	Die-back (cascade stops) Emergency stops	Conveyors stop running and bags are not moving to get to the tracking PEC on time	✓ Yes	✓ Yes	When system resumes, the tracking model ensures correct sorting behaviour.
Jam	Bags stuck in unexpected areas.	Disruption to the position and orientation of the bags on nearby conveyors	✗ No	✓ Yes	Tracking model may fail to accurately identify and track these bags therefore bags upstream/downstream of the jam position are assumed mistracked. The number affected bags is calculated based on the distance from the jam position.
Encoder Fault	Loose connection to encoder Damaged encoder Generates an incorrect pulse frequency.	Tracking reliability is greatly reduced due to incorrect pulse counts	✗ No	✓ Yes	For conveyor outside the security tracking zones, the encoder can be overridden.
Controller Failover	Active PLC fails	Standby PLC takes over	✓ Yes	✗ No	Only if warm backup PLC configuration
Missing Bag Jam	Configurable number (default set to 3) consecutive missing bags at tracking PEC.	Affected conveyor stops immediately. Local beacons and sounders activate. Please refer to 16 Beacon And sounder Indication for information. SCADA alarm is enunciated. Jam Reset push button indication is illuminated.	✗ No	✓ Yes	
Consecutive Tracking Error	Configurable number (default set to 3) consecutive bags that	SCADA warning is enunciated	✗ No	✗ No	SCADA warning will be raised to alert the operator.

	failed to appear within the tracking window are immediately followed by an unknown bag.	SCADA Conveyor status is changed**			
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Table 14 Fault Handling

*All bags on affected conveyor are considered mis-tracked.

**For additional information refer to SCADA FDS.

6.6. CBS Tracking Failsafe

In the security tracking zone, which extends from the exit of an EDS machine to its CBS decision point, bags are tracked with a tighter slip window tolerance, and additional failsafe logics are implemented to ensure the reliability of CBS Operation.

To ensure that alarmed bags do not mistakenly enter the cleared line at the security CBS decision point, via a VSU, the baggage handling system flags the following scenarios as Track Failsafe Faults:

Failsafe Fault	Cause	Effect	Fault Resolution
Unknown bag	Non-cleared status at first tracking PEC of clear line	-> Immediate downstream and next downstream conveyors on the clear line stop immediately -> VSU lower, middle and upper conveyors stop immediately -> Local beacons and sounders activate. Please refer to Section 16 for further beacon and sounder indication information. -> SCADA alarm is enunciated -> CBS VSU Fault reset push button indication is illuminated	-> Operator must remove the triggering bag -> Put CBS VSU into manual mode -> Press CBS VSU reset pushbutton to acknowledge fault -> Put CBS VSU into auto mode -> System can resume normal operation (after issuing a pre-start warning)
Suspect bag	Updated at CBS decision point but not tracked to the first tracking PEC of reject line	-> Immediate downstream and next downstream conveyors on the clear line stop immediately -> VSU lower, middle and upper conveyors stop immediately -> Local beacons and sounders activate. Refer to Section 16 for further beacon and sounder indication information. -> SCADA alarm is enunciated -> CBS VSU Fault reset push button indication is illuminated	-> Operator must remove the triggering bag -> Put CBS VSU into manual mode -> Press CBS VSU reset pushbutton to acknowledge fault -> Put CBS VSU into auto mode -> System can resume normal operation (after issuing a pre-start warning)

Table 15 Failsafe Fault Cause & Effect

Force Reject

Any unknown bag that appears on the CBS security tracking zone, is without any CBS status and regarded as an uncleared bag and is sent to the reject line.

6.6.1 There is a possibility that a bag can get caught on conveyor equipment or slide towards the upstream while traveling through an inclined conveyor. From the tracking model perspective, this would first cause a "Missing Bag" event, as the bag did not reach the tracking PEC within the correct timing. Subsequently, it would become an "Unknown Bag" when it finally arrives at the tracking PEC in a delayed manner. This can result in a cleared bag being mis-tracked and sent to the reject line.

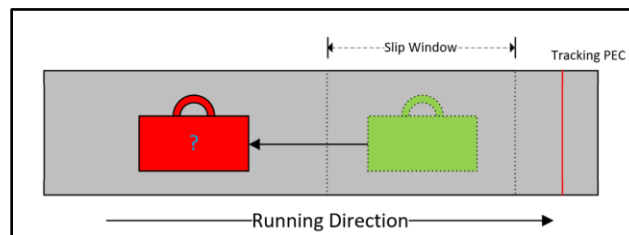


Figure 18: Clear Bag Moved Outside of Windows and Mis-tracked becomes Unknown

In certain scenarios however, a bag can become stranded until a following bag runs into it and pushes it forward, causing the two bags to essentially become one entity.

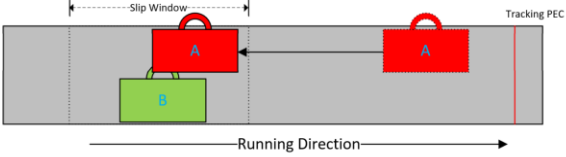
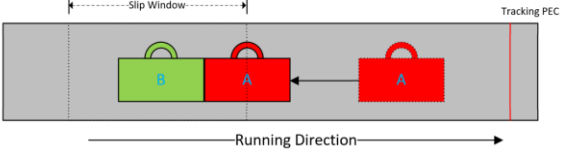
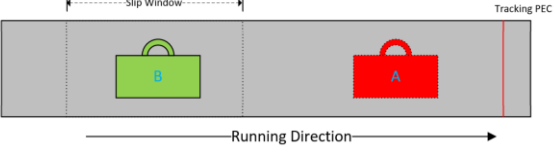
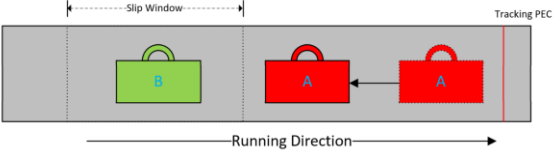
Scenario	Cause	Effect	Diagram
Scenario 1	Rejected Bag A gets caught on sideguard and gets incorporated - Bag B but the entity of both bags is inside the tracking window of Bag B	Combined entity is to be force rejected and is sent to reject line	
Scenario 2	Rejected Bag A gets caught on sideguard and gets incorporated into Bag B but the entity of both bags is outside the tracking window of Bag B	Combined entity becomes an unknown bag and is sent to reject line	
Scenario 3	Rejected Bag A disappeared from tracking model and Bag B remains tracked	Cleared Bag B is to be force rejected and sent to reject line	
Scenario 4	Rejected Bag A moved out of its tracking window but does not interfere with cleared Bag B.	No force reject required. Bag A sent to reject line and Bag B sent to cleared line	

Table 16: Force Reject Cause & Effect

7. ROLLER DOOR

There are two types of roller doors in the BHS:

- **Fire/Security Doors:** Respond to both operational logic and fire signals from the building alarm system.
- **Security-Only Doors:** Operate based solely on subsystem commands and do not respond to fire alarms.

Both door types share identical control logic under normal operating conditions. The only variation lies in the fire signal response unique to Fire/Security doors. See Section [23.3.1 Door types](#) for CBS for a table of fire/security doors for the 4 lane strategy.

7.1. Control Devices and I/O

Each roller door incorporates the following I/O points:

- Motor isolator feedback
- Motor open/close output signals
- Motor run feedback
- Door open proximity sensor
- Door close proximity sensor
- Upstream Bag Present PEC
- Downstream Door Obstruction PEC
- Under-door pressure sensor strip (Optionally installed)
- Motor overload relay

7.2. Control Panel

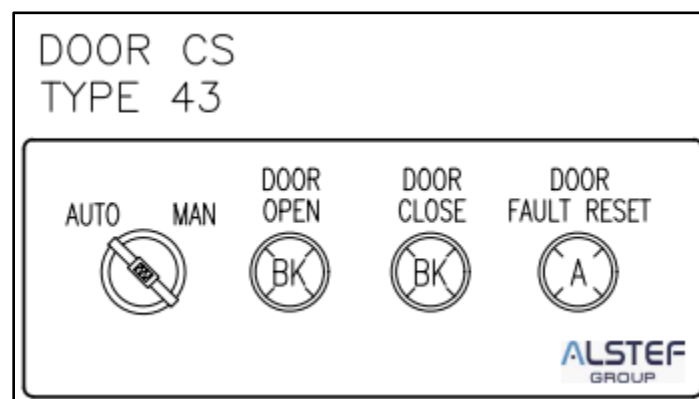


Figure 19 Roller Door Control Station

The door control station contains the following operator controls:

Control	Device	Colour
Mode	2- position switch Auto/Manual	
Door Open	Solid push button	Black
Door Close	Solid push button	Black
Door Fault Reset	Illuminated push button	Amber

Table 17: Door Control Station Type 43

- **Auto Mode:** PLC controls door.
- **Manual Mode** (Maintenance only): Door moves while pushbutton is held (either open or close pushbutton to open or close the door). Movement stops when released.
- Overrides are disabled during fire alarms and faults.
- **In Fault:** Door Fault pushbutton will illuminate. Once the fault is cleared, the pushbutton is pressed, the pushbutton LED extinguishes.

7.3. Normal Operating Modes

7.3.1. Subsystem Start

- Door opens automatically on subsystem start.
- Conveyors upstream and downstream interlocked to start only when door is fully open and fault-free.

7.3.2.

Subsystem Stop

7.3.3.

- Door closes automatically after subsystem stops and adjacent conveyors are stationary.

Energy Conservation Mode (ECM)

The roller door behaviour during Energy Conservation Mode (ECM) varies based on door type:

- **Fire-Only Doors:** The door **remains open** when ECM is active on the adjacent conveyors. This prevents unnecessary cycling and supports energy-saving operations without compromising fire compartmentalization, as these doors are only triggered by fire alarms.

7.3.4.

- **Fire/Security Doors:** The door **automatically closes** when ECM is active on adjacent conveyors. This ensures that the security integrity of the subsystem is maintained during idle periods by physically isolating secured areas.

Manual Operation

- Enabled via 2-position selector switch (Auto/Manual) on local panel.
- In Manual mode, pushbuttons allow open/close control.

7.4. Fire Alarm Mode (Fire/Security Doors Only)

- On receipt of fire signal from building system:
- If the PEC beneath the door is not detecting a bag, the conveyor underneath will stop and the door will close immediately. If the PEC beneath the door detects a bag, the conveyor will continue to run until the bag clears the sensor (gap detected). The conveyor will then stop and the door will close. If the bag does not clear within the preset time, a fault will be activated, and the door will close automatically. Door remains closed until system is restarted manually via [4.3.1 Subsystem Start](#).

7.5. Manual Override (SCADA)

Available via SCADA interface for maintenance or special scenarios:

- Enables operator control of door and override of bag under door detection PECs.
- Conveyor operation beneath the door remains active.

7.6. Emergency Stop Response

- Door halts immediately when E-Stop is activated.
- If door is moving, it stops at the current position.
- System must be fully reset before resuming door activity.

7.7. Fault Management

7.7.1.

Detected Faults

- Bag under door detected (PEC active).
- Door failed to open (proximity not reached in 15s).
- Door failed to close (proximity not reached in 15s).
- Both open and close proximity sensors active >1s.
- Both open and close proximity sensors off >1s.
- Motor feedback not detected.
- Motor isolator feedback missing.
- Motor Overload.
- Fire system signal fault (Fire/Security door only).

All faults are visible on SCADA (visually and in the alarm banner) and trigger physical visual/audible alarms via local beacons & sounders.

Fault Trigger Conditions

Fault	Trigger Condition
Bag Under Door Fault	Downstream PEC blocked during door closure for more than 10 seconds.
Door Failed to Open	Door Open Proximity Sensor not activated within 15 seconds of an open command.
Door Failed to Close	Door Close Proximity Sensor not activated within 15 seconds of a close command.
Motor Feedback Fault	No motor feedback detected during active door motion command.
Motor Isolator Fault	Motor isolator feedback not detected while system is in ready/running state.
Fire Fault	Fire signal received from building alarm system (Fire/Security door only).
Emergency Stop Condition	E-Stop activated in the relevant zone.
Door Positioning Fault	Both Open and Close Proximity Sensors active or off simultaneously for more than 1 second.

Table 18: Roller Door - Fault trigger conditions

Fault Reset and Indication

Fault Type	Reset Method	Indications on Fault
Bag Under Door Fault	Local Reset Station or CCP or FCP	SCADA Alarm, Sounder, Amber Beacon, Illuminated Reset Button Please refer to 16 Beacon And sounder Indication for further information.
Door Failed to Open	Local Reset Station or CCP or FCP	SCADA Alarm, Sounder, Amber Beacon, Illuminated Reset Button Please refer to 16 Beacon And sounder Indication for further information.
Door Failed to Close	Local Reset Station or CCP or FCP	SCADA Alarm, Sounder, Amber Beacon, Illuminated Reset Button Please refer to 16 Beacon And sounder Indication for further information.
Motor Feedback Fault	CCP or FCP Technical Fault Reset	SCADA Alarm, Door/Conveyor Stops, Amber Beacon, Sounder Please refer to 16 Beacon And sounder Indication for further information.
Motor Isolator Fault	Restore isolator	SCADA Alarm, Door/Conveyor Stops, Illuminated Reset Button Please refer to 16 Beacon And sounder Indication for further information.
Fire Fault	Fire Fault auto resets once the Fire signal is healthy	SCADA Fire Alarm, Sounder, Amber Beacons. Please refer to 16 Beacon And sounder Indication for further information.
Obstruction PEC Fault	Manual Reset via Station/CCP/FCP	SCADA Alarm, Sounder, Amber Beacon Please refer to 16 Beacon And sounder Indication for further information.
Emergency Stop Condition	Release all E-Stops + System Reset	SCADA Emergency Indicator, Door Halts, Illuminated Reset Buttons Please refer to 16 Beacon And sounder Indication for further information.

Fault Type	Reset Method	Indications on Fault
Door Positioning Fault	Local Reset Station or CCP or FCP	SCADA Alarm, Amber Beacon, Sounder, Illuminated Reset Button Please refer to 16 Beacon And sounder Indication for further information.

Table 19: *Fault Reset & Indication*

7.8. Cause and Effect Matrix – Roller Door Operation

Cause/Input	Effect/Output	Notes/Remarks
Start Command (Auto Mode)	Door opens → adjacent conveyors enabled	No faults, no fire/E-Stop
Stop Command (Auto Mode)	Door closes after flush and stop	Normal shutdown
Manual Mode - Door Open PB Held	Door opens while held	Only if no fault and/or fire alarm
Manual Mode - Door Close PB Held	Door closes while held	Only if no fault and/or fire alarm
Fire Alarm	Door closes immediately	Overrides faults, conveyors run briefly
E-Stop	Door halts	Freezes in place
Bag Present PEC blocked	Door does not close	Triggers fault
Obstruction PEC detects bag	Door Obstruction alarm	Conveyors stop, alarm in SCADA
No Door Closed signal	Door Failed to Close alarm	After the 15 sec timeout
No Door Open signal	Door Failed to Open alarm	After the 15 sec timeout
Both sensors active	Positioning Fault alarm	Sensor issue
No motor feedback	Motor Feedback Fault	Stops system
Isolator signal missing	Motor Isolator Fault	Stops system
SCADA Manual Override	Manual open/close allowed	Use with caution
Fault Reset	System resumes	Fault cleared
Flush and stop done	Door closes	Normal shutdown
Flush incomplete	Door remains open	Avoids trapping bags

Table 20: Roller Door Cause & Effect

8. PLOUGH DIVERTER

8.1. General

The plough diverter is a device which extends or retracts to form a plough across a conveyor belt. The plough arm is housed in an enclosure that is situated alongside a straight conveyor. The plough includes a vertical belt which runs when the plough arm is extended and is used to positively divert a bag.

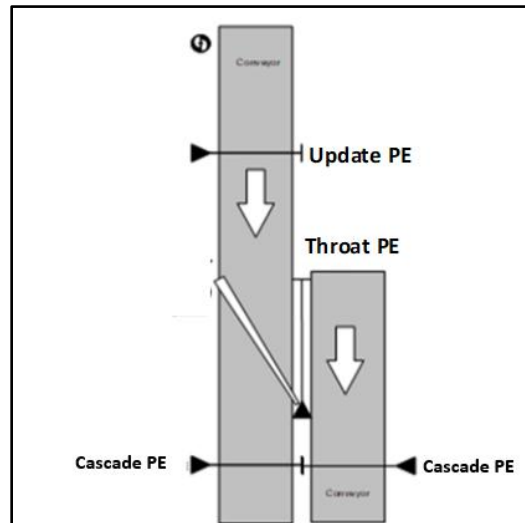


Figure 20: Plough Diverter

Plough Cycle Process		
Status	Cause	Effect
-> Auto key position. -> No faults. -> Arm is retracted.	-> Update PEC is triggered.	-> PLC sends divert signal based on encoder feedback model. -> Plough Arm extends.
-> Auto key position. -> No faults. -> Arm is extended.	-> Update PEC is triggered.	-> PLC sends divert signal based on encoder feedback model. -> Plough Arm retracts.

Table 21: Plough Cycle Process

- Update PEC is placed sufficiently upstream to measure inter bag gap to determine if bag can be diverted safely.
- Throat PEC is used to detect jams and incomplete diverted bags. See Figure 20 above for PEC positions.

8.2. Control Panel

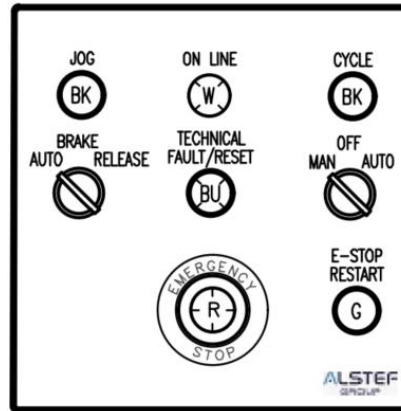


Figure 21 Plough Control Station

Control [Device]	Cause	Effect
Mode [3-position switch] Auto/Off/Manual	Key position in Auto	Plough is operational and the control of the cycle signal and arm belt movement is controlled via the PLC.
	Key position in Off	Plough is not operational and if arm is extended, the arm retracts before plough renders inoperative. SCADA warning message reflects ploughs state. Bypasses the plough faults and releases the conveyor interlock.
	Key position in Manual*	Plough is operational and the control of the cycle signal and arm belt movement is controlled via the manual pushbutton press. SCADA warning message reflects ploughs state.
Cycle [Pushbutton]	Press pushbutton	If Manual Mode: -> Alternate the plough arm position. (retracted to extended or extended to retracted). Otherwise, no effect.
Jog [Pushbutton]	Press and hold pushbutton	If Manual Mode -> The plough arm moves incrementally. -> Will not hold at either the extend or retract position, continues to cycle. -> To change direction, press and hold the cycle button while jogging. Otherwise, no effect.
Brake** [2-position switch] Auto/Release	Key position in Auto	Actuator brake engagement is controlled by the VFD.
	Key position in Release	Actuator brake is disengaged and arm remains unlocked. Arm can be moved by hand if the power of the panel is off.
Online [Indicator]	Plough in Auto mode	Indicator has a steady white illuminated light.

Control [Device]	Cause	Effect
	Plough in Off mode	Indicator remains OFF.
	Plough in Manual mode	Indicator flashes at 1 second interval.
Technical Fault/Reset [Illuminated push button indicator]	Diverter in fault	Indicator has a steady blue illuminated light.
	Press pushbutton	If fault has been corrected: -> Plough returns to operational state. Otherwise, no effect.
Emergency stop [Red illuminated latching mushroom pushbutton]	Press pushbutton	Plough goes to non-operational state and triggers associated e-stop zone fault.
Emergency stop Restart [Pushbutton]	Press pushbutton	If estop latched pushbutton has been released: -> Plough returns to operational state. Otherwise, will only mute sounders in the affected emergency stop zone.

Table 22 *Plough Control Station*

*Plough manual mode is reserved for maintenance purposes only and cannot be used for diverting bags under any circumstances. When manual mode is selected, the conveyor underneath of the plough is interlocked.

** Arm must not be in retracted position to test this functionality of the brake.

8.3. Collision Detection

Bag gap of 1400mm is required to successfully divert a bag on a conveyor running at 0.75-1.4 m/s. Upstream conveyors to the plough ensure this bag gap requirement is met. To handle any reduced bag gaps, the encoder-based bag movement model maps the bag presentation pattern in the plough area.

Cause	Effect
Bags A and B are too close, and plough arm extend will result in collision with previous bag (Bag A)	The divert attempt will be aborted, and the number of unable-to-divert events will be recorded in the SCADA daily report.
Colliding Bag A travels out of exclusion zone while Bag B to be diverted is still within ideal position	Plough arm will extend and complete divert
Bag A has exited the exclusion zone, but Bag B is not yet within the ideal divert position.	Request to extend the plough arm is held until Bag B is in position.

Table 23 Plough Collision Cause & Effect

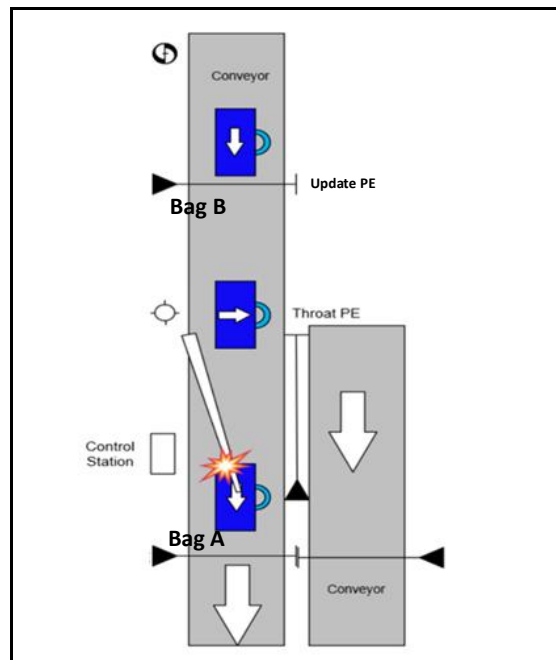


Figure 22 Collision Detection

8.4. Plough Diverter Faults

The fault reset button operates as any other fault reset on a RE station. The indicator will illuminate continuously on the following faults:

Fault	Cause	Effect	Fault Resolution
Plough Arm Failed to Retract	Plough retracted proximity sensor fails to detect the plough paddle within two seconds (adjustable) of the retract request	-> Plough operational is disabled -> Failed to Retract Fault is triggered on SCADA -> Sounder and amber warning beacons near the plough activate. Please refer to 16 Beacon And sounder Indication for further information. -> Technical Fault/Reset button on plough control panel and CCP illuminate	-> Investigate and correct the root cause (e.g., inspect and test the retracted proximity sensor). -> Plough Arm needs to be retracted manually via enabling the Brake release and pushing the Arm back to home position. -> Press Technical Fault/Reset button on plough control panel
Plough Arm Failed to Extend	Plough extended proximity sensor fails to detect the plough paddle within two seconds (adjustable) of the extend request	-> Plough operational is disabled -> Failed to Extend Fault is triggered on SCADA -> Sounder and amber warning beacons near the plough activate. Please refer to 16 Beacon And sounder Indication for further information. -> Technical Fault/Reset button on plough control panel and CCP illuminate	-> Correct the cause of the fault. -> Press Technical Fault/Reset button on plough control panel.
Motor Feedback Fault	Plough actuator or vertical belt motor feedback signal is not detected during diverting operation	-> Plough operational is disabled -> Actuator or Vertical Belt Motor Feedback Fault is triggered on SCADA -> Sounder and amber warning beacons near the plough activate. Please refer to 16 Beacon And sounder Indication for further information. -> Technical Fault/Reset button on plough control panel and CCP illuminate	-> Correct the cause of the fault. -> Press Technical Fault/Reset button on plough control panel.
Motor Isolator Fault	Plough panel isolator is turned off	-> Plough operational is disabled -> Plough Isolator Motor Feedback Fault is triggered on SCADA -> Sounder and amber warning beacons near the plough activate. Please refer to 16 Beacon And	-> Turn the Isolator on. -> Press Technical Fault/Reset button on plough control panel and CCP.

Fault	Cause	Effect	Fault Resolution
		sounder Indication for further information. -> Technical Fault/Reset button on plough control panel and CCP illuminate	
Plough Arm Positioning Fault	Plough remains in a state that is neither retracted or extended for longer than two second (adjustable). Plough arm should always be in either retracted or extended position except during the transition movement	-> Plough operational is disabled -> Plough Arm Positioning Fault is triggered on SCADA -> Sounder and amber warning beacons near the plough activate. Please refer to 16 Beacon And sounder Indication for further information. -> Technical Fault/Reset button on plough control panel and CCP illuminate	-> Correct the cause of the fault. -> Plough Arm needs to be retracted manually via enabling the Brake release and pushing the Arm back to home position. -> Press Technical Fault/Reset button on plough control panel and CCP.
Safety Hatch Fault	Plough maintenance hatch open. Safety interlock switch is triggered.	-> Plough operation is disabled, and underlying conveyor is stopped -> Safety Hatch Fault is triggered on SCADA -> Sounder and amber warning beacons near the plough activate. Please refer to Refer to 16 Beacon And sounder Indication for further information. -> Technical Fault/Reset button on plough control panel and CCP illuminate	-> Maintenance access door must be fully closed -> System reset via the Technical Fault/Reset pushbutton on the plough control panel. -> Pre-start warning and then both the plough and conveyor resume operation.

Table 24 Plough Diverter Faults Cause & Effect

8.5. Additional Features

These additional features allow for safer and more effective operation:

Control	Cause	Effect	Notes
Conveyor Interlock	Plough fault	Conveyor beneath ceases operation until plough is fully retracted. This in turns disables the plough operation.	The plough arm alternation is triggered based on the approaching bag's destination, tracked via a pulse wheel counter. Therefore, if the conveyor under the plough ceases operation, it effectively disables the plough's operation. In automatic mode, under no circumstances can the plough change its paddle position if the conveyor beneath it is halted.
<u>SCADA Override</u> Each plough arm position can be overridden on SCADA, effectively achieving "Divert All" and "Divert None" functionality.	Divert All using SCADA is activated	The diverter extends once the exclusion zone is clear. If a bag is within the exclusion zone, the plough waits for it to exit before extending	This option is used when operators configure a manual load-sharing path for a certain section of the system, bypassing the HLC load balancing algorithms.
	Divert None using SCADA is activated	The plough ensures that any bag currently being diverted completes its transition to the secondary line before initiating the retraction process.	
Load Sharing	Load sharing algorithms are processed at the HLC end. The LLC receives instructions from the HLC to determine the path for each bag at the plough decision point. Additionally, operators can override these decisions via the SCADA if necessary. For details, please refer to DTB2 Auto Routing FDS.		
Destination Unable to Receive	Please refer to DTB2 Auto Routing FDS.		

Table 25 Additional Plough Features

8.6. Sortation Diverter Fallback

All sortation ploughs to the carousels will be setup as divert all for Stage 6 & 8 in the PLC and in SCADA.

9. VSU AND VMU

9.1. General

The Vertical Sort Unit (VSU) and the Vertical Merge Unit (VMU) both have identical hardware elements encompassing a vertical sorter device, a PLC control algorithm, and three conveyors: middle, lower, and upper. Each conveyor is equipped with two PECs located at the upstream and downstream tips respectively. An actuator can alternate the connection of the Middle Conveyor, allowing it to link to either the upper or lower path as needed. Where a VSU uses the middle conveyor as the upstream infeed, with the upper and lower as outfeed downstream conveyors, a VMU uses the upper and lower as the upstream infeed to merge with the downstream outfeed middle conveyor.

VSUs are primarily utilised in screening and sorting systems to bifurcate a conveyor line into an upper and lower path. VMUs are primarily utilised in general sortation, merging the upper and lower conveyors into one conveyor line. The conveyors within a vertical sorter and vertical merge function similarly to standard conveyors, but they can link and interlock with one or two other conveyors. See [Figure 23: Vertical Sort/Vertical Merge Unit](#) for visual representation. For the 4-lane strategy, all VSUs are CBS-VSUs and the one VMU is a non-CBS, sortation only VMU.

Additionally, each VSU and VMU are housed within a secure fencing structure, designed to keep the moving mechanisms out of reach for safety reasons. Access is available for maintenance purposes, and the opening of these access doors is monitored by safety switches. These switches are integrated into the safety circuit and automatically disconnect the power to the VSU/VMU when the doors are opened.

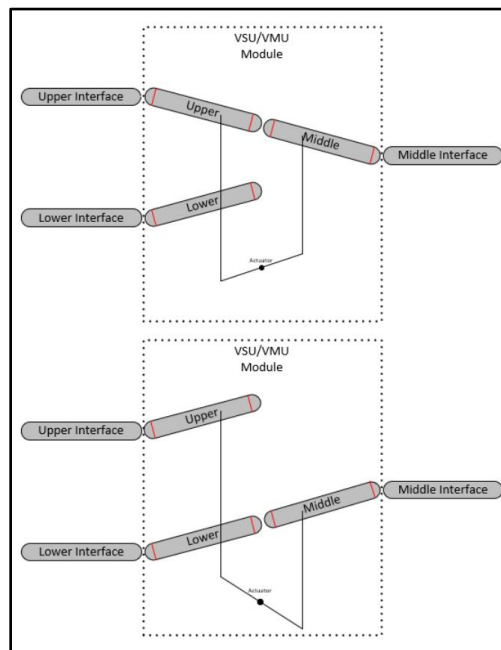


Figure 23: Vertical Sort/Vertical Merge Unit

9.2. Control Panel



Figure 24: Vertical Sorter & Vertical Merge Control Panel

Vertical sorter Local/Auto mode selector

9.2.1.

Control [Device]	Cause	Effect
Mode [2-position switch] Auto/Local	Key position in Auto	VSU/VMU is operational, and the bags are sorted via the PLC based on their tracking and security screening status. Jog controls are inactive in this state
	Key position in Local	VSU/VMU is operational, and the control of the movement is via manual intervention. For further details on manual operation, Please refer to 9.2.1.1 Local Mode .
	Key Switch Local to Auto*	-> A pre-start warning is given before any equipment operates. -> A "flush" operation is implemented by the BHS control system to assist in clearing all bags from the VSU/VMU. Please refer to 9.3 Non-CBS VMU Flush for further information -> Infeed interface conveyor starts sending bag to VSU/VMU once flushing is completed. VSU Local mode including flush operate differently when used in CBS Operation. Please refer to 9.6.4 CBS VSU Local Mode for more details.
	Key Switch Auto to Local	-> The actuator and all conveyors halt instantly. -> Any fault status of the VSU/VMU is retained upon transition to Local mode and must be resolved and reset before manual mode can operate fully.
VSU/VMU Tilt* [Spring-loaded selector switch] Up/Down	VSU/VMU Tilt Switch held in Up	If Local Mode: -> Slowly drive the sorter mechanism so that it tilts toward the upper exit path -> Sorter mechanism moves up until it hits the mechanical limit -> Releasing the button will leave actuator in its current position
	VSU/VMU Tilt Switch held in Down	If Local Mode: -> Slowly drive the sorter mechanism so that it tilts toward the lower exit path -> Sorter mechanism moves down until it hits the mechanical limit -> Releasing the button will leave actuator in its current position
Top Conv Jog [Spring-loaded selector switch] Forward/Reverse	Switch position held in Forward	If Local Mode: -> Upper conveyor will be driven forward (bags can move out of the vertical sorter/vertical merge) -> Releasing this switch will stop the conveyor immediately.
	Switch position held in Reverse**	If Local Mode: -> Upper conveyor will be driven in reverse (bags move towards

Control [Device]	Cause	Effect
		upstream of the vertical sorter/vertical merge) -> Releasing this switch will stop the conveyor immediately.
Middle Conv Jog [Spring-loaded selector switch] Forward/Reverse	Switch position held in Forward	If Local Mode: -> Middle conveyor will be driven forward (bags can move out of the vertical sorter/vertical merge) -> Releasing this switch will stop the conveyor immediately.
	Switch position held in Reverse**	If Local Mode: -> Middle conveyor will be driven in reverse (bags move towards upstream of the vertical sorter/vertical merge) -> Releasing this switch will stop the conveyor immediately.
Bottom Conv Jog [Spring-loaded selector switch] Forward/Reverse	Switch position held in Forward	If Local Mode: -> Lower conveyor will be driven forward (bags can move out of the vertical sorter/vertical merge) -> Releasing this switch will stop the conveyor immediately.
	Switch position held in Reverse**	If Local Mode: -> Lower conveyor will be driven in reverse (bags move towards upstream of the vertical sorter/vertical merge) -> Releasing this switch will stop the conveyor immediately.
Mode [2-position switch] Auto/Manual	VSU/VMU in Manual mode	Indicator flashes at 1 Hz SCADA displays a warning message and updates the graphical status
Fault Reset [Illuminated pushbutton]	VSU/VMU in fault or estop condition in or near the VSU/VMU	Indicator has steady illuminated light
Fault Reset [Illuminated pushbutton]	Press pushbutton	If fault has been corrected: -> Visible and audible pre-start warning -> VSU/VMU returns to operational after actuator automatically returns to desired position If estop fault has been corrected and system start is activated: -> VSU/VMU returns to operational Otherwise: -> Audible alarm will be silenced
Emergency stop [Red illuminated latching mushroom pushbutton]	Press pushbutton	VSU/VMU goes to non-operational state and triggers associated e-stop zone fault.

Table 26 VSU/VMU Control Station Cause & Effect

*To switch to Auto Mode, the Middle Conveyor must be aligned with either the upper or lower conveyor. If this alignment is not achieved, a positioning fault will be triggered for both the VSU & VMU.

**For a VSU utilised in CBS Operation, JOG REVERSE selector switch also drives Cleared Line Outfeed conveyor in reverse direction. Please refer to [9.6.4 CBS VSU Local Mode](#).

9.2.1.1. Local Mode

Local mode provides operator control over the actuator position and individual conveyor belt operation (upper, lower, middle) in both directions within mechanical limits. If the system starts while a VSU or VMU is in manual mode, the startup sequence bypasses it, leaving it inoperative. Upstream systems perceive a VSU or VMU in manual mode as unready to receive, halting bags at the cascade PEC of the interface conveyor preceding the VSU or the VMU.

When the actuator is connected to either the upper or lower conveyor, and if the PEC near the switching edge of the middle conveyor is blocked, the actuator locks itself, preventing a change in position. If the middle conveyor is not connected to either the upper or lower conveyor, jogging a bag beyond the edge PEC on the middle conveyor is not possible. These measures safeguard bags as well as the VSU/VMU from potential damage due to incorrect manual operation.

A bag jogged to the downstream end can exit the VSU/VMU and be taken away by the outfeed interface conveyor. However, a bag cannot be reversed beyond the upstream end and is halted at the upstream PEC of the conveyor (the upper or lower conveyor for a VMU & the middle conveyor for a VSU).

9.2.1.1.1. Tracking in Local Mode

Activating local mode on the VSU or the VMU when there are bags inside of the module results in the immediate clearance of tracking information for all bags within the unit, regardless of whether conveyor reversing operation is utilised or not. The system's tracking model doesn't support the reverse operation of conveyors, and as a result, **all bags within the unit become unidentified or "unknown" bags upon the activation of manual mode.** This means that once manual mode is deactivated and the system returns to automatic operation, these bags would need to be re-identified, and their tracking data restored before they can be correctly sorted and routed to their designated destinations. A VSU for CBS Operation is required to send any affected bags to the reject line to avoid failsafe fault. Please refer to section [9.6.4 CBS VSU Local Mode](#) for more details.

9.3. Non-CBS VMU Flush

For non-CBS VMUs, upon transitioning from local mode back to automatic mode, the VMU activates all VMU conveyor components along the full length of the VMU, to ensure that no bags remain in the module. If the outfeed is unavailable to receive the flushed-out bags, the flushing process will remain pending until all bags are confirmed to have exited. Once the flushing process is completed, the VMU returns to operational status, and the infeed conveyor can resume sending bags into the module.

Please refer to [9.6.2 CBS VSU Flush](#) for flushing control for the CBS VSUs.

9.4. VSU And VMU Faults

The fault reset button operates to reset the technical fault on the VSU or VMU. The indicator will illuminate continuously on the following faults:

Fault	Cause	Effect	Fault Resolution
VSU/VMU failed to switch up	VSU/VMU upper proximity sensor fails to detect the Middle conveyor in the up position within two seconds (adjustable) of the switch up request	-> VSU/VMU operation is disabled. -> VSU/VMU Failed to Switch Up Fault is triggered in SCADA. -> Sounder and amber warning beacons near the VSU/VMU activate Please refer to 16 Beacon And sounder Indication for further information. -> Fault reset button on the VSU/VMU control panel, CCP & FCP illuminate.	Press the fault reset button on VSU/VMU control panel, CCP or the FCP.
VSU/VMU failed to switch down	VSU/VMU lower proximity sensor fails to detect the Middle conveyor in the lower position within two seconds (adjustable) of the switch down request	-> VSU/VMU operational is disabled. -> VSU/VMU Failed to Switch Down Fault is triggered in SCADA. -> Sounder and amber warning beacons near the VSU/VMU activate. Please refer to 16 Beacon And sounder Indication for further information. -> Fault reset button on the VSU/VMU control panel, CCP & FCP illuminate.	Press the fault reset button on VSU/VMU control panel, CCP or the FCP.
Circuit Breaker Fault	VSU/VMU switching actuator motor circuit breaker signal is not detected	-> VSU/VMU operational is disabled. -> VSU/VMU Motor Actuator Circuit Breaker Fault is triggered in SCADA. -> Sounder and amber warning beacons near the VSU activate. Please refer to 16 Beacon And sounder Indication for further information. -> Fault reset button on the VSU/VMU control panel, CCP & FCP illuminate.	Press the fault reset button on VSU/VMU control panel, CCP or the FCP.
Motor Isolator Fault	VSU/VMU panel isolator signal is turned off	-> VSU/VMU operational is disabled. -> VSU/VMU Motor Isolator Fault is triggered in SCADA. -> Sounder and amber warning beacons near	Press the fault reset button on VSU/VMU

Fault	Cause	Effect	Fault Resolution
		the VSU/VMU activate. Please refer to 16 Beacon And sounder Indication for further information. -> Fault reset button on the VSU/VMU control panel, CCP & FCP illuminate.	control panel, CCP or the FCP.
VSU/VMU Positioning Fault	VSU/VMU remains in a state that is neither upper or lower position for longer than two second (adjustable). VSU/VMU should always be in either retracted or extended position except during the transition movement	-> VSU/VMU operational is disabled -> VSU/VMU Positioning Fault is triggered in SCADA -> Sounder and amber warning beacons near the VSU/VMU activate. Please refer to 16 Beacon And sounder Indication for further information. -> Fault reset button on the VSU/VMU control panel, CCP & FCP illuminate.	Press Fault reset button on VSU control panel and CCP
Safety Latch Activated	Safety door surrounding the VSU/VMU is opened	-> VSU/VMU operational is disabled. -> VSU/VMU Safety Latch Activate Fault is triggered in SCADA. -> Sounder and amber warning beacons near the VSU/VMU activate. Please refer to 16 Beacon And sounder Indication for further information. -> Fault reset button on the VSU/VMU control panel, CCP & FCP illuminate.	Please refer to Section 9.4.1 Safety Latch Activated .
Failsafe fault (CBS-VSU only)	Please refer to 9.6.1.5 Failsafe Fault Reaction .		

Table 27 VSU & VMU Fault Cause & Effect

*Upper, Middle, Lower conveyor motor feedback signals are monitored by the conveyor logic and any issues related to these signals are considered conveyor faults rather than VSU or VMU related fault. Please refer to [Section 4.9 Fault and Recovery](#) for conveyor faults.

Safety Latch Activated

Each VSU and VMU contains 2 RFID safety switches, one for monitoring each maintenance access door. If either VSU/VMU access door is opened, the module for that door will lose its power connection, triggering the 'Safety Latch Activate' fault coupled with SCADA and local visible and audible warning, which indicates that a breach in the protective structure has been detected.



Figure 25 VSU & VMU Safety Switch

Each switch is made of 2 parts; one attached to the door and one to the VSU/VMU frame. Each switch is programmed to pair with only its other half, making it challenging to bypass. When triggered, to resolve this fault, once all maintenance access points of the VSU or the VMU have been safely secured, the fault reset button on the VSU/VMU control panel can be used to reset the fault. Following this, a 5-second pre-start warning process must be completed before operations can resume. If either safety switch is found to be faulty, both parts of the faulty switch must be replaced.



Figure 26: VSU and VMU Safety Switch Positions on each Maintenance Access Door

9.5. VSU Operation

VSU

9.5.1

Before a bag proceeds to the Middle Conveyor of the VSU, the divert decision must already be determined. If the VSU is connected to the correct path, the bag is released to the VSU without any stoppage. If the VSU is connected to the other path, a switch over request is issued. If the VSU is cleared to switch, the bag is immediately released. If not, the VSU waits until the previous bag has left the switching zone by clearing the PEC at the beginning of the upper/lower conveyor. Then, the VSU starts switching and releases the held bag to the VSU.

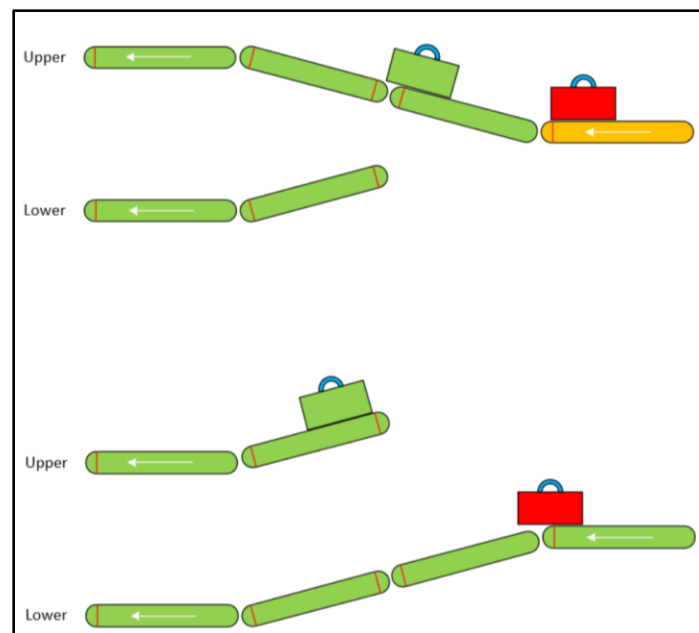


Figure 27 *Diagram of VSU Operation*

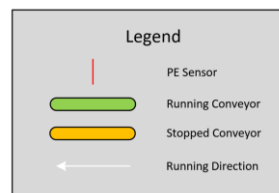


Figure 28: *Legend for the VSU Operation*

VMU

Bags are held before the VMU module if they do not have the right of way. They will wait until the VMU is cleared to switch over before proceeding.

9.5.2.

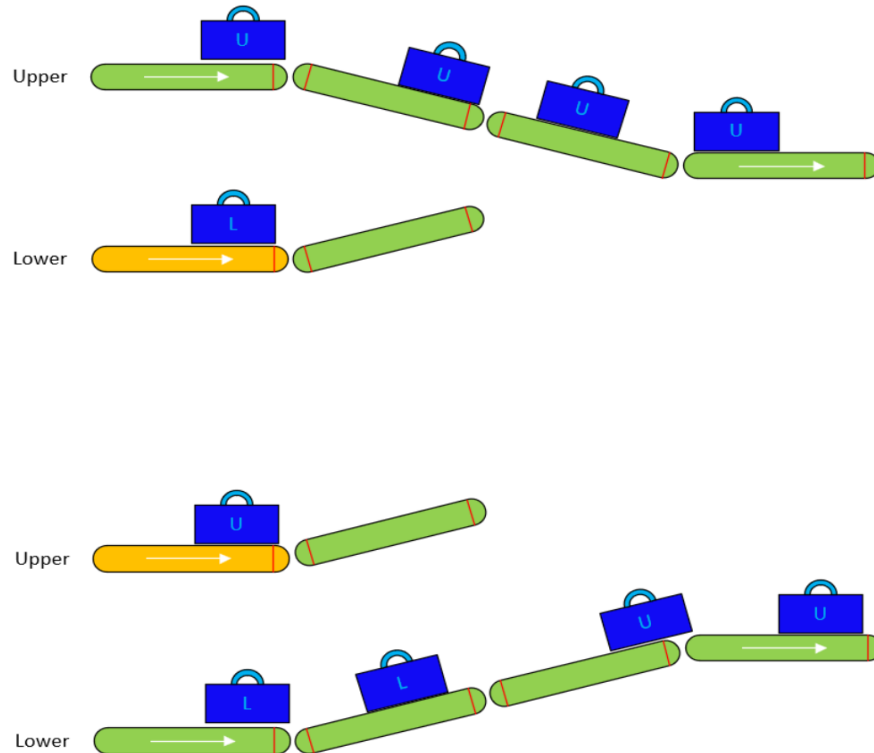


Figure 29 VMU Operation Overview

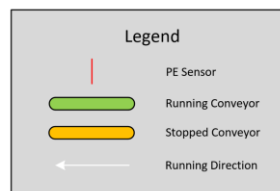


Figure 30 Legend for the VMU Operation

9.5.2.1. VMU Merge Priority

The following merge priority configuration is available for VMU:

- Upper/Lower line priority:
 - Upper or Lower infeed is permanently granted priority.
 - Line with lower priority holds a bag until a suitable window can be found.
- One on One Merge:
 - Whichever bag arrives at the infeed conveyor PEC is granted priority and have the right of way -VMU switch if required.
- Equal priority (N on N merge) – N typically set to 3 – 5:
 - Each line has a fixed number of bags that can be sent to the VMU before it switches to the other infeed.
 - Aims to achieve balance between infeed during high traffic but less switching frequency compared to One-on-One Merge.
 - When the traffic load between the two infeed inputs is highly skewed, the priority of the line with lower traffic can be reduced.

The 4-Lane strategy will use the non-CBS VMU to merge the CL05 conveyor line and transfer line TX03 onto the AS01 conveyor line. As one of the two main lines, upper path CL0511 will be the priority line for the two VMU inputs, and bags from lower path TX0309 will be held upstream of the VMU until a suitable window is found.

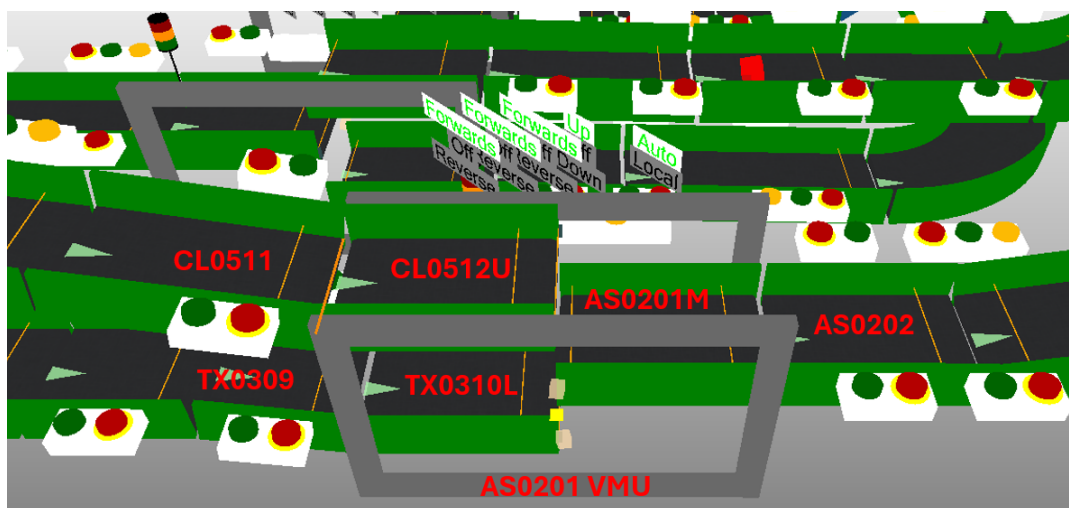


Figure 31 AS0201 VMU

Bag Gap Requirement

A bag gap of 1250mm is required to successfully divert a bag without affecting adjacent bags on the VSU or the VMU and surrounding conveyors running at 1.25m/s. Conveyors upstream of the VSU or the VMU ensure that sufficient gaps are maintained in the approaching bag stream.

9.5.3. SCADA Override

Each non-CBS VMU actuator position can be overridden on SCADA, effectively achieving "Divert Up" and "Divert Down" functionality. This option is used when operators configure a manual load-sharing path for

9.5.4 a certain section of the system, bypassing the HLC load balancing algorithms.

Destination Unable to Receive

9.5.5 If the receiving conveyors of the exit path are unable to receive bags, the conveyors of the VSU or the VMU enter a cascade stop state. This behaviour is similar to the stoppage state experienced by normal conveyor lines when they cannot continue to effectively move bags.

For VSUs in the CBS Operation, **a timeout setting is not permitted** as there is no alternative delivery option for bags that are destined for the reject line.

9.6. CBS Operation

Failsafe

9.6.1.1. General

Conveyor components within the VSU feature full bag tracking. A bag's diverting decision is made when it reaches the tracking PEC of the infeed conveyor. Once the bag enters the VSU, its presence is verified upon arrival at each tracking PEC along its passage.

For a cleared bag, it is expected to maintain intact tracking through:

- From infeed tracking PEC to Middle Conveyor tracking PEC.
- From Middle Conveyor tracking PEC to Cleared Conveyor tracking PEC.
- From Cleared Conveyor tracking PEC to Cleared Line Outfeed conveyor tracking PEC.

For a reject bag, it is expected to maintain intact tracking through:

- From infeed tracking PEC to Middle Conveyor tracking PEC.
- From Middle Conveyor tracking PEC to Reject Conveyor tracking PEC.
- From Reject Conveyor tracking PEC to Reject Line Outfeed conveyor tracking PEC.

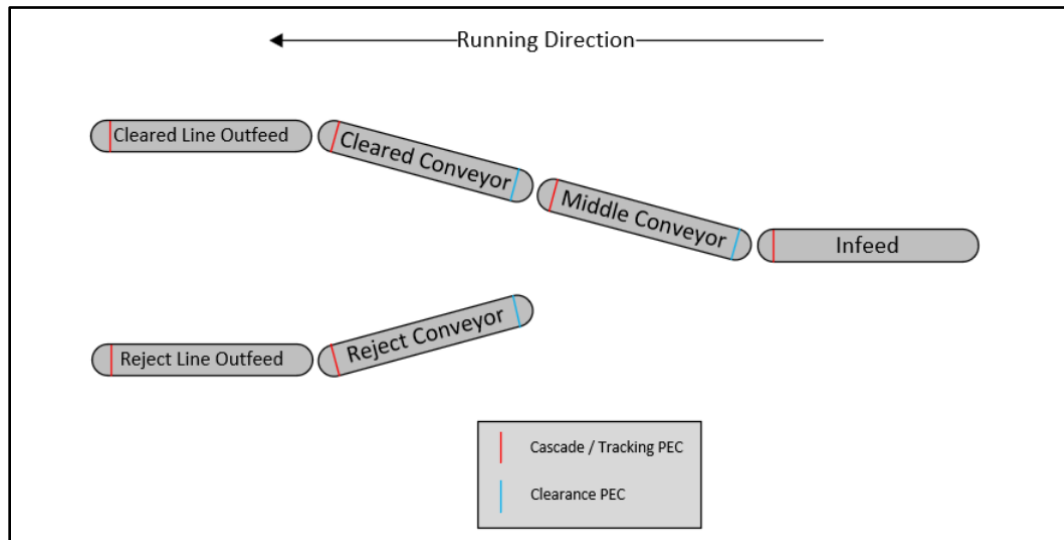


Figure 32: VSU CBS Operation Component

When a tracking PEC fails to detect a bag within the slip window in which the bag is expected to arrive, this is classified as a "**missing bag**" event. Conversely, when a tracking PEC detects a bag outside of the slip window in which a bag is expected to arrive, this is termed an "**unknown bag**" event.

The failsafe algorithm of the VSU is designed to prevent bags from erroneously escaping the reject line under various circumstances. Generally, every bag exiting the VSU via a cleared line must be verified to have a cleared CBS status. Similarly, every bag directed towards the reject line must be confirmed to have correctly arrived at the reject line.

There are two types of failsafe alarm:

- Uncleared Bag Breach Fault.
- Missing Uncleared Bag.

9.6.1.2. Uncleared Bag Breach

A cleared bag is transferred from the tracking PEC of the infeed conveyor through to the tracking PECs of the Middle and Cleared Conveyors while the VSU is switched to the upper position. During this process, if an "unknown bag" or "missing bag" event occurs between Middle and Cleared Conveyor, or if any tracking PEC detects a bag with a CBS status other than "cleared," an "Uncleared Bag Breach" fault is triggered.

9.6.1.3. Missing Uncleared Bag

A reject bag is transferred from the tracking PEC of the infeed conveyor through to the tracking PECs of the Middle and Reject Conveyors while the VSU is switched to the lower position. During this process, if a "missing bag" event occurs between the Middle and Reject Conveyors, a "Missing Uncleared Bag" fault is triggered.

9.6.1.4. Failsafe Scenarios

9.6.1.4.1. Rolling Bags

A cleared bag struggles to climb the incline and potentially becomes stuck.

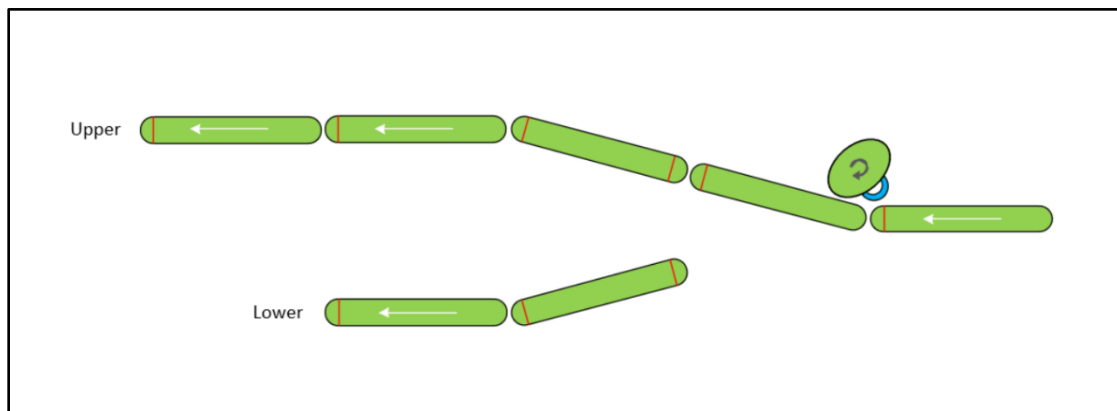


Figure 33: VSU Stuck Bag

As the bag will not arrive at tracking PEC of the Middle Conveyor within in the correct timing, "Uncleared Bag Breach" fault will be triggered.

9.6.1.4.2. Bag Movement Hindrance

Bags with irregular shapes or those with loose straps can become ensnared on the side guards of the conveyor or within crevices, leading to obstructions in the conveyor system. This issue may occur with uncleared bags on the Middle Conveyor of the VSU. In the absence of subsequent bags, if the Middle Conveyor or Reject Conveyor's PEC identifies a "missing bag" event, it will immediately trigger a "Missing Uncleared Bag" fault.

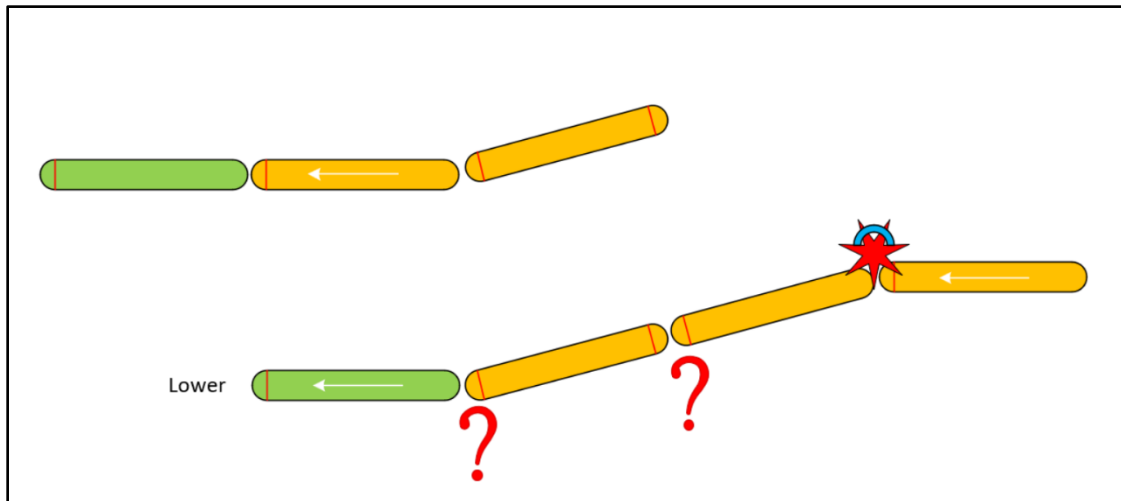


Figure 34: VSU Bag Hindrance

If an uncleared bag gets stuck on the Middle Conveyor of the VSU and the subsequent bag is a cleared bag, it could trigger a change in the VSU to divert to the cleared line destination. However, the "Missing Uncleared Bag" fault would still be activated due to the previous, uncleared bag that went missing. As a failsafe, the cleared bag that was diverted following this fault would need to be rerouted to the reject destination, ensuring that all potential threats are adequately addressed.

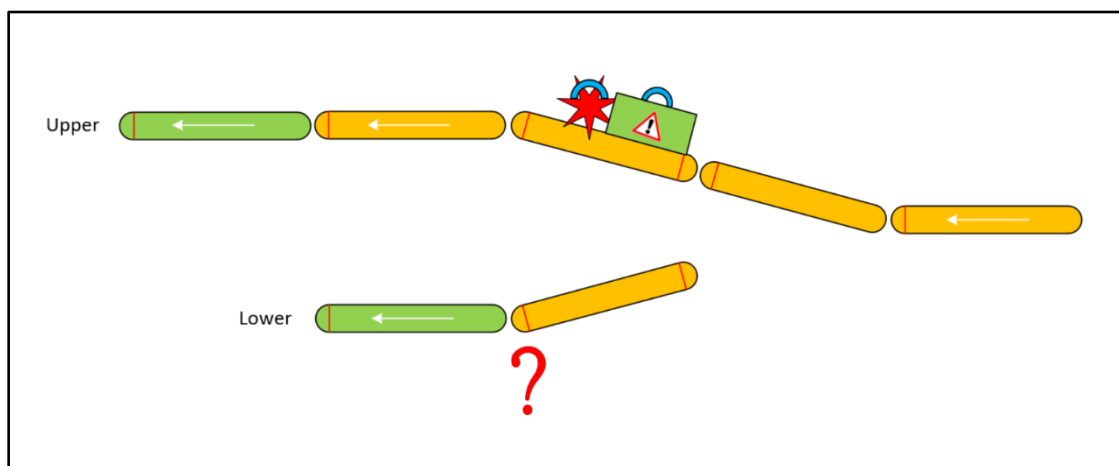


Figure 35: VSU Missing Uncleared Bag Fault

"Missing Uncleared Bag" fault can be triggered even if the following bag is also destined for the reject destination and a "Missing Bag" event is detected on the Reject Conveyor. However, since both bags have already been routed to the reject destination, the operator can swiftly reset the fault. There is no need for reordering the bags.

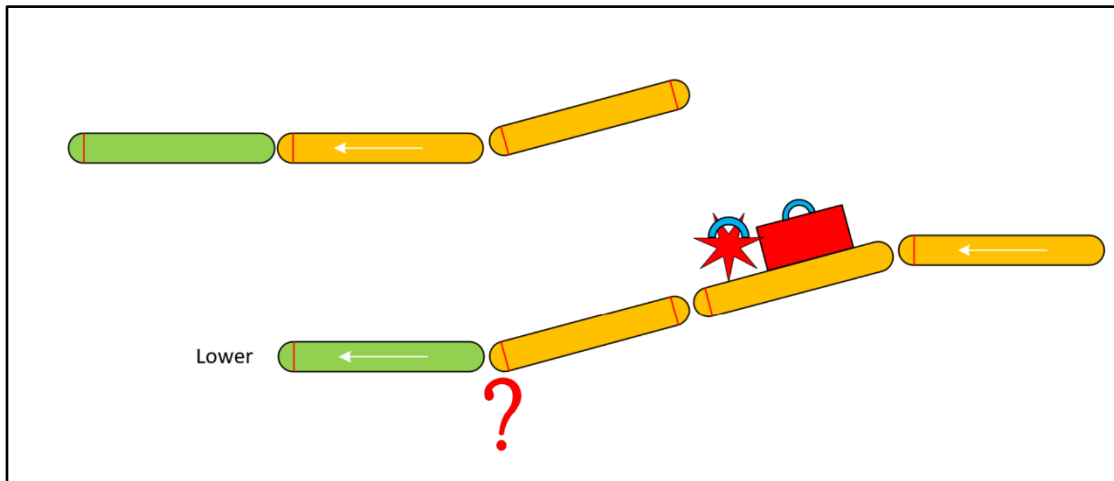


Figure 36: VSU Missing Bag Event

9.6.1.5. Failsafe Fault Reaction

When a "Uncleared Bag Breach" or "Missing Uncleared Bag" fault occurs, the following actions are implemented:

- The infeed conveyor of the vertical sorter stops feeding.
- The Cleared Line Outfeed Conveyor stops.
- The sounder and the local blue beacons are activated. Please refer to [Section 16.6 CBS Status Indication](#) for further information.

This fault is signalled in the SCADA system by the following actions:

- The icon for the VSU shows the fault condition.
- The status dialog for the VSU reports the fault detail.
- The icons for the warning beacon and sounder at the vertical sorter flash when the devices are activated. Please refer to [Section 16.6 CBS Status Indication](#) for further information.

9.6.1.6. Recovering from Failsafe Fault

When the VSU and adjacent conveyors halt due to a failsafe fault, all bags within the VSU module and on the Cleared Line Outfeed Conveyor must be redirect to the reject line. The bag on the Cleared Line Outfeed conveyor must be manually removed and placed on the Reject Line.

For the bags that are still in the VSU, the VSU must be switched to local mode to remove the bags (Please refer to [Section 9.6.4 CBS VSU Local Mode](#)). When completed, the operator switches the position of the Middle Conveyor of the VSU to connect to the Reject Conveyor, and the fault can be reset using the fault reset button on the VSU control panel and change the VSU back to Auto Mode.

Before resuming operation, the BHS control system initiates a "flush" operation to clear all bags from the reject bag path through the vertical sorter (Please refer to [Section 9.6.2 CBS VSU Flush](#)). Once all bags are flushed out of the system, operations can resume, allowing the infeed conveyor to send new bags into the VSU.

9.6.1.7. VSU Fault Recovery for CBS Operation

Jam faults within the components of the VSU are detected using standard conveyor logic. However, if a jam occurs within the VSU, the process for recovery differs from standard procedures. Specifically, any bags affected by such a fault must be redirected to the reject destination. This rule aligns with the failsafe mechanisms in place, prioritising the security aspect of the system by ensuring that no uncleared or suspicious bags are mistakenly routed to the cleared destination during or following a fault condition.

In addition to the bags affected by the fault, other bags within the "threat zone" are required to be extracted and sent to the reject line. The threat zone extends from the end of the Middle Conveyor in the VSU to the outfeed conveyor of the cleared bag line as highlighted in red below:

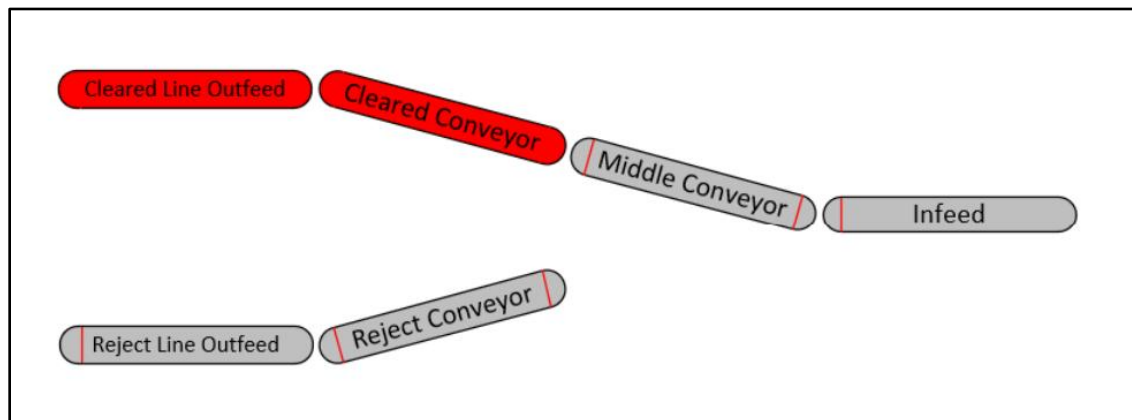


Figure 37: VSU Threat Zone

VSU Manual mode can be used to redirect bags to the reject line if necessary.

CBS VSU Flush

The function described in this section only applies to the VSU used in CBS operation. For details of how VMU flush works in non-CBS operation, please refer to [Section 9.39.3 Non-CBS VMU Flush](#).

9.6.2 The purpose of this function is, in case there are remaining bags within the VSU, to flush all bags from within the VSU to the reject line. Any bags detected on the cleared path during the flush operation cause the operation to halt and trigger “Uncleared Bag Breach Fault”.

A prestart warning followed by a flush operation can be initiated after one of the following scenarios:

- The VSU is switched from manual to automatic mode when the Middle Conveyor is connected to the reject path.
 - **The VSU needs to be put in manual mode first** because the VSU needs to have the middle conveyor manually moved to the reject path position otherwise the flush operation won’t occur.
- A failsafe fault has been resolved and reset.
- A jam or encoder fault within the VSU has been resolved and reset.
- A positioning, failed to switch down, or a failed to switch up fault has been resolved and reset.

The blue reset indicator on the VSU control panel flashes at 2Hz until the flush operation has been successfully completed. The conveyor upstream of the VSU resume feeding bags into the VSU.

The CBS VSU flush will operate as follows:

1. The Prestart warning will operate for 5 seconds. Please refer to [Section 16.3 Pre-Start Warning](#).
2. The Control Panel Fault indicator will flash at 2Hz.
3. The VSU will switch to the Lower position.
4. The Middle and Lower conveyors will run for a time to flush all bags to the Reject line.
5. The VSU will switch to the Upper position.
6. The Upper and Middle conveyors will run in Reverse to for a time, to bring any bags on the Upper conveyor onto the Middle conveyor.
7. The VSU will switch to the Lower position.
8. The Middle and Lower conveyors will run for a time to flush all bags to the Reject line.
9. The Control Panel Fault indicator will extinguish after the flush is complete.

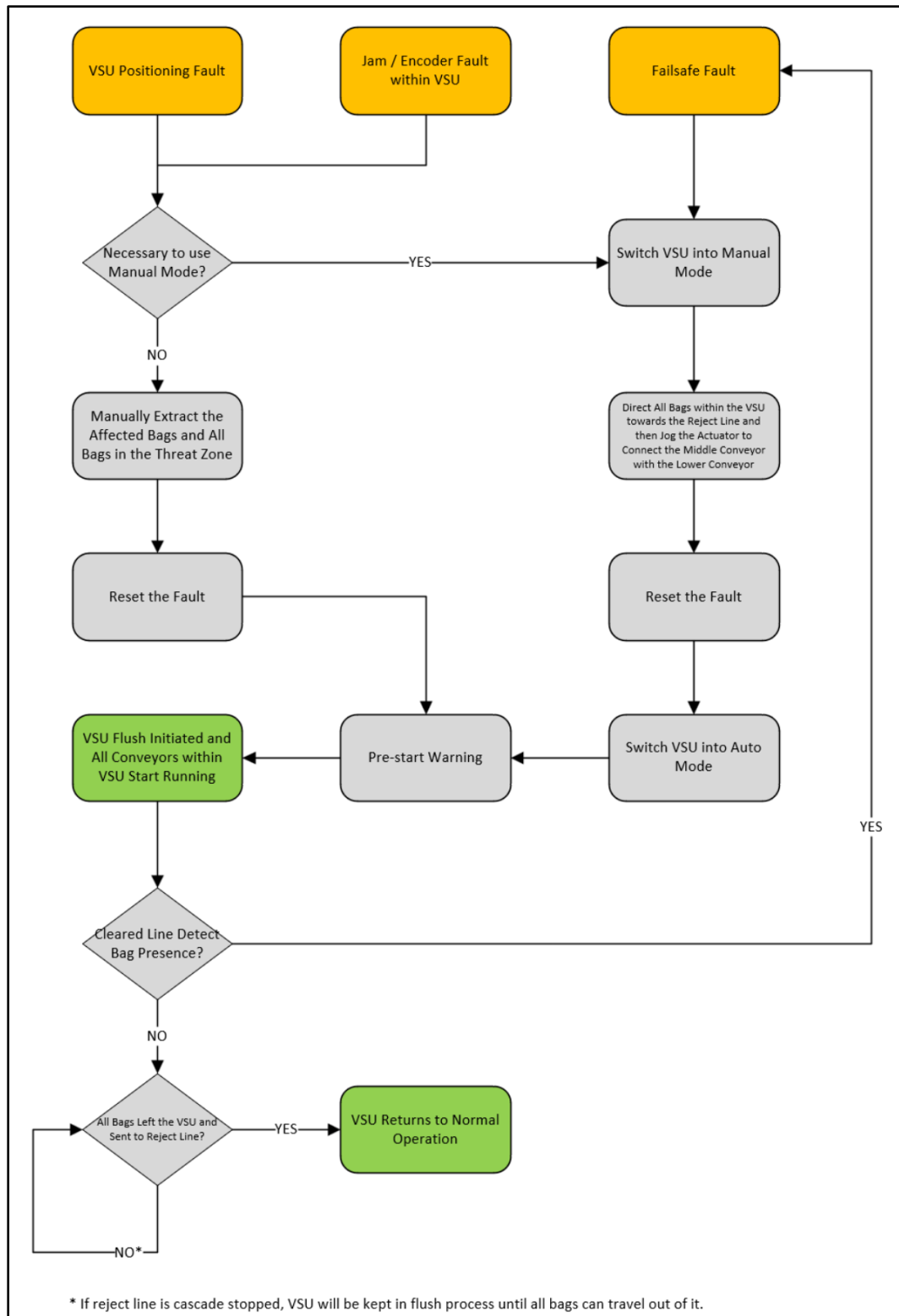


Figure 38: CBS VSU Fault Recovery and Flush Procedures

Decision Pending

The layout of the CBS line ensures that a bag's travel time between the exit of the EDS machine and the CBS divert decision point is greater than the CBS Level 2 decision time limit. Therefore, it's not likely that a tracked bag reaches the CBS VSU without having a Level 2 decision however, any Level 1 Reject bag without Level 2 decisions is regarded as uncleared and sent to the reject line.

CBS VSU Local Mode

In CBS operation, the VSU's local mode has extra functions and constraints.

9.6.4. In addition to the upper belt (Cleared Conveyor), the JOG selector switch can also control the clear line outfeed conveyor, which can only run in reverse. This allows for any bags that exited to the cleared line during a failsafe fault to be redirected to the reject line. The upper conveyor's JOG forward function can only be activated if its PEC is unblocked, preventing any bags from being sent out of the VSU to the cleared line while in manual mode.

Switching to manual mode erases all tracking data and CBS status for bags in the VSU and on the Cleared Line Outfeed.

Before the VSU can return to automatic mode and initiate the flush process, all PECs in the "threat zone" (Please refer to 9.6.1.6 Recovering from Failsafe Fault) must be cleared and the VSU must be switched to the lower path (reject line).

In CBS operation, the VSU's Manual Mode includes additional functions and constraints to ensure safe and compliant handling of suspect baggage.

When in Manual Mode:

- The JOG selector switch can control both the upper belt (Cleared Conveyor) and the Cleared Line Outfeed Conveyor. However, the outfeed conveyor is restricted to reverse direction only to allow any bags that were sent to the cleared line during a failsafe fault to be redirected to the reject line.
- The JOG Forward function for the upper conveyor is only permitted if the associated PEC (Photoelectric Cell) is unblocked, ensuring that no bags are unintentionally released from the VSU to the cleared line while in manual mode.
- Upon switching to Local Mode, all tracking data and CBS status associated with bags located within the VSU and on the Cleared Line Outfeed are erased.

To return the VSU to Automatic Mode and enable the Flush process:

- All PECs within the defined "threat zone" (see 9.6.1.6 Recovering from Failsafe Fault) must be clear.
- The VSU must be in the lower path position (i.e., aligned to the reject line). The operators are required to have manually moved the middle conveyor to the lower path prior to switching to automatic mode, the system will not start if the VSU is in the incorrect position.

CBS VSU Default Position

In its default state for CBS operation, the VSU is configured to switch the Middle Conveyor to the lower conveyor. When the subsystem has completed the flush and stop process, or when all local conveyors have entered ECM mode, the actuator will tip down if it is not already in the lower position.

9.6.5.

10. ATR

10.1. General

An ATR utilises a camera \ RFID hybrid system (by SICK) to provide bag tag reading for CBS and Sortation. Metering conveyors upstream of the ATR's provide correct bag spacing for effective tracking and diverting.

Each ATR is equipped with two redundant MSC800 controllers. Each controller is connected to the PLC via its own EtherNet/IP port and assigned a unique IP address. The PLC communicates with both controllers; however, only the active controller transmits data to the PLC. Within the PLC software, the ATR controllers are configured as Primary and Secondary, with the active status monitored to determine controller availability. Communication with each controller is also continuously monitored, and an alarm is generated in the event of a communication failure.

When the barcode on the bag tag is read successfully, the barcode number is sent to the High Level Control system (HLC), which holds and verifies bag and tracking information. The HLC directs the bags at specific divert points depending on the bag status and sortation requirement. The PLC can route bags appropriately if connection to HLC is lost.

If the connection to HLC is lost, sortation can continue if bags have either of the following:

- Fallback tags.
- A pre-existing sortation destination assigned to the bag.

If there is a problem reading a tag at an ATR or resolving an unknown sort of destination, the bag is directed to manual encode (ME).

The ATR/RFID scanner is the primary scanner including 10 optical scanning heads and a four-antenna RFID scanner. This will be configured as a hybrid unit similar presenting a single interface to the PLC. The scanner will have an extended read zone, at the end of which will be returned any optical tags or RFID tags discerned for that bag. The ATR allows for up to eight tags total. They may consist of up to a maximum of 5 paper tags and 3 RFID tags (And any combination under these maximums). The tags are transmitted from the reader to the PLC in a single message, identifying the source of the tag (paper and/or RFID). Only 10-digit IATA compliant tag numbers are transmitted. Any other data contained in the RFID tag is NOT transmitted. All ATRs in DTB have RFID capability.

Bag Tag Type	ATR Pre-EDS	ATR Post-EDS	CBS Decision	ATR to PLC interaction	Result
Paper Tag only – read	Yes	Yes	Clear	Barcode sent to PLC, PLC sends barcode to HLC	Bag goes to destination
Paper Tag only – no read	Yes	No	Clear	No barcode received in the PLC	Bag goes to ME

Paper Tag only – no read	No	Yes	Unknown Bag	Barcode sent to PLC, PLC sends barcode to HLC	Bag goes to Level 3
RFID Tag only – read	Yes	Yes	Clear	Barcode sent to PLC, PLC sends barcode to HLC	Bag goes to destination
RFID Tag only – no read	Yes	No	Clear	No barcode received in the PLC	Bag goes to ME
RFID Tag only – no read	No	Yes	Unknown Bag	Barcode sent to PLC, PLC sends barcode to HLC	Bag goes to Level 3
Paper & RFID tags – read	Yes	Yes	Clear	Barcode sent to PLC, PLC sends barcode to HLC	Bag goes to destination
Paper & RFID tags – no read	Yes	No	Clear	No barcode received in the PLC	Bag goes to ME
Paper & RFID tags – no read	No	Yes	Unknown Bag	Barcode sent to PLC, PLC sends barcode to HLC	Bag goes to Level 3
Paper or RFID tag – read	Yes	Yes	Threat bag	Barcode sent to PLC, PLC sends barcode to HLC	Bag goes to Level 3

Table 28 ATR Tag Type Decision Matrix

Sick will provide the Logistics Analytics Performance software for the ATRs, this is not in Alstef scope but part of the ATR interface with BAC directly.

Please refer to the ATR Airport Luggage Identification System - Standard Interface Specification (EIP) and to the Logistics Analytics User Manual for further information.

10.2. Power configuration

The ATR control panel gets its power from the BHS PDP UPS via BHS CCP. If the ATR experiences a power loss, this is detected by the PLC through heartbeat monitoring, which renders the ATR inoperative and PLC to update its status via SCADA. The conveyor function and overall system operation are not disrupted by the loss of power to the ATR, and all bags passing through are effectively classified as "no-read" for this instance. Any bag tracked up to this point retains all its existing information, undisturbed by this malfunction. The ATR is designed to automatically recover once power is restored, resuming operations without requiring any reset procedures from the LLC.

10.3. Tag Reading Process

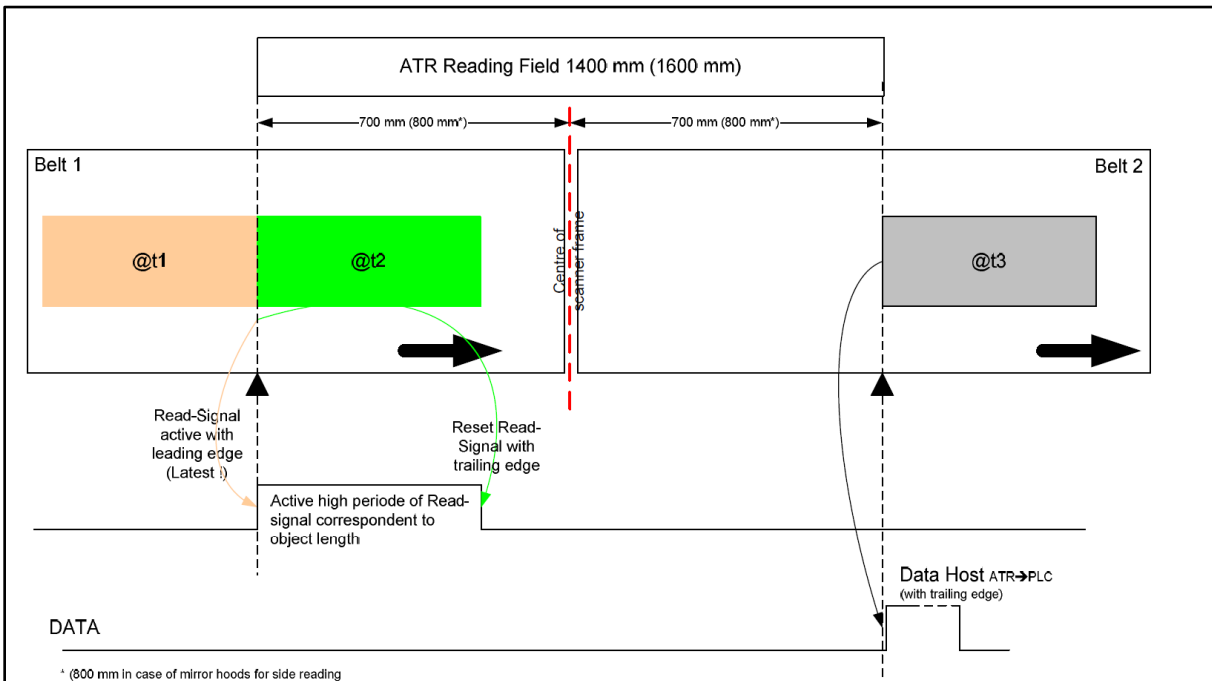


Figure 39: ATR Reading Field

Conveyor belt 1 and 2 run synchronously and with the same speed. The Read-Signal is generated at the rising edge of the object (@t1), a maximum of 700mm (800mm for mirror hoods side reading models) before the scanner's centre. The position of the signal's rising edge remains consistent for all objects. The trigger signal's length correlates with the object length. The LLC bag ID must be sent while the read signal is active (@t2).

The object then fully enters the Reading Field. The Reading Field extends to a minimum of 1400mm. The data transmission point is approximately 1400mm-1600mm after the deactivation of the Read-Signal (@t3), regardless of the object length. More than one object can exist simultaneously in the reading field, provided a minimum gap of > 150mm is maintained between two objects and/or tags.

10.4. RFID

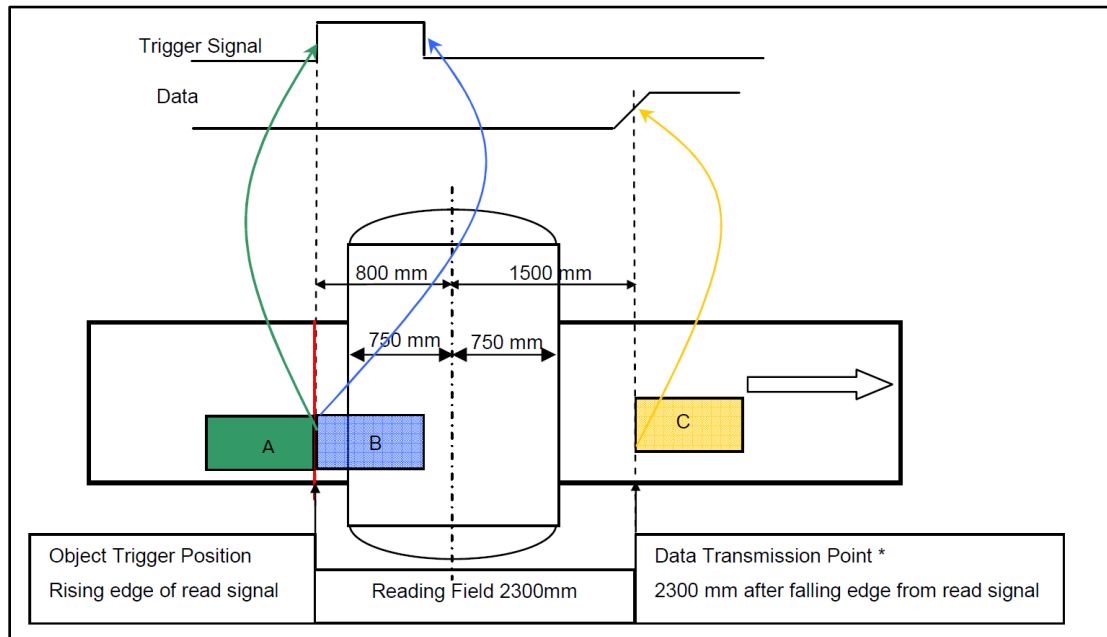


Figure 40: ATR RFID Tunnel

Radio Frequency Identification (RFID) functionality is included in the scope of the DTB BHS. All ATRs installed with the new BHS are fitted RFID modules.

The RFID tunnel is integrated into the camera gantry and reads the bag's identity simultaneously with the optical system. The RFID module does not require additional bag gaps compared to the optical mechanism and completes its reading before the optical counterpart as well.

The ATR controller verifies the flight date information contained in the RFID tag. Only if the date matches the day of operation will it extract and send the IATA code to the LLC.

10.5. Barcode Format

Each item of baggage that is to be handled by the BHS must be labelled with a bag tag containing a unique bag identifier (IATA bag ID) or coupled with a non-unique pier tag for fallback operation. The content and layout of all tags must conform to the relevant IATA standards.

The High Level Control system (HLC) is designed to work with demand-printed baggage tags that conform to the format defined by IATA recommended practice IATA RP 740. The HLC is designed to work with fallback (pier) tags that conform to the format defined by IATA recommended practice IATA RP 1740b. Items checked-in via the out-of-gauge facility must also be identified with an IATA format bag tag containing a unique bag identifier as described above.

10.6. Barcode Prioritisation

The ATR is capable of reading barcodes on all surfaces of a bag. It internally filters out invalid tags and transmits up to five IATA-compliant paper tag reads to the PLC. The ATR can also transmit up to three RFID tag reads per bag.

In total, a single ATR scan may contain up to five paper tags and three RFID tags.

The PLC forwards the read results to the HLC. With access to the BSM, the HLC determines the most relevant tag(s) and responds to the PLC accordingly via a SAM message.

10.7. Tag Handling

No-Read

10.7.1. The ATR returns “?????????” to indicate that no valid IATA tag can be read from the bag. No-read bags on the cleared line are sent to the MES for manual scanning.

Multi Read

10.7.2.

Bags with 2 or more tags of a single type are termed multi-read tags. If there are more than 5 paper tags and/or 3 RFID tags, the ATR returns “!!!!!!!!” to indicate that too many different IATA tags can be read from a bag. Multi-read bags are processed in the same way as no-read bags.

10.7.3.

Pier/Flight Tag

10.7.4. Pier Tags and Flight Tags are used in fallback scenarios only. They are not unique across the system as they are coded with destination information only.

Q-Tag

Q-Tags are airline provided tags attached to bags. They are unique RFID tags that only contain flight information the same as a flight tag.

The ATR unit does not check paper/RFID tag duplication, the PLC must do this check and eliminate any duplicates (between paper and the RFID tags). Each bag record in the PLC can hold a maximum of 8 unique tags, however the bag record does not mark the tag as paper or RFID therefore the PLC does not differentiate between a paper tag or a RFID Q-Tag on a bag, either that is scanned by the ATR will uniquely identify the bag and its destination.

By default, the PLC will report to HLC tags as paper tags unless the tag(s) returned by the ATR are exclusively RFID Q-Tag(s) (i.e. no paper tags are scanned by the ATR).

Read Result Response Matrix

Each bag should have a single, unique, and valid BSM tag to establish its identity. In fallback mode, a bag may also possess one valid Pier tag, serving as an additional attribute. Please refer to the Airport (BNE) DTB HLC Phase 2 Functional Design Specification, section **XXX** for the ATR barcode matrix.

10.7.5.

CBS Decision Reconciliation

The ATRs prior to each CBS decision point are designed to reconcile any mis-tracked bags in the security tracking zone.

10.7.6 The PLC, upon retrieving the IATA barcode, initiates a search through the local database to find the original scan result for the bag. Simultaneously, the HLC provides the CBS status along with the sort destination. Reconciliation of a mis-tracked clear bag requires clear status confirmation from both sources. In the event of HLC going offline during fallback operations, the BHS will exclusively refer to the local search results.

The goal of this feature is to minimise the number of clear bags erroneously directed to the reject line due to mis-track. However, for security reasons, force rejected bags cannot be reconciled.

It is expected that the use of BSM tags in tandem with Pier tags when operating in fallback mode. While the Pier tag overrides the sort destination, the BSM tag is maintained by the BHS for operations necessitating the unique identification of a bag, like CBS reconciliation. Please note, CBS reconciliation cannot be carried out if only the Pier tag is utilised.

10.8. Consecutive No-Reads

If the ATR reads a number of consecutive No-Reads, this means that there is potentially a group of bags entering the Subsystem that do not have tags.

When the ATR detects 10 No-Reads in a row, an Alarm will be displayed on the SCADA (ATR/<name> Alarm Multiple No Read). There will be no further action.

The Alarm will be reset when the ATR reads the next barcode.

10.9. Low Read rate

The ATR Read Rate Low Alarm helps monitor the performance of the Automatic Tag Reader (ATR) system. It alerts the operator when the ATR is not reading baggage tags at an acceptable rate, indicating a potential issue that may require attention. The alarm (TR/<name> ATR Read Rate Low) will activate if both of the following conditions are true:

- The ATR read rate drops below 90%.
- At least 100 bags have passed through the ATR since the beginning of the shift.

The alarm will not trigger if fewer than 100 bags have been processed, even if the read rate is low. This is to prevent false alarms early in the shift.

This alarm will be automatically reset under the following conditions:

- The ATR read rate improves and rises above the threshold value (90%).
- A manual or system-initiated statistic reset is performed.

- A midnight pulse occurs, resetting the daily bag count.

10.10. Low Low Read rate

The ATR Read Rate Low Low Alarm is a high-priority alert that signals a critical drop in the performance of the Automatic Tag Reader (ATR). This alarm indicates a potential configuration issue or a batch of untagged bags entering the system.

This alarm (ATR/<name> ATR Read Rate Low Low) is triggered when both of the following conditions are true:

- The ATR read rate drops below 75%.
- At least 100 bags have passed through the ATR since the beginning of the shift.

This alarm is designed to capture more serious issues than the standard “Read Rate Low Alarm.”

This alarm will be automatically reset under the following conditions:

- The ATR read rate improves and rises above the threshold value (75%).
- A manual or system-initiated statistic reset is performed.
- A midnight pulse occurs, resetting the daily bag count.

10.11. ATR Read Failure Fallback

10.11.1. Pre-EDS ATR Read Failure

If a Pre-EDS fails, bags will continue to be processed as follows:

1. Bags will be screened by the EDS system using the assigned BHS ID number.
2. When a bag is read at a subsequent Post-EDS ATR, the barcode will be reconciled with its screening results, the HLC will assign a sortation destination, and the bags will be processed accordingly. If a bag

10.11.2. is mis-tracked after the EDS Machine, the Secondary ATR will not be able to reconcile the bag with its screening results. This bag will be diverted to the Level 3 line.

Post-EDS ATR Read Failure

If a Post-EDS ATR fails, bags will continue to be processed as follows:

1. Bags will continue to be processed according to the current information held in the PLC tracking database.

11. CHECKED BAGGAGE SCREENING SYSTEM

11.1. General

Each inline EDS consists of a single conveyor that is uninterrupted throughout the length of the machine. The scanned object does not need to stop or slow down during the scanning process, which operates at a constant speed of 0.5m/s. The EDS has a tracking PEC inside the scanning tunnel to run its own tracking model.

As a bag enters the EDS conveyor, the bag object will be inserted into the EDS bag model, and the EDS will request the Bag ID from the BHS. The BHS does not need to track the bags on the EDS conveyor, as the EDS controls the tracking process and passes the correct information to the BHS when the bag exits the machine.

The Overall Bag Disposition Field will use Set B.

The Last Known Decision Field will use Set B.

For further details on the CBS process please refer to the RTT 110 BHS Protocol Specification – Airport-TSA and the DTB Simulation FDS, section 4.6..

11.2. Control Panels

An Exit Reset pushbutton is installed on the infeed side of the EDS machine and has the following layout:

Control	Device	Colour
Exit Reset	Illuminated Green Pushbutton	Green
Test / Normal	Two-Position Key Switch	N/A

Table 29 EDS Infeed Control Station

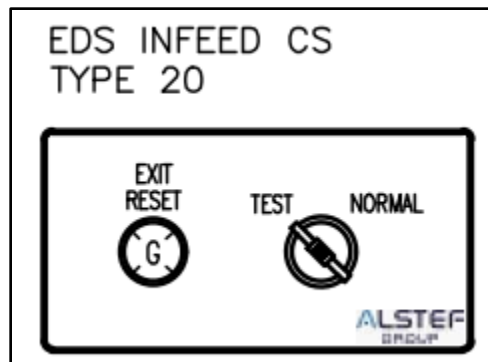


Figure 41 EDS Infeed Control Station

A control panel is fitted on the outfeed side of the EDS machine and has the following layout:

Control	Device	Colour
Emergency Stop	Red Illuminated Mushroom-Type Pushbutton	Red
Restart	Green Illuminated Pushbutton	Green
Jam Reset	Amber Illuminated Pushbutton	Amber
Exit Reset	Green Illuminated Pushbutton	Green

Table 30 EDS Outfeed Control Station

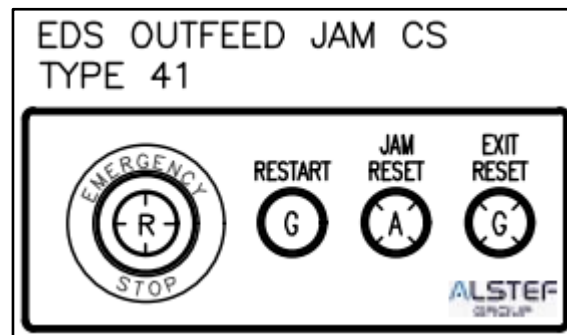


Figure 42 EDS Outfeed Control Station

11.3. Operation Mode and Status

An EDS has the following modes that the PLC can interact with, during a fully integrated configuration:

- **Standby:** EDS should be in this mode when the BHS is inactive for more than 15 minutes. To switch back to screening mode it will take:
 - 5 sec if in Standby
 - 30 sec if in Standby/Lowered Ready
 - 60 sec if in Standby/Filaments off

To minimise bag waiting time and potential dieback, The BHS will force the EDS out of the standby mode by issuing the 'Keep Alive' signal when a bag is detected upstream of the CBS Line.

- **Scanning:** EDS should be in this mode when the BHS is enabled.
- **Image Quality Test:** EDS test mode, refer to [Section 11.7 Image Quality Test Mode](#).
- **Path Through:** EDS becomes a conveyor line with screening disabled.
- **Low Power:** Not recommended for BHS operation.
- **Lock Down:** Radiation component disabled, maintenance only.

And the following operational statuses:

- **Power Off:** For maintenance purposes only.
- **Starting Up:** Startup from Power Off state, may take up to 15 minutes.
- **Shut Down:** Power turned off except for some control circuit, maintenance only.
- **Calibration:** Takes place during scanning state, takes a few seconds to complete.

11.4. Bag Spacing and Presentation

The EDS machine can handle the following bag profile:

- Bag length from 80mm up to 2300mm.
- Minimum bag gap of 300mm.
- Minimum bag weight of 5kg.

Please refer to the RTT 110 BHS Protocol Specification – Airport- TSA Following conditions are to be avoided:

- Standing bag. (Bag upright on one of the short sides of the bag)
- Bag with telescopic handle extended.
- Bag placed severely off-centre and rubbing against the conveyor side wall.
- Bag with loose straps or excessive free moving object protrusions.
- Non-conveyable items without a carrier tub.
- Bags placed side by side. (Bag gap of less than 300mm).

11.5. Bag ID and Decision Communication

A bag identification is required for each screening instance. This identification has two parts:

- A unique Pseudo/Screening ID is generated for each bag entering the EDS machine - this ID serves as a reference for the screening instance.
- IATA Code of the bag previously read by an ATR.

The Pseudo/Screening ID is a mandatory component, while the availability of the IATA code relies on the readability of the barcode. Should a no-read situation arise, the system can still use the Pseudo/Screening ID exclusively for the CBS decision update. However, in a mis-track situation, only bags with a readable IATA code can have their CBS results reconciled instead of necessitating a rescan.

When the bag's identification is requested at the entrance, both the Screening ID and IATA tag (if available) are sent to the EDS. Throughout the rest of the CBS process, the EDS returns the bag's identification with Level 1 and Level 2 (if needed) decisions attached.

If a bag's identification is not successfully received at the EDS machine entrance, the EDS assigns a dummy ID to the bag, allowing it to complete the screening process. The BHS can recognise this dummy ID and create a new bag instance so it can still be processed through the CBS.

Scenario	Pseudo ID	IATA (BSM)	Returned at the EDS Exit	Valid CBS Decision	Reconcilable	Result
Bag with read IATA	Yes	Yes	Pseudo + IATA	Yes	Yes	Fail or Clear
Bag with ATR No-Read	Yes	No	Pseudo	Yes	Yes, at downstream ATR	Fail or Clear
Bag ID transaction unsuccessful	No	No	EDS Dummy ID	Yes	Yes, at downstream ATR	Fail or Clear
Mis-tracked in EDS machine	N/A	N/A	EDS Dummy ID	No	Yes	Rescan
Bag Stoppage while in EDS machine	Yes	Yes	Pseudo + IATA	No	Yes	Rescan

Table 31: Bag Identification

A read, single, unique, and valid BSM prior to the EDS machine is required for CBS reconciliation in the case of a mis-track.

11.6. Bag Flow Control

The BHS system interfaces with the EDS machines, updating their outfeed availability to receive bags. In the unlikely event of an unavoidable die back/EStop causing the EDS tunnel conveyor to halt while bags are undergoing the screening process, the reliability of the screening results is compromised. The BHS system strives to prevent this from happening by proactively monitoring downstream bag flow patterns to predict potential cascade stop scenarios and regulating the EDS infeed accordingly. However, should the EDS conveyor stop, the affected bags could be marked with a no-scan status and rerouted for rescan via the reject line.

11.7. Image Quality Test Mode

This section outlines the operational procedure for executing an Image Quality Test on the Explosive Detection System (EDS) prior to the start of daily operations. This procedure ensures that test bags can be safely introduced, scanned, and retrieved under controlled conditions.

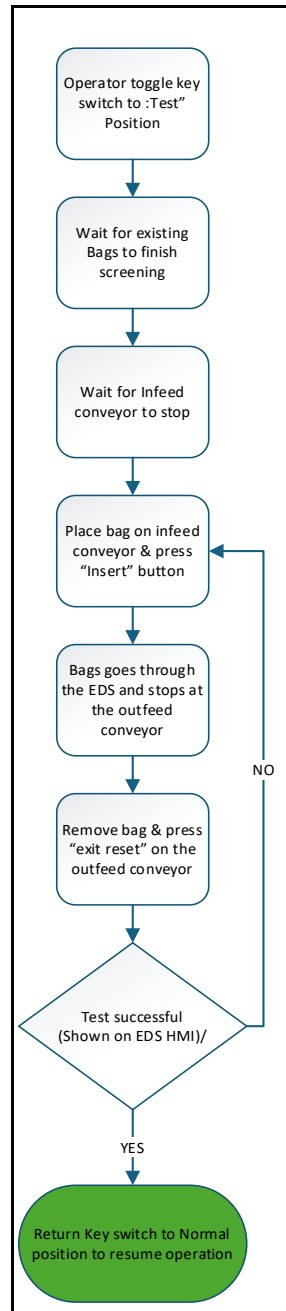


Figure 43 Image Quality Test Mode

Test Mode Activation

- A security operator enables Test Mode by turning the physical key switch on the local control panel to the “IQ Test” position.
- In this mode, the system halts the release of new bags upstream of the EDS infeed conveyor.
- 11.7.1. Bags already in transit are allowed to pass through and be processed to completion.
- Once the test zone is clear of all bags, the PLC sends a request to the EDS to enter Image Quality Test Mode.

Test Mode Readiness

- The operator verifies readiness via the local EDS HMI.
- 11.7.2. The Insert Bag button becomes available (illuminated), and the EDS infeed conveyor is stopped.
- SCADA updates the EDS icon to indicate Test Mode, and a corresponding warning message is displayed.

For easy access to load/unload bags, side guards on the following conveyors have been permanently removed:



Figure 44: EDS Infeed Conveyor Sideguard and Control Station

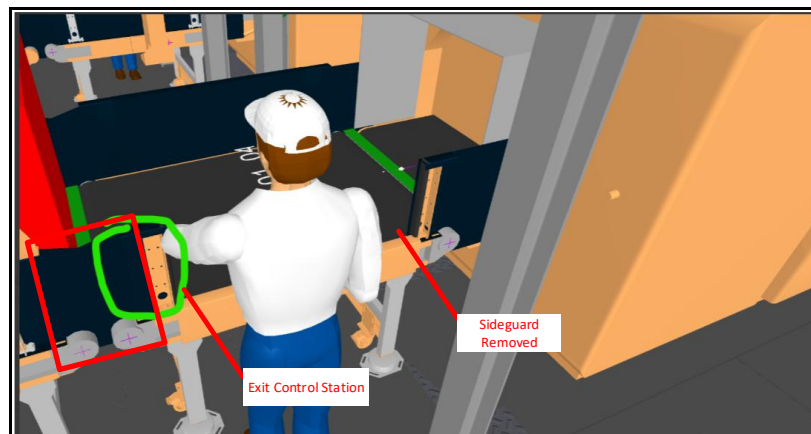


Figure 45: EDS Exit Conveyor Sideguard and Control Station

Bag Insertion and Processing

- The operator places the test bag on the infeed conveyor and presses Insert Bag on the EDS HMI.
- This starts the conveyor, which conveys the bag into the EDS for scanning. The infeed conveyor continues to run until the test bag has fully cleared (cascade PE cleared), then stops. After scanning, the bag exits onto the outfeed conveyor and stops at the PEC, causing the Exit Reset button to illuminate.

Bag Retrieval and Exit Reset

- Once the bag is removed from the conveyor, the Exit Reset indicator begins flashing at 1Hz.
- Pressing the Exit Reset button will:
 - Restart the outfeed conveyor, and
 - Extinguish the flashing indicator.

Test Completion and Normal Mode Restoration

- After reviewing the results on the local EDS HMI, the operator may:
 - Repeat the test procedure as required, or
 - Turn the key switch back to the "Normal" position
- Upon return to Normal mode:
 - The PLC sends a command to the EDS to exit Test Mode and return to Scanning Mode.
 - Once confirmation is received, the PLC will initiate the startup warning and the conveyor section will resume normal operation.
 - SCADA shows the EDS machine as healthy and available.

11.8. Transit Mode

The EDS machine can be switched to "Transit Mode" or "Pass Through Mode," via its built-in HMI, allowing the EDS to act as a conveyor without any screening capability. During this mode, no CBS decisions can be issued, and all bags are sent to the reject lines and subsequently direct to rescan route to other EDS machines. This mode may be used during maintenance, calibration, or other non-operational situations where the screening function is temporarily not needed, while still allowing the baggage to move through the BHS.

11.9. SCADA Indication

The PLC consistently updates the indication tags communicated to SCADA, which then displays the real-time status, operational mode, and alarm states of each EDS machine. For further details, please refer to “SCADA / HLC Function Specification”.

11.10. Screening Result between ITB and DTB

CBS results are accessible and can be polled between both the ITB and DTB. Transfer bags moving from the ITB to the DTB can obtain their screening status by reading their IATA barcode, eliminating the need for additional screening processes. For details, please refer to “SCADA/HLC Function Specification”.

11.11. Minimum Screening Level Requirement

Information for minimum screening level requirement is retrieved from the BSM and this message sent to the BHS via SAM Response message at an ATR. With this requirement, BHS/HLC can only accept “Clear” result made from the required level. Detailed screening result processing is as below:

11.11.1. Unknown Required Screening Level

This is the scenario where a bag mistracked or had “No Read” at the Primary ATR and screening requirement level is unknown. The following steps will occur:

- BHS will set required screening level to 1 (default) when bag enters the EDS machine.
- Security Routing Requirement code 11 will be sent to the EDS.
- The EDS will send L1 result to the BHS.
- If the decision is “Reject”, the image will be sent to a L2 operator.
- The EDS will then send a L2 decision to the BHS.
- Bag arrives at the Secondary ATR and attempts to receive screening requirement level information.
- Bag will travel towards a CBS VSU, the BHS will verify screening status against the required screening level. If the minimum required screening level is not met, bag will be sent towards L3 for inspection at a higher level. If the minimum required screening level remains unknown, the BHS will use L1 as the default setting.
- For a bag with unknown required screening level, the bag will be routed towards a ME line as the barcode is unknown.
- The bag will arrive at a Pre ME ATR and attempts to recover screening requirement level information. If this information is still unknown or minimum screening level is not met, the bag will be diverted to ME for manual process.

Minimum Screening Requirement Level 1

This is the scenario where a bag tracks positively and has a minimum screening level of 1. The following steps will occur:

- The BHS will set the required screening level to 1 when the bag enters the EDS machine.
- 11.11.2. • Security Routing Requirement code 11 will be sent to the EDS.
- The EDS will send L1 result to the BHS.
- If the decision is “Reject”, an image will be sent to a L2 operator.
- The EDS will then send a L2 decision to the BHS.
- Bag will travel towards the CBS VSU. A bag with “Clear” screening result from Level 1 or higher will be routed to the clear line, otherwise will be routed to the L3 line for higher level inspection.

Minimum Screening Requirement Level 2

11.11.3. This is the scenario where a bag tracks positively and has a minimum screening level of 2. The following steps will occur:

- The BHS will set the required screening level to 2 when the bag enters the EDS machine.
- Security Routing Requirement code 13 will be sent to the EDS.
- The EDS will send a L1 result to the BHS.
- If the L1 Decision is Clear, the BHS will send the L1 Reject Result to the HLC with a Reason Code of “Minimum Screening Level Not Met”.
- Regardless of the L1 result, the EDS will send an image to a L2 operator.
- The EDS will then send the L2 decision to the BHS.
- Bag will travel towards the CBS VSU. A bag with “Clear” screening result from Level 2 or higher will be routed to the clear line, otherwise, will be routed to the L3 line for higher level inspection.

Minimum Screening Requirement Level 3

This is the scenario where a bag tracks positively and has a minimum screening level of 3. The following steps will occur:

- The BHS will set the required screening level to 3 when the bag enters the EDS machine.
- 11.11.4. • Security Routing Requirement code 13 will be sent to the EDS.
- The EDS will send a L1 result to the BHS. If the L1 Decision is Clear, the BHS will send the L1 Reject Result to the HLC with a Reason Code of “Minimum Screening Level Not Met”.
- Regardless of the L1 result, the EDS will send an image to a L2 operator.
- The EDS will then send the L2 decision to the BHS. If the L2 Decision is Clear, the BHS will send the L2 Reject Result to the HLC with a Reason Code of “Minimum Screening Level Not Met”.
- Bag will travel towards the CBS VSU. Bag will be rejected at the VSU and routed to the Level 3 OSR room for higher level inspection by an operator.
- An operator will search the bag, if the Operator gives the bag a L3 decision of “Clear”, the bag will be routed to the clear line (CL05).

Minimum Screening Requirement Level 4

11.11.5.

This is the scenario where a bag tracks positively and has a minimum screening level of 4. The following steps will occur:

- The BHS will set the required screening level to 4 when the bag enters the EDS machine.
- Security Routing Requirement code 13 will be sent to the EDS.
- The EDS will send a L1 result to the BHS. If the L1 Decision is Clear, the BHS will send the L1 Reject Result to the HLC with a Reason Code of “Minimum Screening Level Not Met”.
- Regardless of the L1 result, the EDS will send an image to a L2 operator.
- The EDS will then send a L2 decision to the BHS. If the L2 Decision is Clear, the BHS will send the L2 Reject Result to the HLC with a Reason Code of “Minimum Screening Level Not Met”.
- Bag will travel towards the CBS VSU. A bag with “Clear” screening result from Level 3 or higher will be routed to the clear line, otherwise, will be routed to the L3 line for higher level inspection.

12. CONVEYOR SWING OVER

12.1. General

Swing overs are in various locations around the BHS to provide personnel safe access to both sides of the conveyor system. Each swing over has 2 control panels on both sides of the conveyor; both with a red illuminated Stop push button and a green illuminated Start push button.

12.2. Control Panel

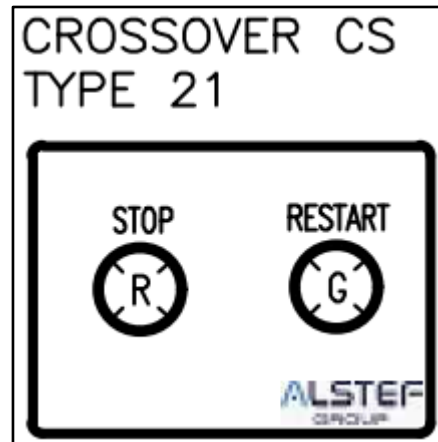


Figure 46 Swing-Over Control Station

A control panel is fitted on each side of the swing over and has the following layout:

Control [Device]	Cause	Effect
Stop [Illuminated red Pushbutton]	Press pushbutton	-> Conveyor underneath stops and is interlocked from running immediately under a Control Stop (not Emergency Stop), including if the conveyor is already running in SCADA override run mode -> Start pushbutton on both sides of the conveyor illuminate, indicating the conveyor line is safe to cross. -> Affected conveyors and activate swing over stations are displayed on SCADA.
Start [Illuminated green Pushbutton]	Press pushbutton	-> Stop pushbutton flashes red, and a 5 second pre-start warning on local beacon/sounder(s) signals a pending start-up. Please refer to Section 16.3 Pre-Start Warning for further information. -> Stop pushbutton on both sides of the conveyor illuminate red, indicating that the swing over is no longer safe to cross.

Table 32 Swing Over Control Station

12.3. Conveyor Swing Over Faults

Fault	Cause	Effect	Fault Resolution
Swing Over Fault	Swing over is not restarted in a pre-defined time period (adjustable)	-> Conveyor underneath faults -> Swing over Fault is indicated on the SCADA alarm banner and in HLC reports. -> Sounder and amber warning beacons near the swing over activate. Please refer to Section 16.5 System Fault for further information. -> Stop pushbuttons flash red. -> Start pushbutton remains lit.	-> Press Illuminated Start pushbutton. -> 5 second pre-start warning occurs. - Conveyor starts back up

Table 33 Swing over Fault Cause & Effect

13. EMERGENCY STOP

13.1. General

The BHS is divided into electrical zones to ensure that pressing an emergency stop in one zone does not unnecessarily affect other areas of the system. In the event of an emergency stop, the system will still be able to transport baggage in areas not affected by the stoppage. When an emergency stop is activated, the 3-phase power is immediately cut off from all drives within that specific zone. The design of the emergency stop circuits complies with local regulations and is determined by a project-specific risk assessment.

The following warning devices and indicators are activated when any emergency stop device in a zone is triggered:

- If the device is an emergency stop button, the built-in indicator illuminates.
- If the device is an emergency stop lanyard switch, a local lanyard-specific indicator illuminates.
- All red warning beacons illuminate in the zone (including at the CCP & FCP). Please refer to [Section 16.4 E-Stop](#) for further information.
- All air-side sounders in the zone activate (including at the CCP & FCP).

Public-side piezo sounders at control panels in the emergency zone are only activated when the emergency stop is triggered by an emergency stop device on the landside. Please refer to [Section 16 Beacon And sounder Indication](#) for further information.

13.2. Emergency Stop Types

Pushbutton E-Stop

The system is equipped with illuminated, mushroom-head Emergency Stop (E-Stop) push buttons.

- **Operation:** The E-Stop is activated by pushing the button and is reset by twisting & pulling it back to the released position.
- **Illumination:** When activated, the E-Stop button remains illuminated to indicate the stop condition.
- **Reset Indication:** Once released, the integrated LED will flash at 2 Hz, indicating that the E-Stop has been released.



Figure 47: Pushbutton E Stop

Lanyard E-Stop

A lanyard E-Stop is a type of emergency stop device widely used in BHS applications. This device features a cable or rope (lanyard) attached to an E-Stop safety switch. The LED indicator on the device provides visual feedback on the operational status of the Lanyard E-Stop.

13.2.2 The lanyard itself can be extended over a significant distance, allowing a single lanyard E-Stop system to cover a large area or multiple sections of a straight conveyor line.

The lanyard E-Stop operates by allowing personnel to initiate an emergency stop from a distance or from an awkward position by pulling the lanyard. When the lanyard is pulled, it activates the E-Stop switch and the LED indicator will illuminate.

To Reset, release the tension on the lanyard by allowing it to return to its original position, then use a “twist-to-reset” motion to reset the E-Stop state.

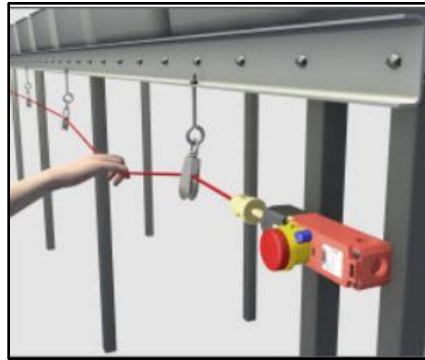


Figure 48: *Lanyard Estop*

EDS E-Stop

The physical E-Stop pushbuttons on the EDS are connected to the BHS via dry contacts and functions in the same way as other E-Stops in the system. The EDS E-Stop can only be reset via a local E-Stop Reset pushbutton provided by Alstef. Please refer to the RTT 110 BHS Protocol Specification – Airport- TSA, [Section 5.1](#) for further details.



Figure 49 EDS E-Stop Locations

13.2.4. ABD E-Stop

Each ABD has an Estop provided by Amadeus, these Estops can only be reset by a special Estop station located on the public side that has a key-switch reset, these estop cannot be reset at the ABD. See [Section 3.2.2 Safety Interface](#) for details.

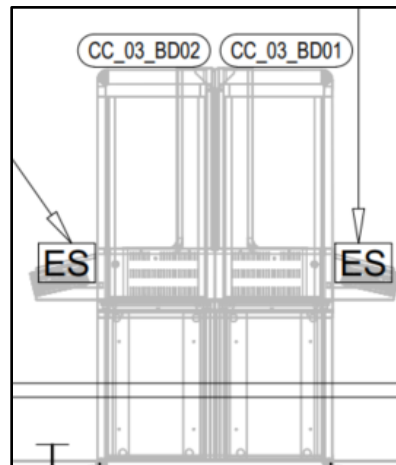


Figure 50 Example Locations of ABD E-Stops

13.3. Activation and Reset

E-Stop activation results in all BHS devices, including EDS machines, coming to a halt, with driving units simultaneously disconnecting from 3-phase power. E-Stop is a latched fault, meaning a separate reset procedure is required. For safety reasons, the emergency stop state can only be reset from the local E-Stop Restart pushbutton associated to the emergency stop device that triggered the condition.

Field Control Station

Condition	Cause	Effect
13.3.1. Emergency stop button is pushed in	Press Restart pushbutton	-> Deactivate any sounders. -> LED on the mushroom pushbutton will remain solid red, indicating that the estop is still active. -> Local emergency stop beacons will remain solid red.
Emergency stop button is pulled out/physically unlatched		-> Power to the motors is restored.
	Press Restart pushbutton	-> Unlatches emergency stop fault. -> Beacons and sounders are deactivated. Please refer to section 16 Beacon And sounder Indication for further information. -> Alarm condition is cleared. -> Pre-Start warning begins. -> System resumes operation.

Table 34 Field Emergency Stop & Restart

CCP/FCP

13.3.2.

Condition	Cause	Effect
Emergency stop button is pushed in	Press E-Stop Reset pushbutton	-> Deactivate any sounders -> Estop Activated lamp on the CCP or FCP will remain solid red, indicating that the estop is still active. -> Local emergency stop beacons will remain solid red.
Emergency stop button key is inserted and turned to release the Emergency stop button		-> Power to the motors is restored.
	Press Estop Restart pushbutton	-> Unlatches emergency stop fault. -> Alarm condition is cleared. -> Beacons and sounders are deactivated. Please refer to 16 Beacon And sounder Indication for further information. -> Pre start warnings begins. -> System resumes operation.

Table 35 CCP/FCP Emergency Stop

13.4. E-Stop Zone Design

All the equipment in the system is divided into emergency stop zones. The arrangement of the zones is based on assessments of the following factors:

- **Consistent Design:** Use a consistent design for all E-Stop devices throughout the BHS to prevent confusion and ensure that workers are familiar with their operation.
- **Functional lines:** Divide the system into smaller, manageable zones based on the specific functions of each section, such as check-in, screening, sortation.
- **Operational Impact:** the need to keep the system operational, such that pressing an emergency stop in one zone does not needlessly inhibit other parts of the system. Considerations include that:
 - bags can continue to be safely processed downstream of the zone.
 - bags can be safely queued upstream of the zone.
 - Minimising the impact to other subsystems.
- For further details on the E-Stop zones layout, please refer to the following documents:
 - Sub-System CCP1 E-Stop Zone Layout
 - Sub-System CCP2 E-Stop Zone Layout
 - Sub-System CCP4AT E-Stop Zone Layout
 - Sub-System CCP5 E-Stop Zone Layout
 - Sub-System CCP6 E-Stop Zone Layout
 - Sub-System CCP7 E-Stop Zone Layout

Interface with Existing E-stops

The remainder of the existing system will be recontrolled at the completion stage of the project. As a result, it will be integrated with the same safety system as the new system. During the various phases, the new PLCs will utilise AS-i safety output modules to interface with either AS-i or hardware-based safety circuits of the existing system.

13.5. E-Stop Failsafe

All emergency stop devices in the system are dual-channel, a safety mechanism that employs two separate and independent channels to monitor the E-Stop circuit. For the system to continue running, both channels must have matching states. If either channel detects a fault or discrepancy, such as a short circuit or an open circuit, the system will still initiate the emergency stop process but cannot be put back into operation until the discrepancy has been resolved. This redundancy in the E-Stop system serves as a preventive measure against potential accidents caused by component failures, wiring issues, or other faults within the E-Stop circuit, thereby ensuring a high level of safety for the operation of the system.

14. CBS LEVEL 3 STATION/LEVEL 4 RECONCILIATION

14.1. General

Purpose and Overview

At the Level 3 station, operators are presented with images of bags that were not cleared during the Level 2 screening process. There is **no time constraint** imposed on the review process at this level. In cases where the screening result for a bag is mistracked but the bag IATA tag has been scanned by an upstream ATR; the information can still be retrieved within the CBS System. However, for mistracked bags that have not been scanned by an ATR and subsequently reach Level 3, an Explosive Trace Detection (ETD) screening is mandated. Once the bag has been cleared at Level 3, it is manually transported (using the extendable workstation table) to the designated re-insertion line for cleared items. Conversely, any bag that fails the Level 3 screening is manually returned to the failed line and forwarded to the Level 4 area for further evaluation.

System Context

The alarmed/failsafe destinations from the four EDS machines converge into a single alarmed line through the four VSUs, one on each CBS line. Bags that fail screening at level 2 (not to be confused with CBS rejected bags) are sent to the re-scan line past AL_01_16P plough. A Level 3 inspection room is located downstream of the plough and consists of ten, metering/queuing conveyors. Adjacent to this line of conveyors, are four Level 3 SMI stations. Near to the Level 3 stations, there is a loading point for the cleared reinsert line (CL0501). Beyond the Level 3 inspection point, the metering/queuing conveyor leads to the Level 4 area, behind a roller door for security, for further examination and processing of the bags.

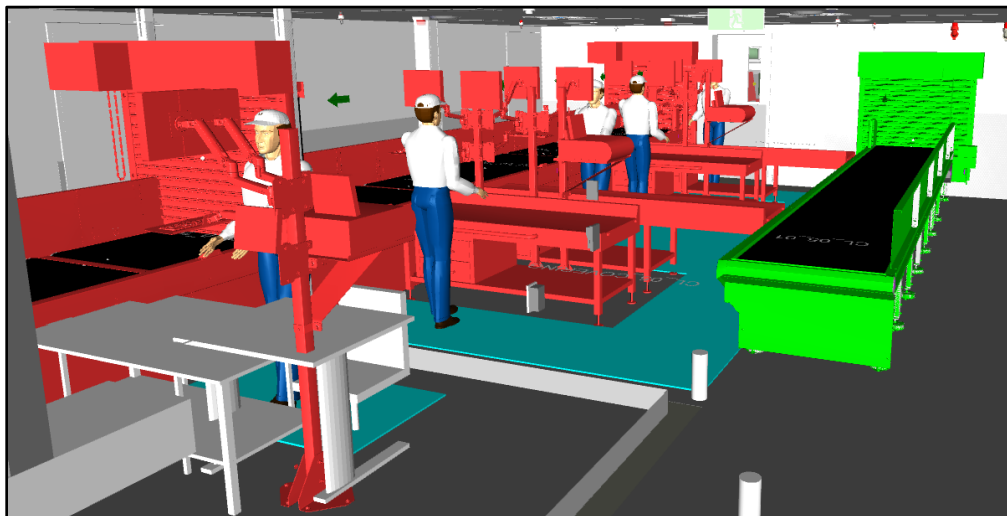


Figure 51 DTB OSR Room

14.2. Level 3 Station Hardware Components

- Security Manual Inspection (SMI) Level 3 workstation – BHS.
- Secondary Viewing Station (SVS) – Security.
- Keyboards and mice – Security.
- Barcode Scanners – BHS.
- Extendable inspection platform.
- Estop/Restart control stations.
- Handheld barcode scanner.
- Encoding point on open conveyor.

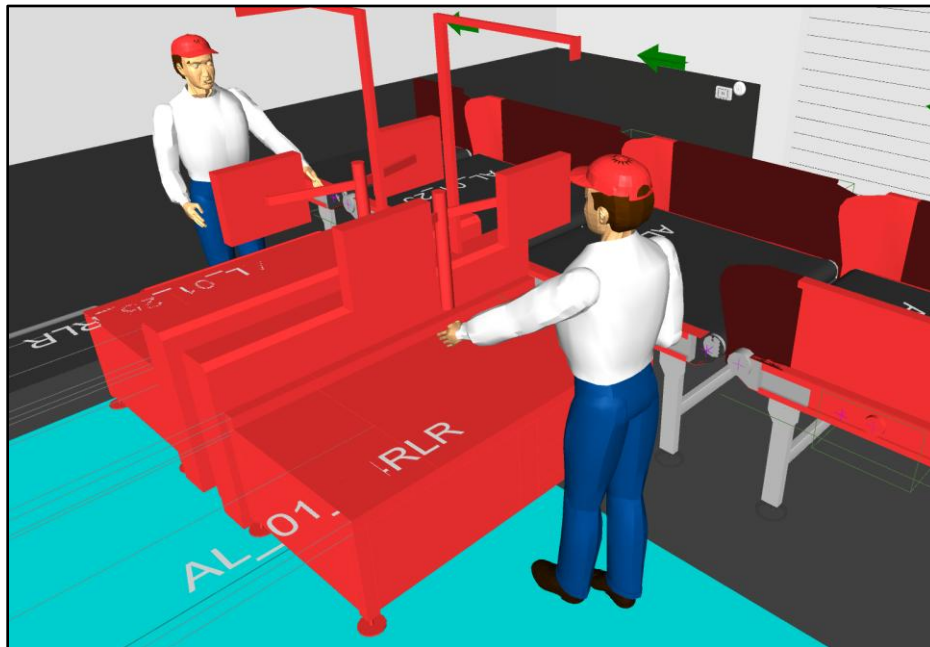


Figure 52: Level 3 Station Stanchion Design (image is only for reference)

14.3. Level 4 Hardware Components

- Security Manual Inspection (SMI) Level 4 workstation – BHS.
- Secondary Viewing Station (SVS) – Security.
- Keyboards and mice – Security.
- Barcode Scanners – BHS.
- Estop/Restart control stations.

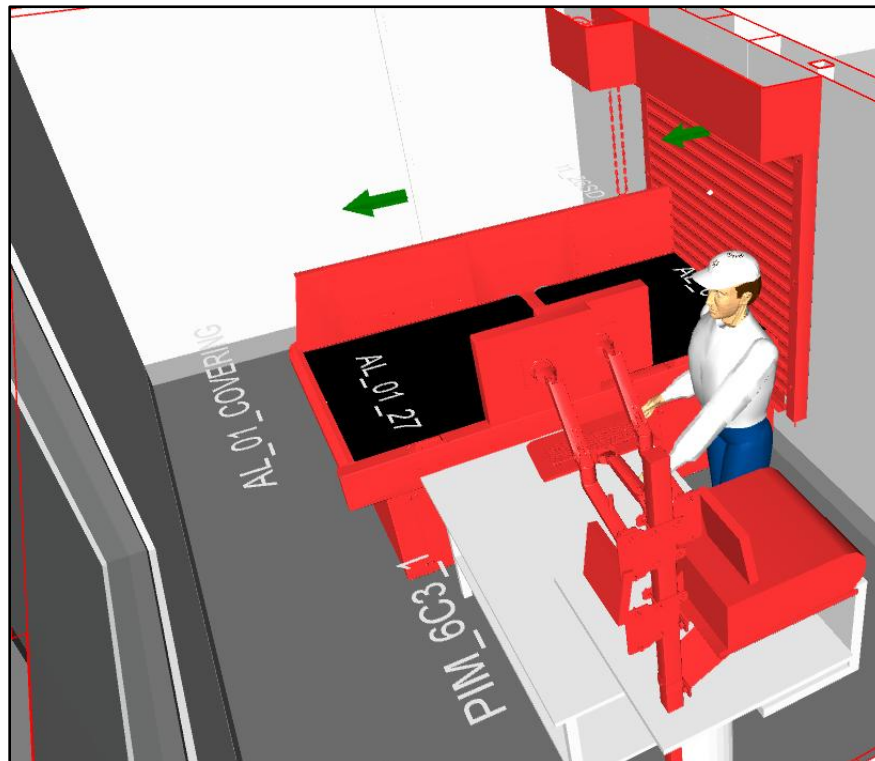


Figure 53: Level 4 Station Stanchion Design (image is only for reference)

14.4.1.

14.4. Operational Procedures

L3 Procedure Overview

When bags arrive at the Level 3 inspection station, they are queued on metering conveyors, waiting for operators to reconcile them, with the assistance of a barcode reader. The operators follow a specific procedure as shown in the following flow chart (Figure 54). For Further detail please refer DTB HLC Phase 2 Functional Design Specification to section 4.4.4 figure 118.

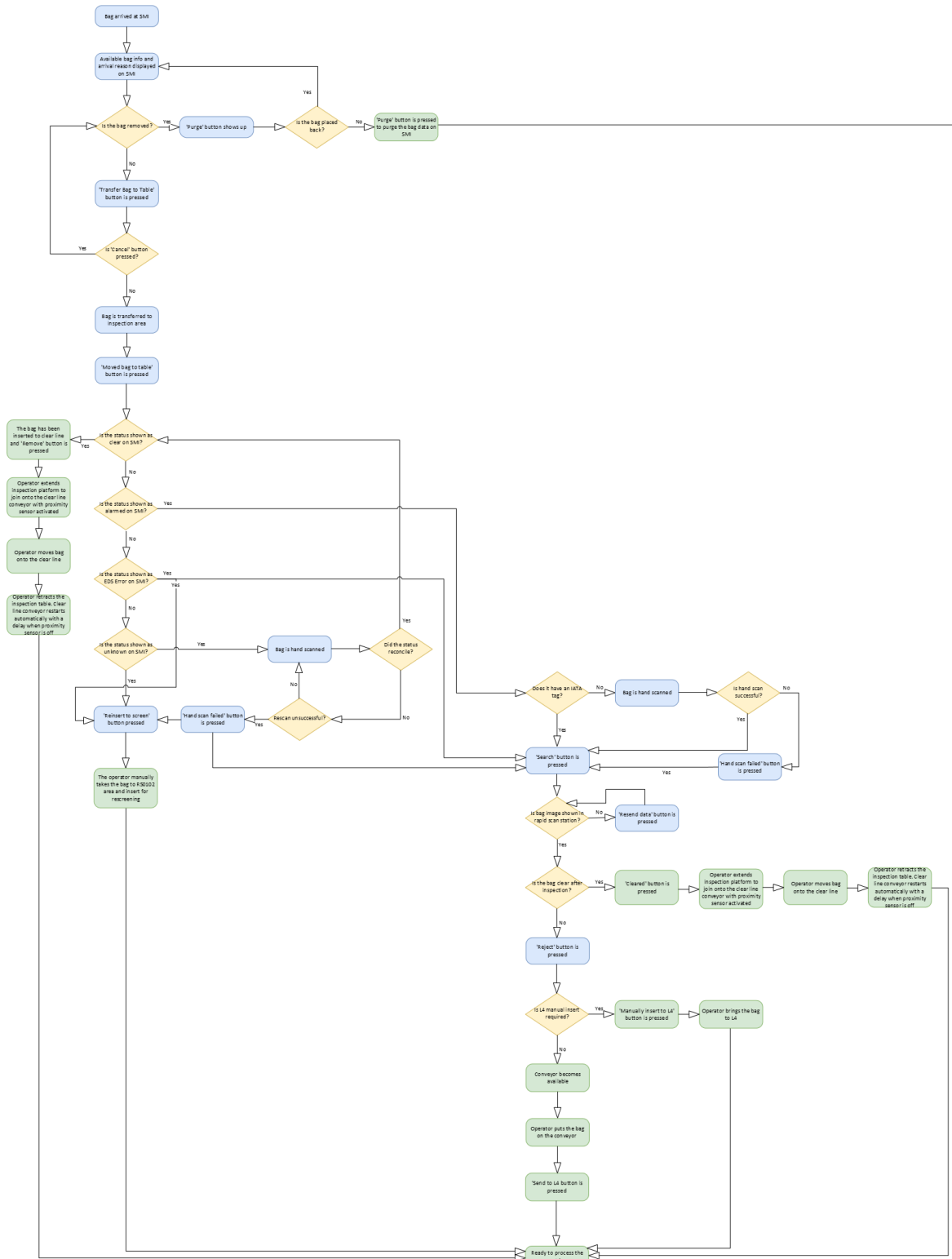


Figure 54 Level 3 SMI Flow Chart (Refer to HLC FDS)

L4 Procedure Overview

When bags arrive at the Level 4 inspection station, they are queued on metering conveyors, waiting for operators to reconcile them, with the assistance of a barcode reader. The operators follow a specific procedure as shown in the following flow chart (Figure 55). For Further detail please refer DTB HLC Phase 14.4.2.

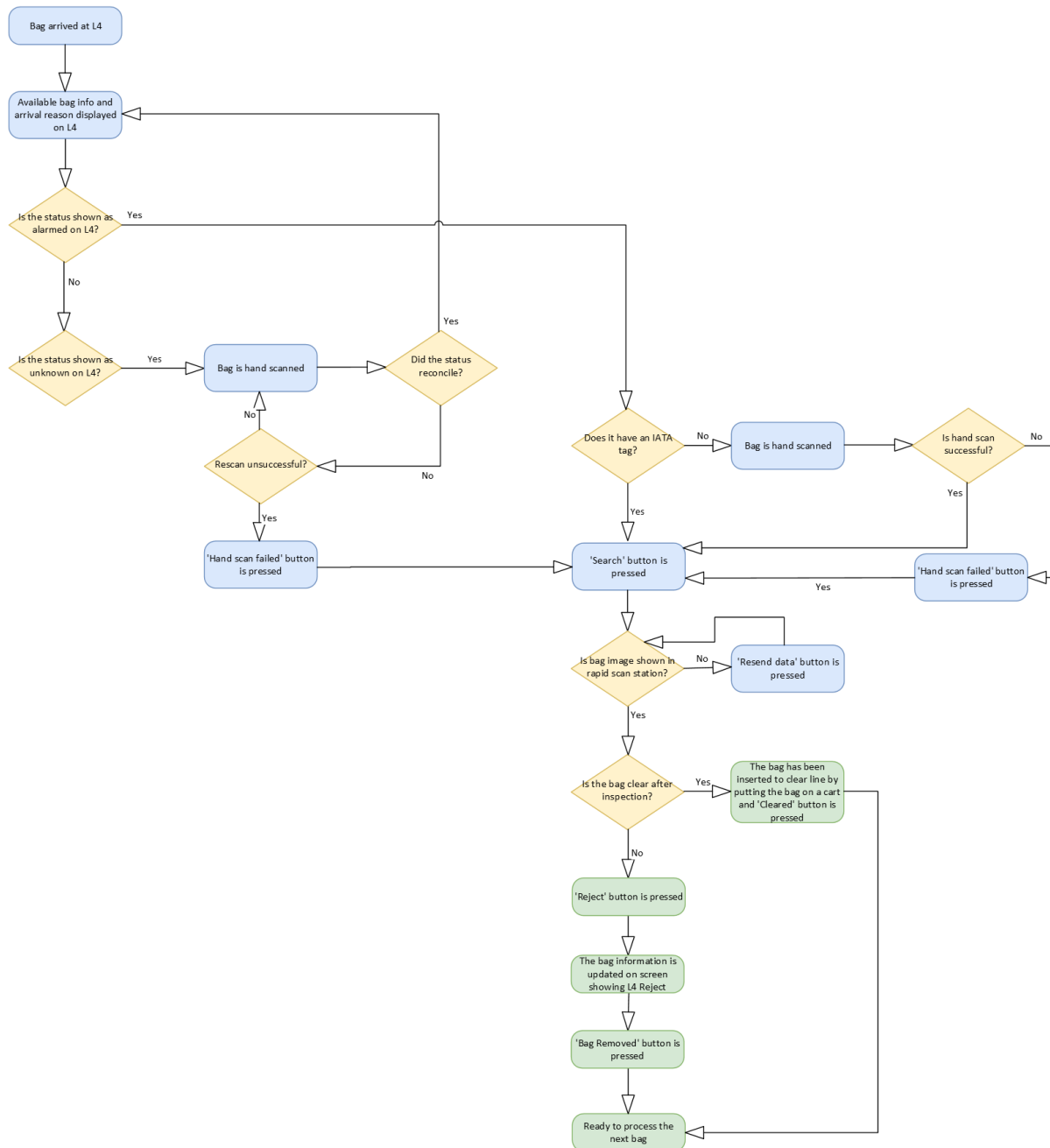


Figure 55 Level 4 SMI Flow Chart (Refer to HLC FDS)

Level 3 Bag Allocation

The system encompasses four Level 3 Stations, each of which can be independently enabled or disabled. When a bag arrives at the decision point (AL01_20), availability of all four workstations is verified, then round robin method is used to allocate bags starting from the most downstream workstation. This is to distribute the workload evenly between the workstations.

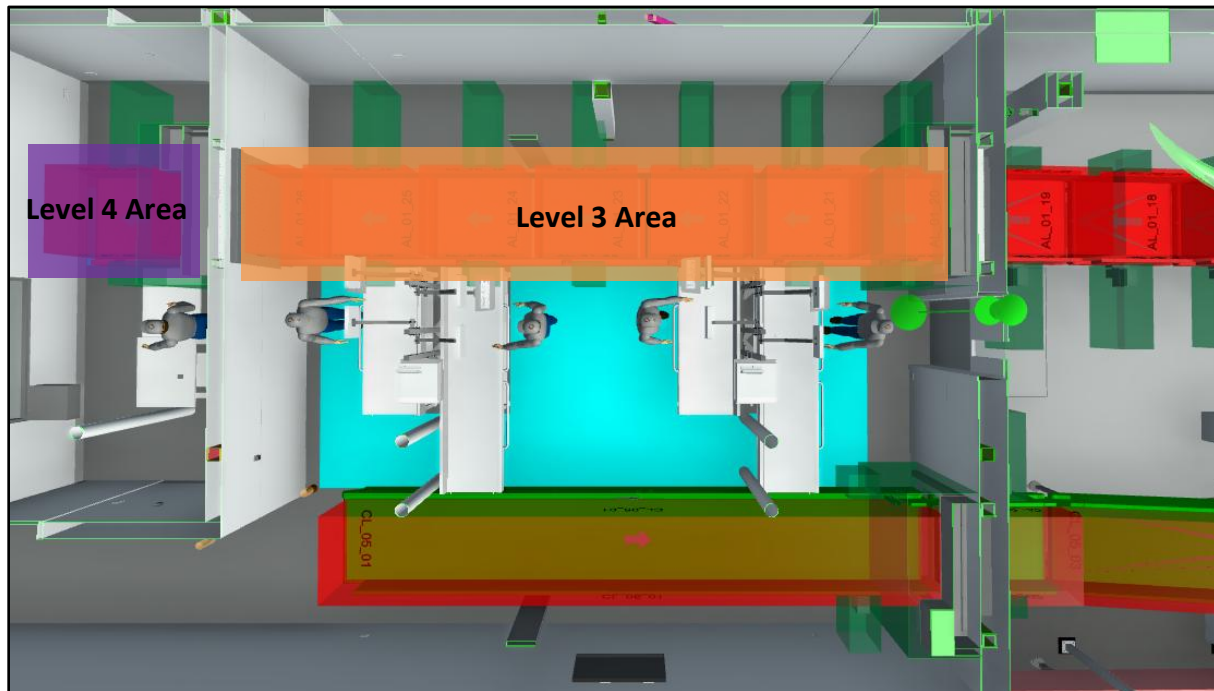


Figure 56: Layout of the Level 3 and Level 4 Area

Round Robin Logic

Detailed Round-Robin Algorithm is shown below:

14.4.4.

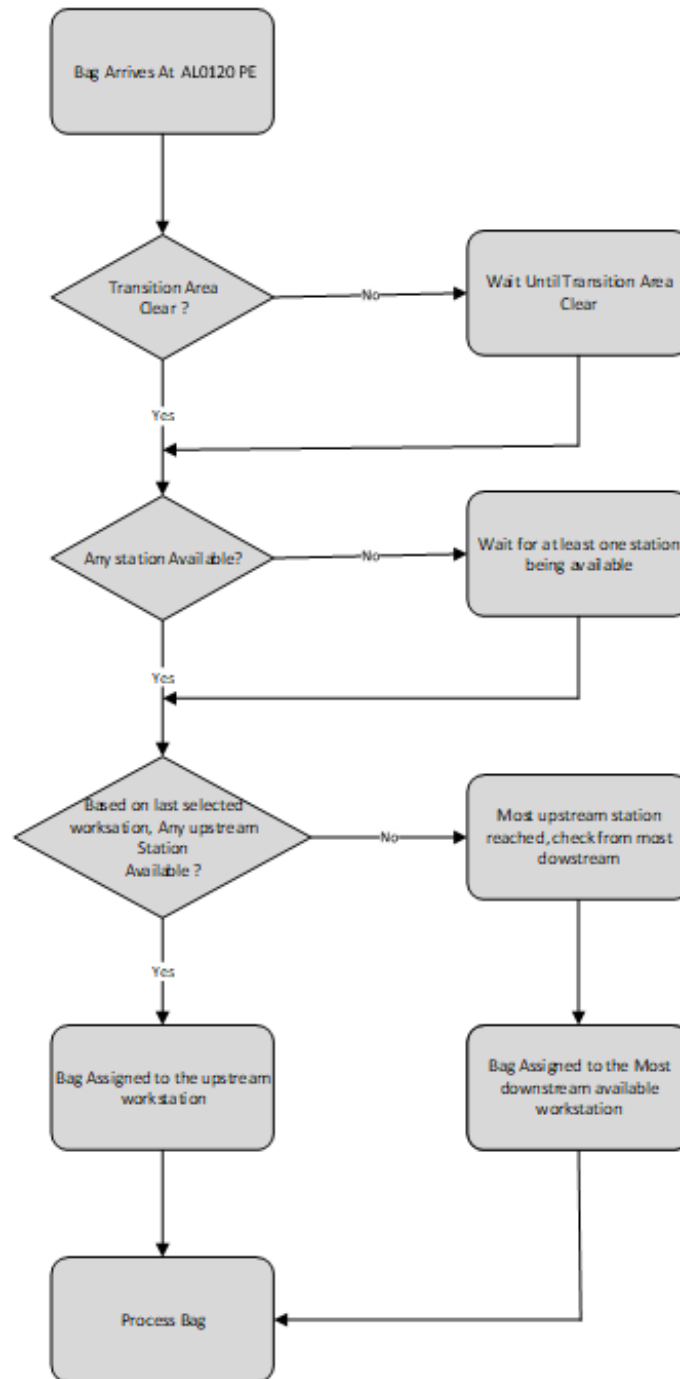


Figure 57 Round-Robin Algorithm

Workstation Re-Allocation

Once bag has assigned to a L3 workstation, it starts moving towards the station. During this period, bag can be re-allocated to next available workstation in the following exceptional scenarios.

14.4.5. Once assigned, the bag begins routing towards the selected workstation. During this process, whether the bag is in transit or has already stopped at the assigned workstation, the PLC continuously monitors the status of that station. If the assigned workstation becomes unavailable (e.g., it is disabled, faulted, or an operator fails to initiate inspection within a defined time), the PLC will automatically reallocate the bag to the next available downstream workstation in the round-robin sequence. This ensures continued system efficiency and avoids unnecessary delays in inspection.

If the bag has already stopped at the most downstream workstation (AL0125) and that station becomes unavailable, the PLC cannot perform automatic reallocation. In this case, the bag must be manually removed by the operator to resolve the issue.

Level 3 Bag Removal

14.4.6. When a Level 3 station is enabled and not currently inspecting a bag, the SMI station will be in a “waiting for bag” state. Upon arrival, a bag will stop at the cascade PEC of the conveyor belt adjacent to the inspection station, with SMI showing the reason for the bag's arrival. If the CBS status can be retrieved via the tracking information, the EDS OSR will automatically display the screening image. The operator first must press the “Transfer Bag to Table” button on the SMI touchscreen to unlock conveyor flow, then manually move the bag from the conveyor to the inspection table. This inspection station cannot receive additional bags while a Level 3 inspection is ongoing. Refer to Section 4.4.4 on the SMI in the DTB HLC Phase 2 Functional Design Specification for further information.

14.4.7.

Level 3 Bag Image Retrieval

Every bag within the BHS can be associated with two forms of CBS identification: an IATA barcode and a Pseudo ID. Each type of identification can be utilised to ascertain the CBS status of the bag. The IATA barcode is preferred if it has been read and tracked before the bag enters the CBS machine. On the other hand, looking up the Pseudo ID necessitates consistent tracking throughout the bag's journey, and can be permanently lost in case of mis-tracking. For more comprehensive information on bag identification methods used for the CBS process, refer to [Section 11.5 Bag ID and Decision Communication](#) for further information.

When a bag arrives on its allocated Level 3 workstation, the PLC will pass the information to the SMI for screening result lookup. Please refer to the DTB HLC Phase 2 Functional Design Specification for a detailed data/image retrieval matrix at a Level 3 workstation.

Level 4 Manual Reconciliation

When a bag fails Level 3 inspection, it must be moved to the Level 4 area. Once a reject decision is made, the operator presses the "Send to L4" button on the touchscreen. The SMI shows "Ready" when the adjacent alarm conveyor halts. The bag is placed onto the conveyor, blocking the cascade PEC, by the operator. After pressing "Send to L4", the system restarts when the path to Level 4 is clear. The Level 3 station is ready for the next bag. The operator can also manually move the bag to the Level 4 SMI and press the "Manually Insert to L4" button on the touchscreen after moving the bag. Refer to Section 4.4.4 of the DTB HLC Phase 2 Functional Design Specification for further information.

SmartSuite Forms Client		AL0518RLR	Log Out	DISABLE
Bag Status: ALARMED		Pseudo ID: 101001		
IATA: 0123456789		Security Level: L2 Reject		
EDS ID: (CBS0305XR)				
SEND TO L4		MANUALLY INSERT TO L4		
✓ PLC Connection	✓ Server Connection	SmartSuite 13.0 © 2011-2025 Book Solutions Inc.	BROCK SOLUTIONS	2025-02-12 14:49:58

Figure 58 Level 4 Insertion SMI screen

The Level 4 station can process one bag at a time. A SMI and hand scanner are equipped at the inspection point for the operator to retrieve and review the image of the bag. At Level 4, after all necessary procedures, any cleared bag can be manually removed to the cleared re-insert line (CL0501). An operator can use the SMI to encode a cleared decision and enable the level 4 station to process the next bag. The operator must do the following on the SMI in order:

- The green “remove” button needs to be pressed by an operator.
- The Operator manually moves the bag to the clear line, CL0501.
- A “Cleared” green button then appears at the SMI.
- The “Cleared” button must be pressed by an operator to allow the level 4 station to process the next bag.

The screenshot displays the SmartSuite Forms Client interface. At the top, the title bar reads "SmartSuite Forms Client" and the device name is "AL0518RLR". Below the title bar, there is a header section with the SmartSuite logo, the device name "AL0518RLR", and two buttons: "Log Out" and "DISABLE". The main content area is divided into two columns. The left column contains the following text: "Bag Status: CLEARED" (highlighted in green), "IATA: 0123456789", and "EDS ID: (CBS0305XR)". The right column contains the following text: "Pseudo ID: 101001" and "Security Level: L2 Clear". Below this information, there is a large green button with the text "REMOVE". At the bottom of the interface, there is a status bar with the following text: "PLC Connection", "Server Connection", "SmartSuite 13.0 © 2011-2025 Brock Solutions Inc.", the Brock Solutions logo, and the timestamp "2025-02-12 14:51:40".

Figure 59 Clear Bag example

If the bag is rejected at level 4, BHS doesn't participate in any further necessary escalation processes beyond this point. Refer to Section 4.4.5 of the HLC FDS for further information.

Automatic Routing to Rescreening Line

Bags will be automatically diverted to the re-screening line if they exhibit any of the following conditions:

14.4.9.

- EDS Machine Error
- EDS Machine Results Unknown
- EDS Bag Spacing Error
- EDS Result No Scan

These bags with any of the above EDS statuses are initially diverted to the reject line to prevent them from continuing through the standard baggage flow. However, before they reach any Level 3 inspection station, the system uses the AL0116 Plough diverter to intercept and reroute these bags onto the rescanning line. If the number of rescanning attempts on the same bag exceeded three attempts, the bag will be sent directly to Level 3 inspection area. When the bag arrived at the Level 3 inspection station, the SMI will identify the bag as "Excess Re-Screen" and the operator process the bag accordingly.

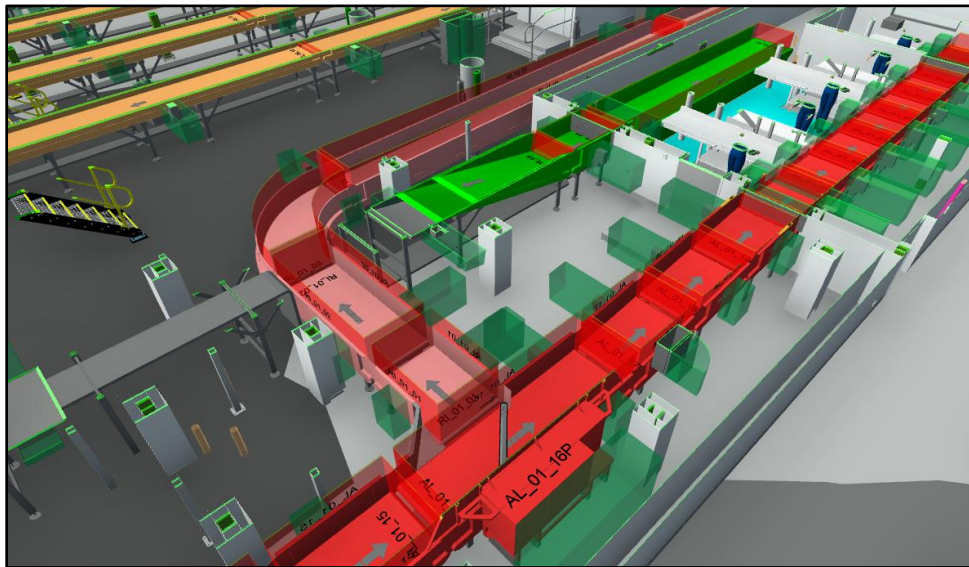


Figure 60: AL0116 Plough for Re-Screening

In circumstances involving an emergency stop or cascade stop, the belt of the Explosive Detection System (EDS) machine may come to a halt, interrupting the screening process. As a result, some bags may fail to complete screening and are automatically assigned a "No Scan" status.

Manual Insert for Rescreening Bag from L3

Bags that are mistracked—meaning they have not been properly read or associated with a tracking ID—cannot be reconciled or processed at Level 3 (L3) inspection stations. Similarly, if the screening image is unobtainable after scanning the IATA tags, the bag is considered invalid for inspection and requires re-screening.

In such cases, the following procedure is followed:

- The operator presses the "Rescreening" button on the SMI. This action clears the bag from the L3 station and re-enables the station to receive the next bag.
- The operator then manually transports the bag to the designated re-scan insert point (RI0105).
- A dedicated reinsertion control panel is installed at this location, equipped with illuminated Stop and Start pushbuttons.
- Using the control panel, the operator presses the Stop button to halt the upstream conveyor and the reinsertion conveyor, ensuring a safe environment for manual placement.
- Once the conveyors have stopped, the operator opens the hinged side guard and manually places the bag onto the re-screening line.
- To resume operation, the operator presses the Start button on the panel. This initiates a 5-second pre-start warning (with audible/visual indicators), after which the conveyors restart.

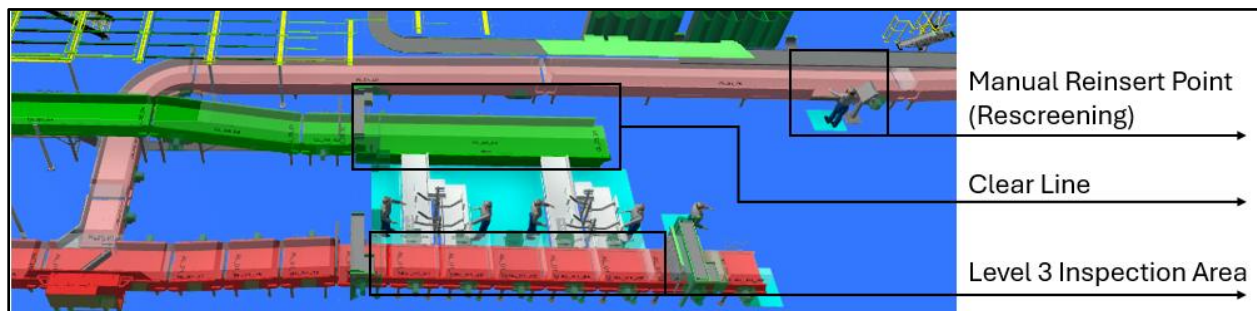


Figure 61 Level 3 Manual Inspection Area



Figure 62 Example of ME/MIS hybrid Station Screen

14.4.10.1. Control Panel

To support manual bag reinsertion operations, a dedicated control panel will be installed near the reinsertion conveyor. This panel enables the operator to safely manage the reinsertion process by providing local control to:

- Stop the upstream conveyor and reinsertion conveyor before placing the bag, and
- Restart the conveyors once the item has been safely reinserted.

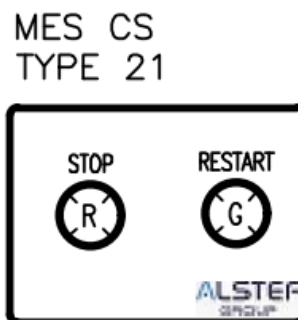


Figure 63 Manual Insertion Control Station

14.4.10.2. Control Panel Functions – Reinsertion Zone

Control	Cause	Effect
Stop [Red Illuminated Pushbutton]	Press pushbutton	-> Immediately stops the reinsertion conveyor and upstream belt via a Control Stop (not Emergency Stop). -> The restart pushbutton illuminates green indicating it is safe to reinsert a bag. -> Status is displayed on SCADA.
Restart [Green Illuminated Pushbutton]	Press pushbutton	-> Initiates a 5-second pre-start warning via local beacon/sounder. -> Reactivates the upstream and reinsertion conveyors. -> Stop pushbutton illuminates red Indicating system is no longer ready for reinsertion.

Table 36 Manual Insertion Control Station

L3 Cleared Bag Reinsert

14.4.11.

To reinsert cleared bags from Level 3 into the system, the operator needs to extend the inspection platform, establishing a connection with the loading conveyor (CL0501) via a proximity sensor. The bag is then manually transferred onto the loading conveyor, followed by the operator clearing the bag on the SMI station, and the operator can retract the platform. Consequently, the Level 3 station is prepared to receive the next bag, and the load conveyor starts after the 5-second pre-start warning. The clear line will then be available for automatic transfers.

However, even if the table remains fully extended after a bag has been cleared, the corresponding SMI station can still receive a new bag for manual processing, ensuring operations can continue.

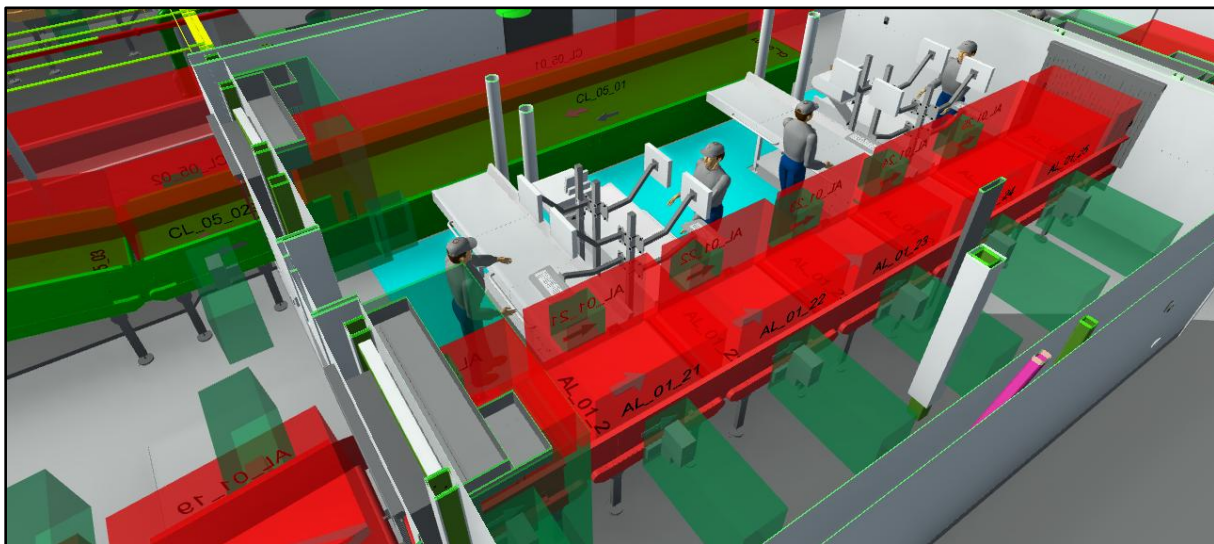


Figure 64: L3 Clear Bag Re-Induct Line -AL_01

14.4.11.1. Clear Line Energy Save Conservation Mode

To reduce energy consumption during periods of inactivity, the clear bag loading conveyor is equipped with an energy conservation mode. If no bags are detected for a predefined duration, the conveyor automatically enters standby mode.

This mode is interrupted, and the conveyor reactivates when any of the following events occur:

- A “Clear” decision is made on any Level 3 workstation.
- The Emergency Stop Restart pushbutton is pressed on any of the following control stations:
 - AL0121CS1
 - AL0123CS1
 - AL0126CS1
 - CL0501CS1
 - CL0501CS3

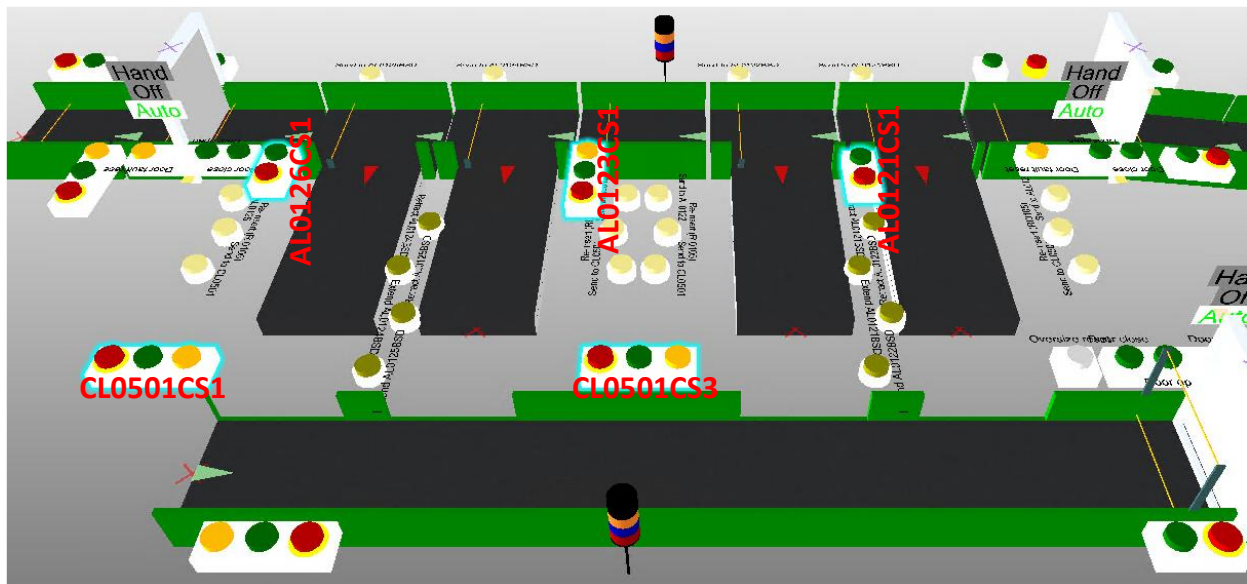


Figure 65 Control Stations to start the CCP6 Subsystem

14.4.11.2. L3 SMI Station Alarms

The L3 SMI station will have two alarms associated with this process:

- Bag waiting Alarm – A bag at the SMI station is waiting longer than 15 seconds; an alarm is triggered in the SCADA alarm banner.
- Table extended Alarm – A SMI station inspection table has been left extended for more than 15 seconds; an alarm is triggered in the SCADA alarm banner.

Operator Visibility:

- The alarm is displayed prominently on the SCADA alarm banner.
- An audible buzzer activation occurs.
- The SCADA graphic for the table changes visual (e.g., flashing) to indicate the extended state under alarm.

Associated Interlocks & Safety Implications:

- Clear Line conveyor (CL0501) start is inhibited while the alarm is active.
- All alarm events are logged with timestamp, operator ID, and resolution actions.

14.4.11.3. Clear Line Conveyor Interlock

The Clear Line conveyor, specifically CL0501, must not start when any of the L3 tables are being extended, as indicated by the proximity sensor being OFF. If the sensor shows the table is retracted (ON), the conveyor may start provided all other safety conditions are met, including the relevant E-Stop Zone is healthy and all guards are closed.

This also applies if any SMI table proximity sensor is damaged.

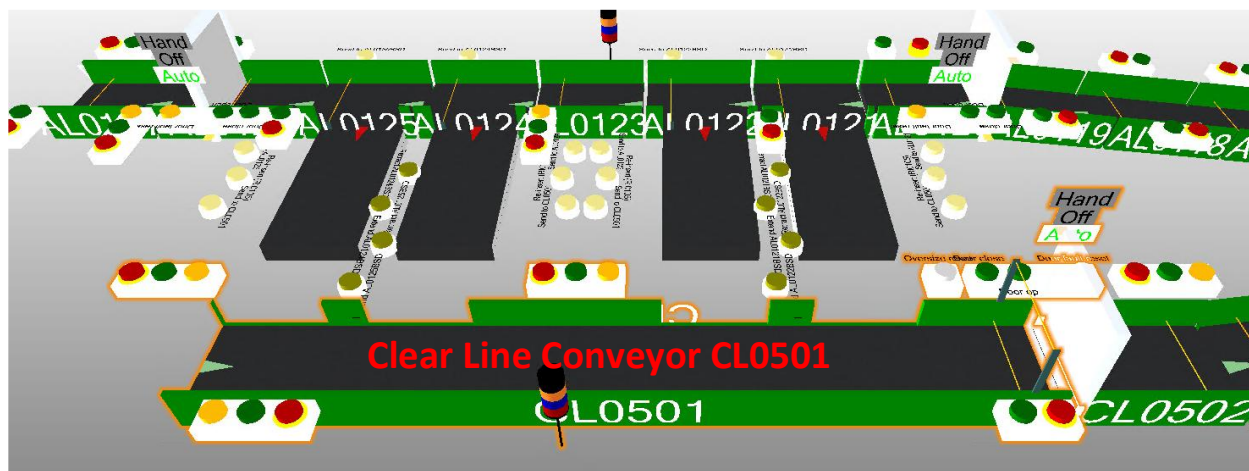


Figure 66 Clear Line CL0501

14.4.11.4. Proximity Sensor Override

Proximity sensor overrides are permitted only under maintenance control, requiring SCADA authorisation, logging of user ID and reason. During override, the SCADA displays a clear status message (on the proximity sensor pop-up and alarm banner).

L4 Cleared Bag Reinsert

To reinsert cleared bags from Level 4 into the system, the operator needs to manually transport the bag from the L4 inspection station to CL0501 Clear Bag Induction Line. If CL0501 conveyor is in Energy Conservation Mode, an operator can use a designated E Stop Restart PB to reactivate the conveyor. See [Section 14.4.11.1 Clear Line Energy Save Conservation Mode](#).

To do this, the operator follows these steps:

1. Manually transport the cleared bag from L4 to CL05 Clear Line.
2. Place the bag onto the CL0501 conveyor for re-inducting into the system.
3. If in Energy Save Mode, press one of the local E-Stop restart pushbuttons.

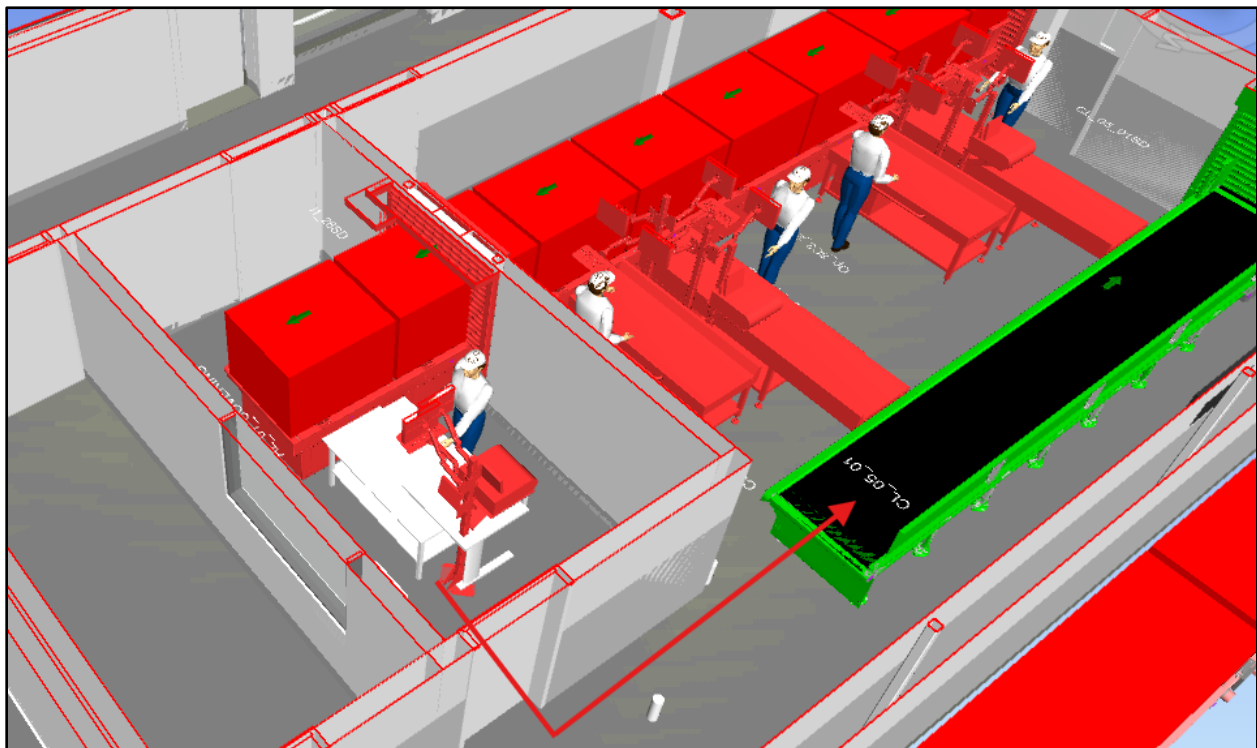


Figure 67 Level 4 Manual Re-insertion Station

Bag Tracking

The BHS is designed to fully track bags throughout the Level 3 and Level 4. Tracking for rescan and cleared bag re-insert route are for statistics purpose only. This tracking system ensures that each bag's security and routing information is consistently maintained, even as they move through inspection and potential Level 4 processes.

Error Resolution and Exception Handling

For bags arriving at Level 3 stations, the operator may scan the bag to retrieve the associated screening image. It is the operator's responsibility to ensure that the screening image corresponds to the presented bag. If the match is uncertain, Explosive Trace Detection (ETD) can be conducted, or, if required, the bag should be escalated to Level 4 for further inspection.

For bags to be re-inserted, to ensure a smooth process, please adhere to the following recommendations:

Replace/Fix tag if damaged or folded to improve ATR readability.

Place a bag with wheel facing up.

Use a tub if necessary.

Unexpected Item at Transition Area

If unexpected item is detected or bag mistracks in Transition Area, the item will be routed to L4 as failsafe procedure.

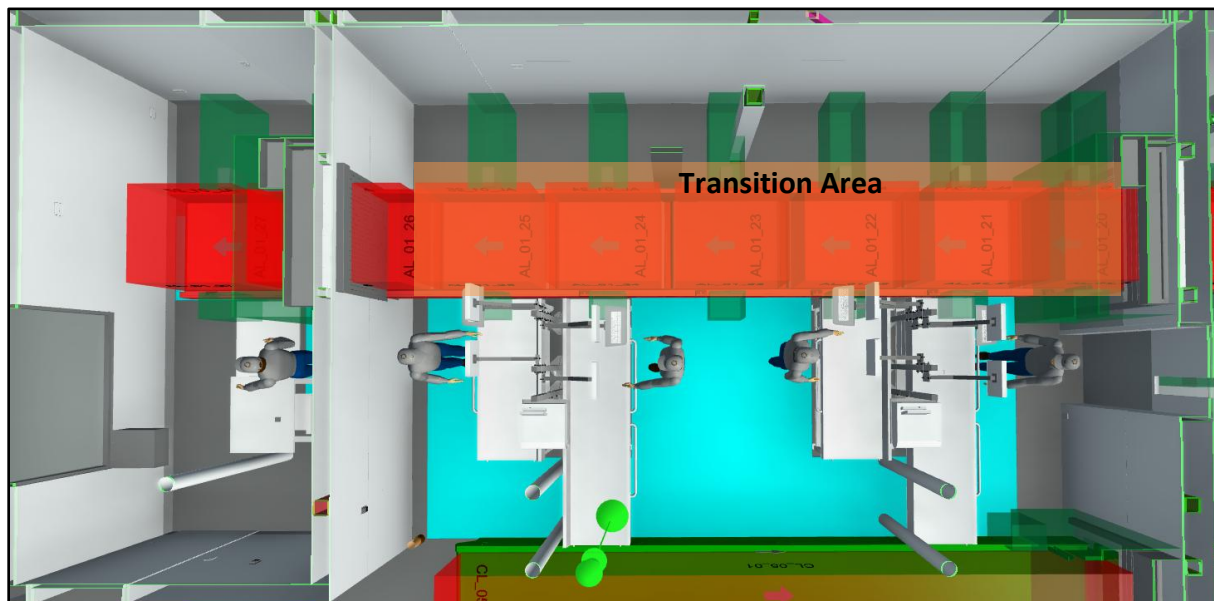


Figure 68: Layout of Level 3 Transition Area

15. MANUAL ENCODING STATION

15.1. General

Purpose and Overview

The Manual Encode Station (MES) is specifically designed to handle bags that are not accurately identified by the ATR or encounter issues that prevent the system from determining their correct sortation destination.

Bags are diverted to the MES per the table below (refer to Appendix L of the HLC FDS).

Reason	Description
Scanner/ATR Timeout	Communication error between the ATR and PLC
Scanner/ATR No Read	No tags read at ATR
Invalid Barcode	A barcode type not accepted at BNE e.g. a fallback tag for another airport
Failed to Get Pier Mapping	Flight was in active flight schedule, but no pier was assigned
No BSM Data Downloaded	No BSM data was found for the scanned IATA tag
Flight Not in Schedule	No flight was found in the active flight schedule
Tag Conflict	Two or more tags of the same priority were read at the ATR
SCADA Divert All Override	An operator has placed the diverter to the Manual Encode into divert all
Lost Tracking	The bag lost tracking between the ATR, upstream of the MES and the MES divert decision point.
BHS cannot confirm a Clear Result	The security result of the bag is unknown or unclear.
Deleted BSM	If configured, a bag with a deleted BSM can be routed to the MES
Unknown Destination	Lateral/Carousel destination cannot be resolved.

Table 37 MES Arrival Reasons

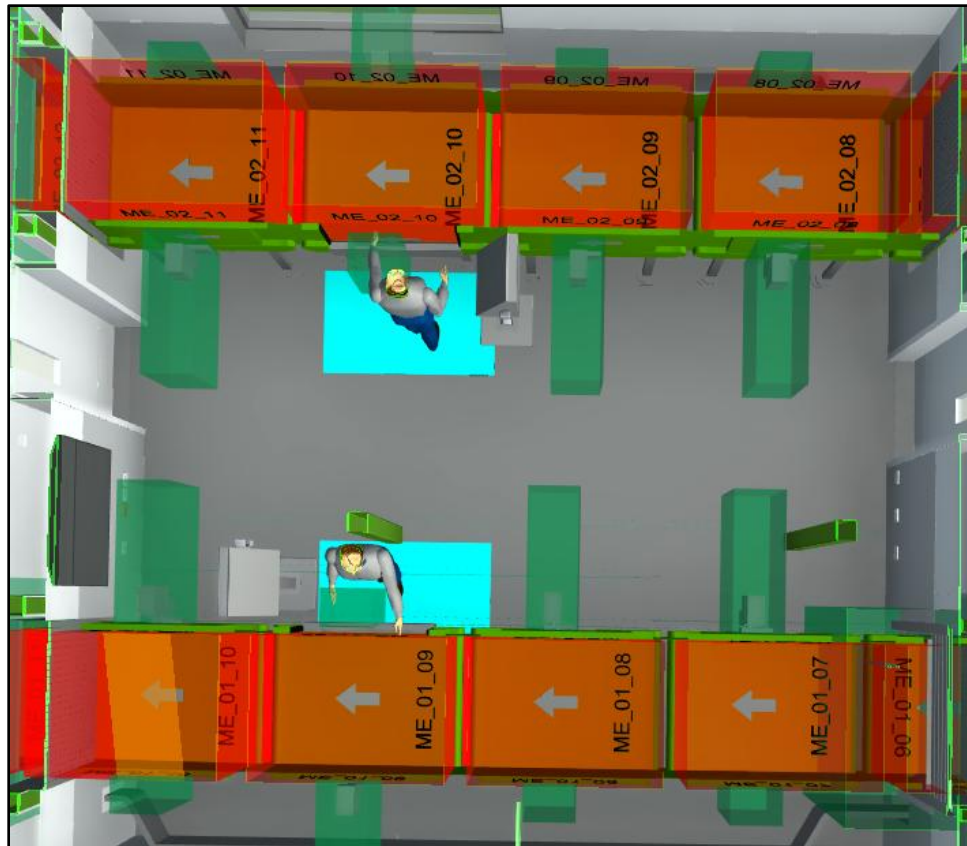


Figure 69: ME01 and ME02

15.1.2.

System Context

The Manual Encoding Station (MES) in the new baggage handling system (BHS) is strategically split between the 4 sortation lines, ME01 is designated for CBS01 and CBS02 screening lines whereas ME02 is designated for CBS03 and CBS04 screening lines. Both ME01 and ME02 are within the Manual Encoding room in the CBS, both stations will be utilised and functional in the 4-lane strategy.

15.2. Manual Encoding Station Components

- MES workstation with touchscreen.
- Handheld barcode scanner.
- Encoding point on open conveyor.
- Estop/Restart control stations.

15.3. Operational Procedures

Encoding Overview

A typical MES layout consists of several metering conveyors leading to the encoding point, parallel to the system's primary transport conveyor line.

15.3.1. Bags that are unable to be correctly identified by ATR are diverted to the MES by a plough diverter. This diversion occurs when a bag's tag is unreadable, damaged, or that causes it to not have a valid sortation destination. All ME arrival reasons are stated in **Table 37** MES Arrival Reasons

Once at the MES, bags are metered onto individualised queuing conveyors. These conveyors regulate the flow of bags; separates joined bags and ensures that the MES has sufficient space to store bags while the operator manually inspects and encodes the details of each bag into the system.

After the information for a bag is successfully encoded at the MES, it is reintroduced into the main flow of the BHS through a 45-degree merge layout. Refer to HLC FDS for MES work flowchart.

Encoding and Validation

15.3.2. At the MES, a bag is stopped at the cascade PEC of the conveyor at the encoding point for easy operator access. Any known information about the bag is forwarded from the PLC to the HLC and then displayed on the MES workstation screen.

The operator then performs an inspection of the bag to find the tag and resolve any issues affecting tag readability. Some bags might need to be placed in a tub to improve tracking accuracy. Once the tag is found, it is scanned by the operator to retrieve the sorting information. If required, barcode numbers can be manually input using an on-screen keyboard.

After successful encoding, the bag is reintroduced into the BHS for routing to its designated destination. Depending on the MES operational mode, the bag's sorting destination can either be assigned automatically or manually overridden. For detailed MES process flow and operation scenarios under different conditions, refer to Section 2.6 of the DTB HLC FDS.

Error Resolution and Exception Handling

At the MES, operators can move a bag during inspection as necessary. However, to release it back into the system, it must be positioned to block the conveyor's cascade PEC at the encoding point. Bag release options aren't available if the bag isn't present at the PEC.

15.3.3 If a bag cannot be resolved, operators can remove it from the system. They can then press the "Bag Removed" button on the touchscreen, so MES can process the next bag.

If a bag arrives at MES with a "no screening" status, the operator is required to remove it from the system. They press the "Bag Removed" button on the station, after that it is operator security procedure for further investigation and/or escalation.

For no-read bags that have cleared the CBS process using a Pseudo ID and arrived at MES with intact tracking, the operator can scan the IATA code and attach it to the CBS status.

HLC regularly updates the flight schedule and sends sorted data to the MES client, which enables MES to operate temporarily if the connection with HLC is lost. With MES directly connected to the PLC, any bag tag found in its local sort plan can be used to determine the destination of a bag. Please refer to HLC FDS section 2.2.7 for details of MES running in HLC fallback mode.

The HLC server periodically synchronises the barcode database with the MES. Therefore, if the HLC is inaccessible, the MES will still be able to reconcile bags by referring to its local archive. In such cases, a warning banner will be displayed on the MES interface to indicate that the system is operating in offline mode.

No-read bags that clear CBS using a Pseudo ID but are mistracked after a CBS VSU, will arrive at the MES without the ability to reconcile their CBS status with the IATA code. To resolve this, the operator must correct the underlying barcode issue then have the bag sent back to the system with "Send to Rescreen" option on the touchscreen. The bag should be physically retrieved from ME_01 or ME_02 and transferred to ME Chute 02 (downstairs) for rescreening.

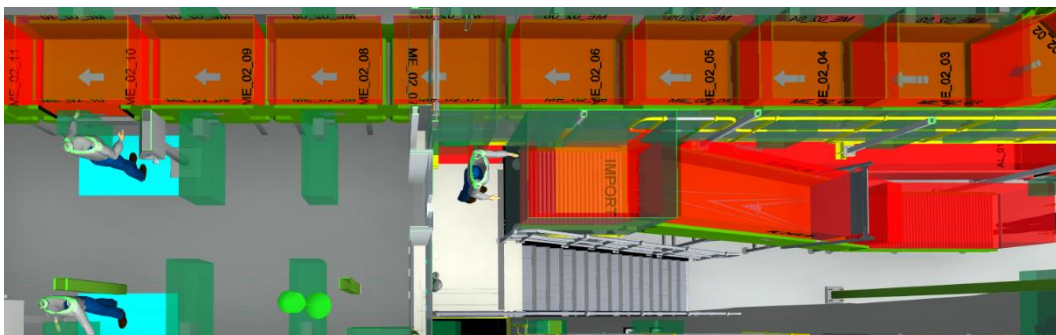


Figure 70 ME Chute Location

15.4. ME Line Fallback Procedures

ME Line Failure Conditions

The following are conditions that can cause the ME lines to become unavailable:

1. The MES Station unavailable – refer to the HLC FDS for reasons why the MES would be unavailable.
- 15.4.1. 2. Any ME line conveyor technical fault.
3. ME Plough/Diverter technical fault.

ME Fallback Procedure

If the ME01 line and/or ME02 line has failed, the following will occur:

- 15.4.2. 1. Bags that are destined for the ME line in fault will bypass that ME line and will remain on the AS01 or AS02 line and be diverted on to the PB03 Carousel.
2. Bags already on the ME line in fault will need to be manually removed and directed to the RI01 rescreening line using the ME_02 chute.

16. BEACON AND SOUNDER INDICATION

16.1. General

A typical beacon stack in the BHS consists of a three-layer of beacons with different colour codes, as well as and a sounder. These beacon stacks are strategically located throughout the BHS to provide visible and audible warnings for faults or issues within the area.

A single-layer beacon and sounder are placed at each over size detection point to provide a local indication of oversize fault. These sounders only activate for an oversize detection.

Certain pushbuttons can illuminate with colour codes that match the beacon stack colours.

Each colour in the beacon stack represents a specific type of faults or status, allowing operators and maintenance personnel to quickly identify the nature of the problem.

- **Red:** E-Stop fault
- **Blue:** Motor Fault/Technical Fault.
- **Amber:** Operational Fault/Pre-start Warning. Economy-Mode.
- **White:** Oversize fault.

CBS Status of the passing bags are to be displayed on the CBS beacon stacks. Please refer to [Section 16.6 CBS Status Indication](#) for further information.

16.2. Beacon Stack Placement

A Three-layer beacon sounder stack is placed in the following locations:

- On all CCPs and FCPs.
- Within maximum intervals of 30 meters along conveyor ROW to ensure that at least one beacon is always within line of sight.
- Adjacent to all merges and diverts.
- Adjacent to all CBS Operation decision points.
- On each side of a PLC control boundary.
- A single stacked white beacon will be placed for oversize items at all oversize checks (overlength or over height per [Section 4.9.8 Oversize Detection](#)).

16.3. Pre-Start Warning

The following points outline the key aspects of pre-start warning:

- When a subsystem transitions from standby to startup, a pre-start warning is required to ensure the safety of personnel working in the area.
- Similarly, if a conveyor recovers from a fault, a pre-start warning must be given before it resumes running, as a precautionary measure.
- The pre-start warning consists of an audible alarm emitted by a sounder, with a beeping frequency of 1 Hz.
- In addition to the sounder, an amber beacon provides a visual cue by flashing at a rate of 1 Hz.
- The pre-start warning is designed to last for **five seconds**, giving personnel sufficient time to react and clear the area before the subsystem or conveyor begins to operate.
- To maintain the high level of safety, the pre-start warning **cannot be muted or disabled**, ensuring that it always remains an effective alert mechanism.

16.4. E-Stop

In the event of an E-Stop fault, red beacons within the E-Stop zone flashes at a rate of 2Hz and a sounder emits a beeping sound at a rate of 2Hz. The Sounder can be muted if the beeping sound becomes a distraction or impedes communication.

16.5. System Fault

In the event of a general fault such as bag jam or pulse wheel fault, amber beacons located in the fault area will illuminate (Solid). Similarly, in the event of a motor fault blue beacons will illuminate. A sounder will also emit a warning sound, but it can be muted if necessary.

16.6. CBS Status Indication

A three-tier beacon sounder stack will be positioned adjacent to every CBS diverting decision point. The CBS beacon stack does not indicate fault states. However, it has an additional function is to provide it visually indicates the CBS status of each bag that passes through. Each bag passing the decision PEC activates a particular beacon colour for one second, reflecting the CBS status and providing visual confirmation for the operator:

- **Green:** Cleared bag.
- **Red:** Rejected bag, Timed out or No Image.
- **Amber:** Unknown bag.
- **Red + Amber:** Force reject.
- **Green + Red + Amber:** Failsafe alarm activated.

16.7. Siren Mute

In the event of a fault within the Baggage Handling System (BHS), an audible warning will be emitted by the sounder to alert operational staff.

The sounder can be temporarily muted under the following conditions:

- By pressing any Jam Reset pushbutton located within the same Emergency Stop (E-Stop) zone as the jammed conveyor, while the fault condition remains active; or
- By pressing the Alarm Silence pushbutton on the CCP & FCP that controls the E-Stop zone the emergency stop was activated in.

If a new fault occurs after the sounder has been muted, the alarm will reactivate automatically, and the mute function may be used again if required.

The mute function is intended to reduce operator fatigue and limit disruption in the workspace due to prolonged audible alarms. However, for safety-critical operations, the pre-start warning alarm cannot be muted and must remain active at all times.

16.8. Pushbutton Indication

The pushbutton indication matches the colour code of the beacon stacks.

An E-Stop pushbutton has a red indicator that illuminates when the E-Stop is activated. When the E-Stop is released, the red indicator flashes at a frequency of 1 Hz to show that the E-Stop is in a resettable state. In scenarios where a E-Stop pushbutton has a key switch or is a lanyard type, a red indicator should be installed near the E-Stop device.

Depending on the fault type the reset pushbutton is associated with, each reset indicator can illuminate in the colour corresponding to the fault type. An amber reset button can be used to reset operational faults, a blue reset button can be used to reset technical faults, and a white reset button can be used to reset oversize faults.

Name	Colour	Device	Status	Position	Description
Reset Button	Amber	Indicator	Pressed	Spring Return	Reset operational faults
Reset Button	Blue	Indicator	Pressed	Spring Return	Fault condition & to reset technical faults
Reset Button	White	Pushbutton	Pressed	Spring Return	Fault condition & to reset oversize faults
Estop	Red	Pushbutton	On/Off	Lamp On/Off	Estop Fault Condition, Lamp flashes (1HZ) ready to reset
Estop with Key	Red	Pushbutton	On/Off	Lamp On/Off	Same as above, lamp is installed near the Estop

17. CONTROL CABINET

17.1. Power Distribution Panel

General

The Power Distribution Panel (PDP) is responsible for distributing power from the service layer to the Central Control Panels (CCPs). The LLC system does not directly control any components within the PDP.

17.1.1 However, remote input modules are housed within the PDP that provide the LLC with the ability to monitor equipment status and performance.

Control Panel

17.1.2 Each PDP features the following panel layout:

Name	Colour	Device	Status	Position	Description
Feed Phase Line 1	White	Indicator	On/Off	Lamp On/Off	Availability of Line 1
Feed Phase Line 2	White	Indicator	On/Off	Lamp On/Off	Availability of Line 2
Feed Phase Line 3	White	Indicator	On/Off	Lamp On/Off	Availability of Line 3
UPS Phase Line 1	White	Indicator	On/Off	Lamp On/Off	Availability of Line 1 to the UPS
UPS Phase Line 2	White	Indicator	On/Off	Lamp On/Off	Availability of Line 2 to the UPS
UPS Phase Line 3	White	Indicator	On/Off	Lamp On/Off	Availability of Line 3 to the UPS
Lamp Test	Black	Pushbutton	Pressed	N/A	Tests all lamps
Power Monitor	N/A	Display	Active	N/A	Active power monitor module
Circuit Breaker Healthy	White	Indicator	On/Off	Lamp On/Off	Availability of the Circuit Breaker
Circuit Breaker Off	Green	Indicator	On/Off	Lamp On/Off	Circuit Breaker is currently off
Circuit Breaker Fault	Blue	Indicator	On/Off	Lamp On/Off	Circuit breaker is unhealthy

Table 38: Panel Layout

*PDPs will have multiple of the last 3 indicators for each of the 3-Phase circuit breakers the PDP has

Panel Isolator

Isolators are designed to disconnect power to a specific section or the entire system in the event of an emergency or maintenance. They are not intended for use during normal operation, as their use could disrupt the operation of the BHS and pose a safety hazard. They should only be operated by authorised personnel and only when absolutely necessary.

Switch Changeover

PDP is supplied by a power source that already includes a switch changeover mechanism coupled with a generator in the event of a power outage.

UPS

UPS units are integrated within the Power Distribution Panel (PDP) to deliver control voltage to the subsystems while transitioning to generator supply during a power outage. Each UPS has a network interface connected to the BHS network, providing real-time operational status and fault diagnostic data directly to the SCADA system. Refer to the HLC FDS and Section 2.2.10 UPS for further information.

17.2. Central Control Panel

General

A CCP is an electrical cabinet that provides the local power distribution to the PLC subsystem and its field equipment. Each CCP houses:

- A PLC unit.
- Network hardware.
- A beacon and sounder stack are located on the top of each CCP. Refer to Section 16 for further beacon and sounder indication information.

Control Panel

All CCPs are fitted with a control panel providing controls and status indicators. These are defined as follows:

Name	Colour	Device	Status & Position	Function
E-STOP	Red	Mushroom Pushbutton	Pushed In	E-Stop activated
			Twist-pull-out with key	E-Stop released
E-STOP ACTIVATED	Red	Indicator	Lamp On	Panel E-Stop is activated/pushed in
			Lamp Flash at 1Hz	Panel E-stop is released

Name	Colour	Device	Status & Position	Function
E-STOP RESTART	Green	Pushbutton	Pressed	Restarts the subsystem once all E-Stops have been released following an E-Stop activation.
PHASE A	White	Indicator	Lamp On	Phase A of 3Φ power available
			Lamp Off	Phase A of 3Φ power unavailable
PHASE B	White	Indicator	Lamp On	Phase B of 3Φ power available
			Lamp Off	Phase B of 3Φ power unavailable
PHASE C	White	Indicator	Lamp On	Phase C of 3Φ power available
			Lamp Off	Phase C of 3Φ power unavailable
START	Green	Indicator	Lamp On	CCP is running
		Pushbutton	Lamp blinking (1 Hz)	CCP is starting up
			Lamp Off	CCP is stopped
			Pressed	Starts the CCP
STOP	Red	Indicator	Lamp On	CCP is stopped
		Pushbutton	Lamp blinking (1 Hz)	CCP is stopping
			Lamp Off	CCP is running
			Pressed	Stops the CCP
TECH FAULT/RESET	Blue	Indicator	Lamp On	Technical fault condition: sensor is blocked.
		Pushbutton	Lamp blinking (1 Hz)	Technical fault condition at CCP; sensor is not blocked. Ready for fault reset.
			Lamp Off	No technical fault.
			Pressed	Resets the technical fault.
ALARM SILENCE	Yellow	Pushbutton	Pressed	Acknowledges the alarm and silences the horn.
LAMP TEST	Black	Pushbutton	Pressed	Illuminates all lamps in the panel and connected control stations
ISOLATOR	N/A	Two Position Switch	Off	The panel is off, power is disconnected
			On	The panel is on

Name	Colour	Device	Status & Position	Function
POWER METER	N/A	Display	Active	Displays power information (voltage, current, power factor and power consumption)

Table 39: CCP Control Panel

CCP Fault

The fault conditions associated with equipment in the CCP are as follows:

- 17.2.3. PLC module fault.
 - PLC battery low.
 - Fire alarm.
 - Network fault.

17.2.3.1. Please refer to the Appendices section 23.8 for farther details on the effect of CCP failure .Controller Failover

PLCs for any subsystem whose failure would affect more than 25% of the capacity of a specific function within the plant are set up in a warm backup redundant configuration.

In case of a failure incapacitating the primary PLC, the SCADA will display an alarm.

The following PLCs will have redundant configurations:

- CCP1
- CCP2
- CCP4A
- CCP6
- CCP7

Refer to 2.2.1.7 PLC Assignment for further details.

17.2.3.2. CPU Battery

A low/dead battery does not stop the PLC whilst running. However, if the PLC is powered down, then there is a danger it will not re-start. When the battery is low, an alarm appears in the SCADA system alarm banner and the CCP popup shows a battery fault.

17.2.3.3. CCP Emergency Stop

The emergency stop, triggered by the CCP pushbutton, impacts all conveyors within the local PLC subsystem. The restart for this emergency stop can only be executed from the EStop restart pushbutton located on the CCP.

17.3. Field Control Panel

General

Each CCP has several FCPs responsible for specific control groups within a subsystem, aligned with E-Stop zones. Unlike CCPs, FCPs do not have PLCs. Instead, they include a network switch connected to the access switch in the CCP. EIP motor drives are daisy-chained out of the FCP switch, and FCPs house E-Stop monitoring and circuitry.

For further details on the FCPs Layout, please refer to the following documents:

- Overall System FCP Splitting Layout
- Sub-System CCP1 FCP Split Layout
- Sub-System CCP2 FCP Split Layout
- Sub-System CCP4A FCP Split Layout
- Sub-System CCP6 FCP Split Layout
- Sub-System CCP7 FCP Split Layout

Control Panel

FCP has similar appearance to CCP and shares the same control panel interface and functionality:

Name	Colour	Device	Status & Position	Function
E-STOP	Red	Mushroom Pushbutton	Pushed In	E-Stop activated
			Twist-pull-out with key	E-Stop released
E-STOP ACTIVATED	Red	Indicator	Lamp On	Panel E-Stop is activated/pushed in
			Lamp Flash at 1Hz	Panel E-stop is released
E-STOP RESTART	Green	Pushbutton	Pressed	Restarts the subsystem once all E-Stops have been released following an E-Stop activation.
PHASE A	White	Indicator	Lamp On	Phase A of 3Φ power available
			Lamp Off	Phase A of 3Φ power unavailable
PHASE B	White	Indicator	Lamp On	Phase B of 3Φ power available
			Lamp Off	Phase B of 3Φ power unavailable
PHASE C	White	Indicator	Lamp On	Phase C of 3Φ power available
			Lamp Off	Phase C of 3Φ power unavailable

Name	Colour	Device	Status & Position	Function
START	Green	Indicator	Lamp On	FCP is running
		Pushbutton	Lamp blinking (1 Hz)	FCP is starting up
		Indicator	Lamp Off	FCP is stopped
		Pushbutton	Pressed	Starts the FCP
STOP	Red	Indicator	Lamp On	FCP is stopped
		Pushbutton	Lamp blinking (1 Hz)	FCP is stopping
		Indicator	Lamp Off	FCP is running
		Pushbutton	Pressed	Stops the FCP
TECH FAULT/RESET	Blue	Indicator	Lamp On	Technical fault condition: sensor is blocked.
		Pushbutton	Lamp blinking (1 Hz)	Technical fault condition at CCP; sensor is not blocked. Ready for fault reset.
		Indicator	Lamp Off	No technical fault.
		Pushbutton	Pressed	Resets the technical fault.
ALARM SILENCE	Yellow	Pushbutton	Pressed	Acknowledges the alarm and silences the horn.
LAMP TEST	Black	Pushbutton	Pressed	Illuminates all lamps in the panel and connected control stations
ISOLATOR	N/A	Two Position Switch	Off	The panel is off, power is disconnected
			On	The panel is on

Table 40: Field Control Panel Operator Station

18. UNSCREENED TRANSFER INPUT

18.1. General

The loading conveyors in the transfer line are specifically designed to facilitate manual bag loading onto the first conveyor, which then transports the bags into the system. These conveyors lack side guards on the loading side for easy access, ensuring efficient and seamless loading by operators.

Upon starting the system, the loading conveyors run continuously, bypassing energy conservation mode, allowing for a smooth and uninterrupted operation during the bag loading process. Once the loading is completed, the system can automatically initiate a flush and stop process after a configurable period of time without detecting a bag enter into the transfer line.

All bags that enter the BHS from this transfer line are to be screened by a CBS machine in the BHS.

18.2. Control Panel

Two control panels are located at each end of the loading conveyor with for following panel layout:

Control	Device	Colour
Fault Reset	Illuminated push button	Amber
System Start	Illuminated push button	Green
System Stop	Illuminated push button	Red
E-Stop	Illuminated mushroom push button	Red

Table 41: Transfer Area Control Panel

18.3. Oversize Detection

Sensors are installed at the end of the loading conveyor to detect over-length and over-height objects and enunciate alarms accordingly. Preventing bags from progressing into the system helps minimise the risk of equipment damage, delay, or other disruptions in the handling process.

Adjacent to each oversize detection location, a dedicated oversize reset station is available, either integrated into the existing control station panel or configured as an independent control station. When an oversize fault is detected, the white indicator illuminates. Upon removal of the oversize bag from the PEC, the indicator flashes at a frequency of 1 Hz to signal that the fault is now resettable. Pressing the "Oversize Reset" pushbutton clears the fault, initiates a pre-start warning, and allows the system to resume normal operation.

Control	Device	Colour
Oversize Reset	Illuminated Push button	White

Table 42: Oversize Reset Input

18.4. Energy Conservation Mode

Energy Conservation Mode (ECM) is enabled on all infeed transfer line conveyors

- The Loading Conveyor can be reactivated from ECM via the Start pushbutton on the Central Control Panel (CCP) or the Local Control Panel located at the Loading Conveyor.
- This is the only way to restart these conveyors out of energy conservation mode, flagging the PEC of the load conveyors will not run the conveyors.

Further information on energy conservation mode is detailed in [4.5 Energy Save Conservation Mode](#).

19. ROUTING

19.1. General

From the perspective of the Low-Level Control (LLC), routing functionality is divided into two core components:

- **Auto Routing:**
Rule-based routing logic executed at diverters, ploughs, and VSUs. This logic dynamically responds to conditions such as segment status (e.g., busy/full) and configured destination groups to ensure uninterrupted and efficient bag flow. Auto Routing will not be done for stage SP5
- **Load Balancing:**
A subsystem-level routing method designed to evenly distribute baggage across parallel processing lines to optimise system throughput and prevent overloading. Thresholds for load balancing ploughs (Busy and full) is detailed in the load balancing Section 4.8.4 of the DTB Simulation FDS.

19.2. Load Balancing

19.2.1. Basic Principle

19.2.1.1. Assumption

Load balancing algorithms are enabled only when both the primary and secondary routing segments are operational. If either segment becomes unavailable due to conveyor faults, SCADA manual override, or other system inhibitions, the fallback re-routing mechanism will be triggered. In such cases, the system will override the initial load balancing decision to maintain flow continuity and prevent gridlock.

19.2.1.2. General Divert Decisions

The following principles govern divert behaviour at load balancing points:

- **Preferred Path Busy:**
If the Preferred Path is flagged as busy, bags will still be permitted to continue this path.
- **Preferred Path Full:**
If the Preferred Path is full, bags will be diverted to an available alternate path that is not full. If both primary and secondary paths are full, bags will remain on their Preferred Path and wait for downstream clearance, the bags will be held prior to the Load sharing Diverter.
- **Custom Logic Conditions:**
Certain load balancing locations may require customised logic to accommodate physical layout constraints or system-specific behaviour. These exceptions will be handled through local PLC rules or SCADA-configured toggles. For further details on the load balancing, please refer to the appendices section 23.1 .

- **Fallback Path Logic:**

In the load balancing layer, fallback paths are not used. When both the primary and secondary routes are full or unavailable—and diverting would risk a cascade stop on the Plough—the system will hold the bag upstream of the plough conveyor until the any of the paths becomes available.

- **Primary Path:**

The path which diverting operation is NOT required.

- **Secondary Path:**

The path which diverting operation is required.

- **Preferred Path:**

The “**Preferred Path**” is defined as the route enabling a bag to reach its destination with the shortest distance or minimal detour. In the case of pre-CBS diverters, they keep track of the number of bags assigned to each EDS machine. If a discrepancy of more than 100 bags (a configurable parameter) is detected, the diverter redirects more bags towards the underutilised EDS machine, thereby designating it as the “**Preferred EDS**” Consequently, the path incorporating the “**Preferred EDS**” is referred to as the “**Preferred Path**”. (Quoted from section 4.8.1 of the DTB Simulation FDS)

This approach ensures balanced usage of parallel lines while preventing unnecessary recirculation or routing into less desirable flow paths.

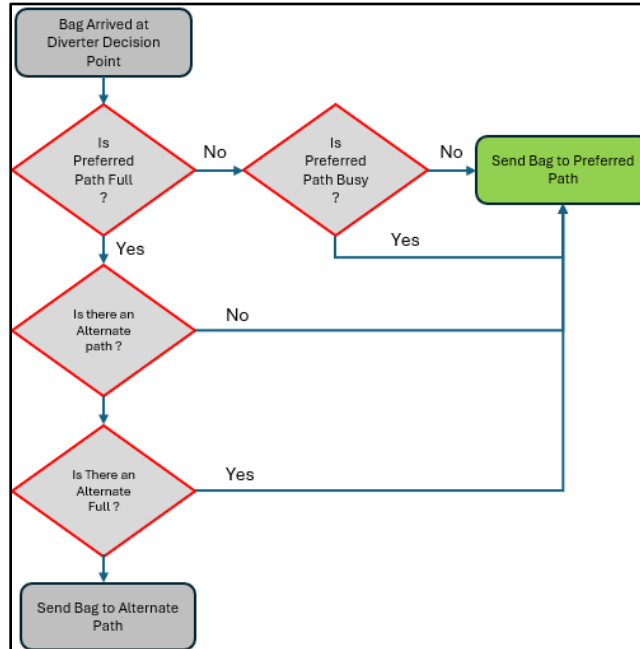


Figure 71: LLC Load Balancing Logic Overview

CBS Load Balancing

CBS machines generally operate at a speed of 0.5m/s and need roughly 300mm gaps between bags to ensure optimal screening quality. To avoid a bottleneck in throughput due to this reduced processing speed (Throughput B and C, Figure 63), the transport mainline divides into two separate CBS lines. This arrangement aims to efficiently distribute the high volume of bags (Throughput A, Figure 51) among each CBS machine. The key is to mitigate any cascade stoppage and maintain a balanced long-term distribution of bags to each CBS machine.

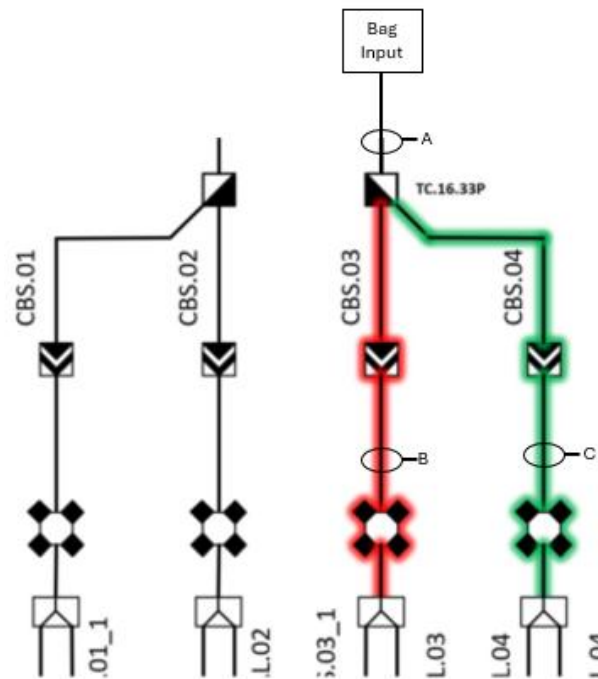


Figure 72 CBS Load Balancing Layout

When both CBS lines are available and have a balanced traffic load, the decision to divert bags is primarily dictated by the density of incoming bags to a CBS load balancing diverter. This density is represented by the distance between approaching bags, known as the 'bag pitch.' If the indicative bag pitch size is small enough to create a throughput rate exceeding a single CBS line's busy threshold, the subsequent bag will be directed to the alternative CBS line.

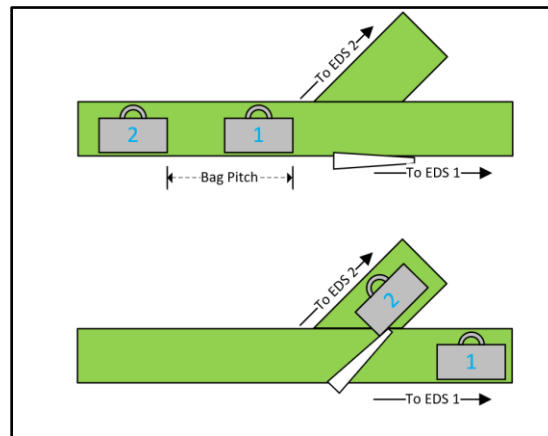


Figure 73: *Alternating Distribution for High Traffic Volume*

Conversely, if the bag pitch size is large enough that the throughput rate is less than the threshold, alternating is not required. Bags are directed to the Preferred EDS.

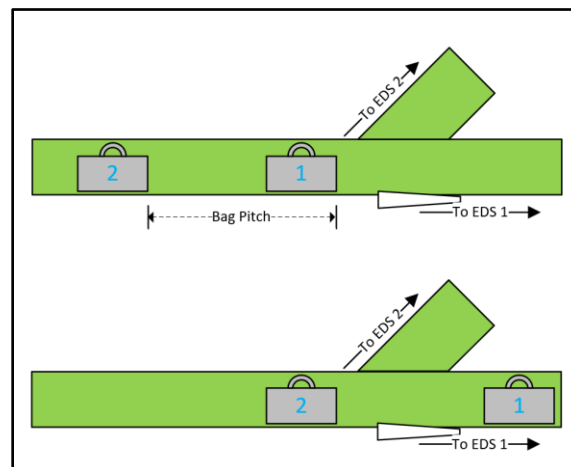


Figure 74: *Sequential Distribution for Low Traffic Volume when EDS1 is the Preferred EDS*

Below is an overview of the CBS Load balancing scheme:

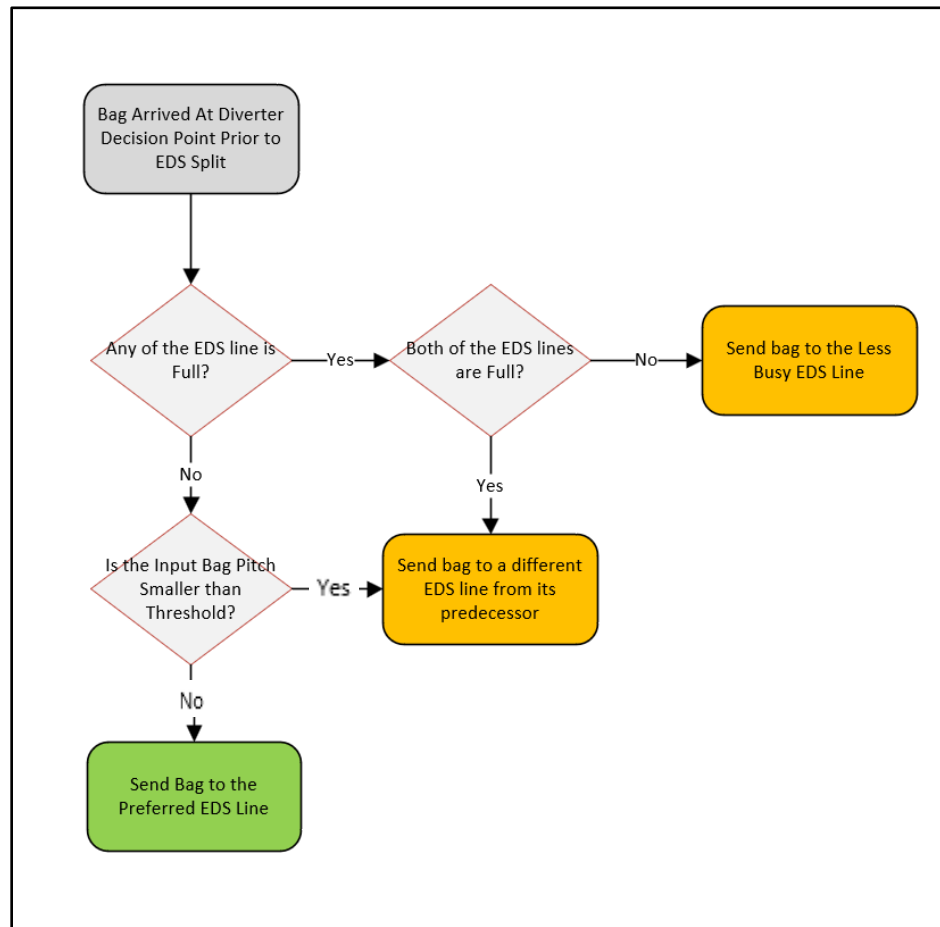


Figure 75: CBS Line Load Balancing Overview

In short, bag pitch is used to calculate the throughput rate. The bag pitch is inversely proportional to the throughput rate, the smaller the bag pitch the bigger throughput rate and vice versa. Only if the throughput rate exceeds or equals the busy threshold will the line be designated busy and if the throughput rate exceeds or equals the full threshold then the line will be designated full.

Fallback and Load Balancing Response to Faults

Faults are categorized as:

- **Operational faults:** Bag Jams, Over Length and Over Height. These are faults that will be cleared by the operator in a short time frame (less than or equal to 15 minutes).
- 19.2.3. • **Technical faults:** These are any fault that will stop the path for an extended period (typically more than 15 minutes).

All faults affecting load balancing, whether operational or technical are treated the same:

- When a fault occurs on a route, the Load Balancing unit will immediately divert all new bags to the non-faulted route or the secondary path, provided the secondary path is available.
- Once the fault is cleared and the route becomes available, normal load balancing will automatically resume.

This approach ensures that any fault condition prioritizes maintaining system throughput and prevents congestion on the faulted route.

Caveat

19.2.4.

Changes to divert conditions will not affect a bag that has already entered the diverter's decision zone.

- The current divert decision remains active for that bag to avoid mid-transfer disruptions.
- The new divert decision takes effect for the next bag arriving at the diverter.

19.3. Technical Fault Fallback

For the 2-lane strategy, there is no recirculation as part of this stage of works. Therefore, if a technical fault (Motor Isolation, carousel fault, conveyor unavailable) occurs upstream of the ME station and the divert decision points, a fallback procedure is implemented. This fall procedure will be removed once the recirculation path is available.

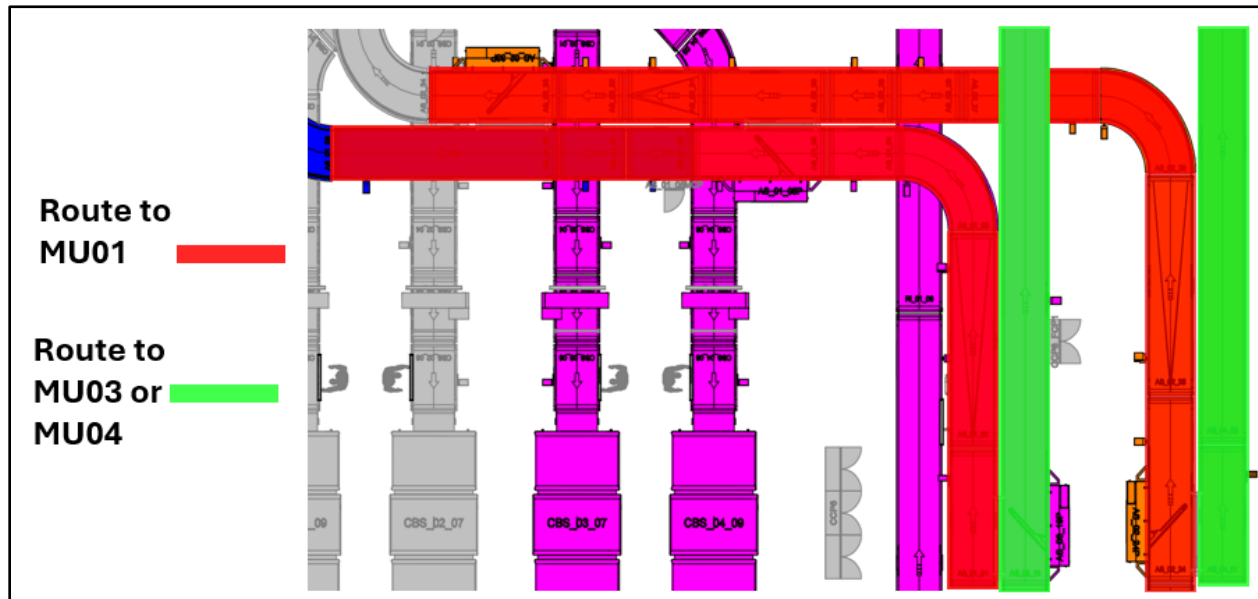


Figure 76 The 2 Key Divert Points for Fallback

19.3.1.

AS_02_24P Fallback procedure

If a fault occurs downstream of this plough, for the route to MU01 or the route to MU03 or MU04, the plough follows the following procedure

1. AS0220 is given a one-shot SCADA override signal to hold bags.
2. This is to give operators time to decide where to send the bags in the HLC.
3. Once the fallback destination is decided, the operators can release the override in SCADA.
4. AS0222 acts as a virtual ATR, taking the known barcodes and rerouting the bags to the new destination set by the operators.
5. The bags are sent to the fallback location.

AS_02_24P Fallback Destinations

The table below shows the ideal destination for bags if a fault occurs within a range of conveyors downstream of AS_02_24P. It should be noted that the operators are manually setting the fallback destination in HLC, this table is for reference of fallback destinations that are ideal for the circumstances.

19.3

Bag Original Destination	Technical Fault Location Range	Fallback Options
MU01	AS_02_25 to MF_01_07	MU03 or MU04
MU03	AS_04_01 to AS_04_05	MU01
MU03	XO_04_01 to PB_03_04	MU01 or MU04
MU04	AS_04_01 to AS_04_05	MU01
MU04	AS_04_06 to MF_08_06	MU01 or MU03

Table 43 AS_02_24P Fallback Matrix

AS_03_19P Fallback procedure

19.3.3 If a fault occurs downstream of this plough, for the route to MU01 or the route to MU03 or MU04, the plough follows the following procedure

6. AS0315 is given a one-shot SCADA override signal to hold bags.
7. This is to give operators time to decide where to send the bags in the HLC.
8. Once the fallback destination is decided, the operators can release the override in SCADA.
9. AS0317 acts as a virtual ATR, taking the known barcodes and rerouting the bags to the new destination set by the operators.
10. The bags are sent to the fallback location.

19.3.4.

AS_03_19P Fallback Destinations

The table below shows the ideal destination for bags if a fault occurs within a range of conveyors downstream of AS_03_19P. It should be noted that the operators are manually setting the fallback destination in HLC, this table is for reference of fallback destinations that are ideal for the circumstances.

Bag Original Destination	Technical Fault Location Range	Fallback Options
MU01	AS_01_01 to MF_01_07	MU03 or MU04
MU03	AS_03_20 to AS_03_21	MU01
MU03	AS_03_22 to PB_03_04	MU01 or MU04
MU04	AS_03_20 to AS_03_21	MU01
MU04	XO_03_01 to MF_08_06	MU01 or MU03

Table 44 AS_03_19P Fallback Matrix

20. GRIDLOCK PREVENTION

For this stage of the project, there is only one loop that could possibly go into gridlock. This is the Re-Screen line RI01.

The RI01 line merges bags back onto the XO01 line at XO0114, which, in turn, merges bags onto the TC16 line to feed back into the EDS machines.

Any bags that need to be re-screened will be directed to the Alarm Lines and diverted to the Re-screen line at Plough AL0116P.

If the RI01 line is unable to merge bags onto the XO01 line at a rate that is equal to or greater than the number of bags entering the RI01 line, then bags will continue to build up on the RI01 line until it is completely full.

If the number of bags on the RI01 line reaches an Alarm threshold (initially set to 6 bags, configurable in the PLC Code only in this stage), an alarm will be raised on the SCADA to alert the operators of an impending gridlock situation.

If the number of bags on the RI01 line reaches a Critical threshold (initially set to 10 bags, configurable in the PLC Code only in this stage), another alarm will be raised on the SCADA and conveyors XO0112 and TC1630 will be set to SCADA Override Stop.

This will allow the bags on the RI01 line to merge onto XO0114 and back into the EDS lines.

When the operators are happy that the Gridlock situation has been resolved, they can remove the SCADA Override Stop on conveyors XO0112 and TC1630 and return them to normal operation mode.

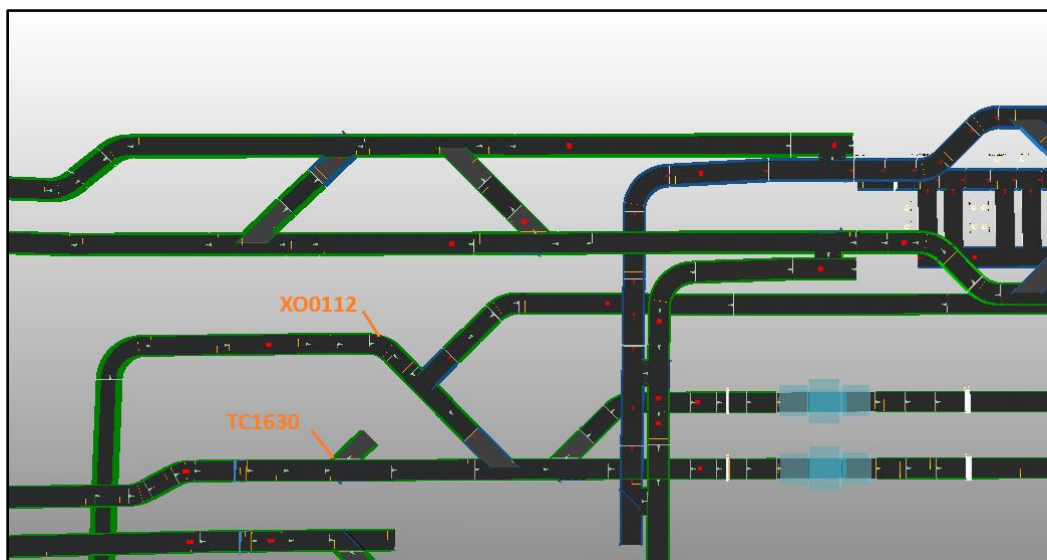


Figure 77 - Gridlock Control Point

21. HLC FAILURE

The HLC regularly (initially set to every 1 minute) sends the Carrier Default Destination table and the Virtual to Physical Destination table to each PLC.

In the unlikely event of a complete failure of the HLC system:

1. The SCADA will display an alarm.
2. The following occurs as a bag passes through an ATR:
 - a. The PLC will extract the Airline Code (digits 2, 3 and 4 of the IATA tag) from the IATA tag.
 - b. The PLC will use the Carrier Default Destination table to look up the Virtual destination for the Airline code.
 - c. The PLC will then look up the Virtual destination in the Virtual to Physical Destination table to assign the Physical destination for the bag.
 - d. The Physical destination is stored in the PLC Tracking
 - e. Bags that arrive at an MES will need to be sorted to either their Carrier Default Destination or direct to a Make Up Carousel. **DO NOT send to the Flight Destination.**

22. MAKE-UP DEVICE

22.1. General

DTB has the following make up devices:

- **South:**

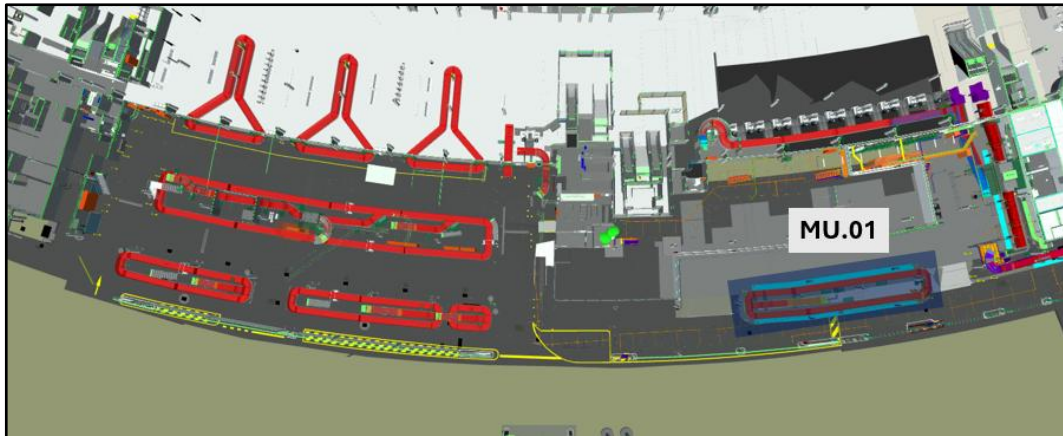


Figure 78: South Carousels

- MU.01 – Carousel.

- **North:**

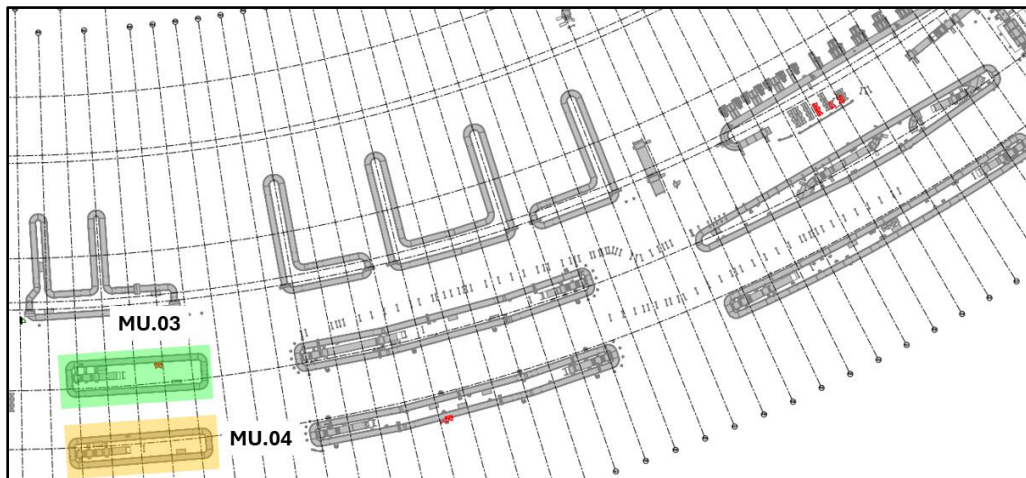


Figure 79: Northern Carousels

- MU.03 – Carousel.

- MU.04 – Carousel.

22.2. Make-up Carousel

A make-up carousel is a large loop made of overlapping rubber slats that form a continuous rotating flat surface. It serves as the terminal device of a system and operates with energy conservation mode while the system is on.

In this stage only conveyor line feeds bags onto the carousel using a 90-degree merge layout.

An update PEC is installed Upstream from the merge point on the carousel, to detect merge windows.

22.3. Control Panel

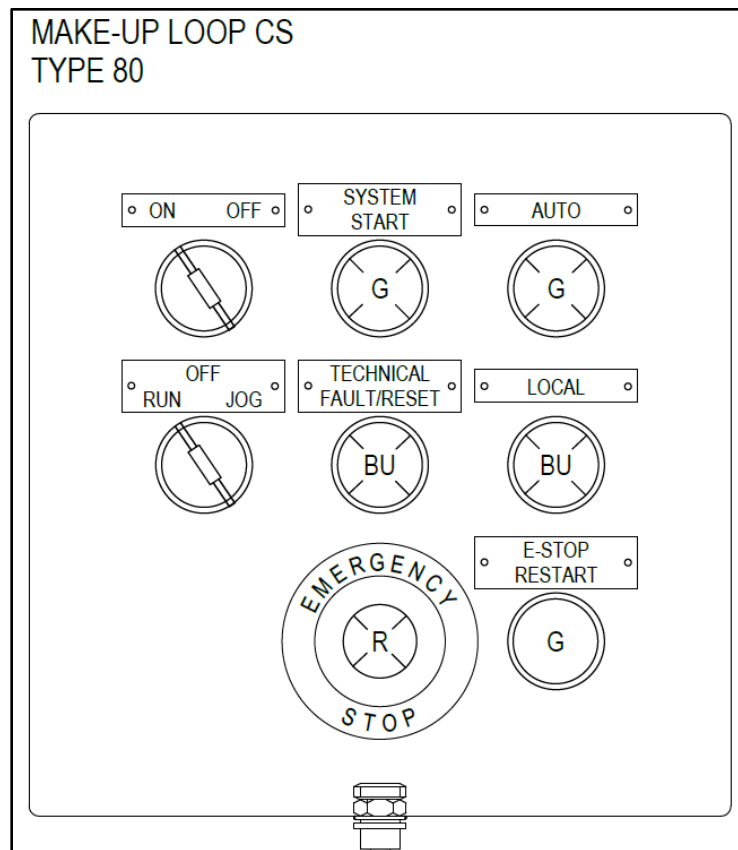


Figure 80 Make-Up Carousel Control Station

Makeup carousels control panel layout:

Control	Device	Colour
Restart	Push button	Green
Start	Illuminated push button	Green
ON/OFF	Two Position Switch	

AUTO	Illuminated push button	Green
LOCAL	Illuminated push button	Blue
Run/OFF/Jog	3 Way Key Switch	
Technical Fault/Reset	Illuminated push button	Blue
Emergency Stop	Illuminated latching mushroom push button	Red
Beacon	Alarm Indicator	Amber
Sounder	Alarm Indicator	

Table 45: Make-Up Carousel Control Station

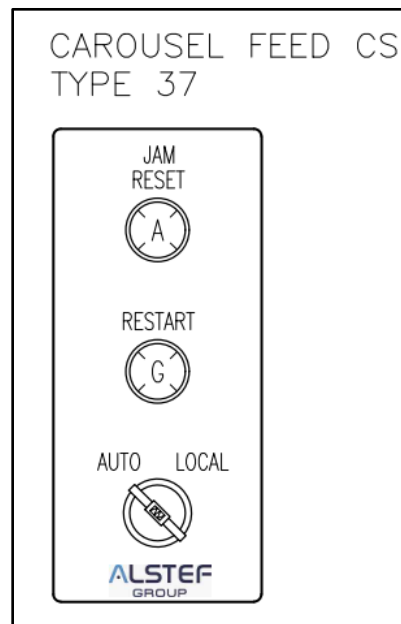


Figure 81 Make-Up Carousel Feed Control Station

Makeup carousel feed control panel layout:

Control	Device	Colour
Jam Reset	Illuminated push button	Amber
Restart	Illuminated push button	Green
Auto/Local	2 Way Key switch	

Table 46: Make-Up Carousel Feed Controls

The beacon/sounder provides start-up warnings and indicates localised system faults.

Emergency stops are located on easily accessible pedestals at 15-metre intervals.

22.4. Carousel Local Operation

Each carousel is equipped with a minimum of two independent motors for redundancy. The mechanical design allows for shared loads across multiple drives or for operation to continue with one drive offline. In carousels with three drives, operation can sustain up to one drive failure without operational impact **(Unless the 2 operating motors are on the same side of the carousel)**.

These drives do not include the SEW local mode selector box.

Local operation is facilitated through a control panel located near the carousel (refer to [Table 45: Make-Up Carousel](#)). Activating the 'Local' pushbutton on the control panel triggers a prestart alarm and initiates the transition to Local mode. If the carousel is in operation, it will halt, and all feeding lines will cease bag delivery. Following the prestart alarm, the 'Local' button lights up in blue, signalling readiness for manual operation. Operators can then use the Run/Off/Jog switch to select the desired motion mode. Local mode synchronously operates all carousel motors, and any motor disconnected via its field isolator will disrupt Local mode, causing the blue indicator to flash at 1Hz. The control panel's isolator switch can also cut power to all carousel motors simultaneously.

To switch back to Auto mode, the operator must set the Run/Off/Jog key switch to 'Off' and press the 'Auto' button. If the key switch is not in 'Off' position, the 'Auto' indicator will flash green at 1Hz as a warning that the transition is not permitted. Following another prestart alarm, the carousel will resume automatic operation, and the feed lines will restart bag delivery if the subsystem is active.

At the merge point where a feed line joins a carousel, there is a control panel equipped with a secondary Local control method. The operator can switch the key to 'Local' to engage local mode. Following a prestart warning, the carousel and all feed line conveyors will begin to operate in local mode simultaneously. Toggle the key switch back to "Auto" will change the operation mode back to auto after a prestart warning. This key switch takes precedence over the 'Local' pushbutton on the carousel's control panel and will accordingly alter the indicators for the 'Local' and 'Auto' pushbuttons. Initiation of local mode via the key switch on the feed line control panel will render the local mode-related controls of the carousel panel, including the pushbutton and 3-position key switch, inoperative. Should conditions arise that prevent the carousel from entering local or auto mode, as previously outlined, the mode selection will be deferred until those conditions are resolved, and the feed line conveyors will remain inactive until the carousel is ready to operate. During this period, the indicators will flash at 1 Hz on the 'Local' or 'Auto' pushbuttons of the carousel control panel serve to notify the operator that the transition to the selected mode is currently on hold.

22.5. Merge Window

A bag on the merge line scans the carousel for a suitable gap (adequate space) into which the bag can merge. When a sufficient gap is found, the bag is injected onto the carousel. The bag on the merge conveyor waits until an appropriate gap is identified before being merged into the carousel.

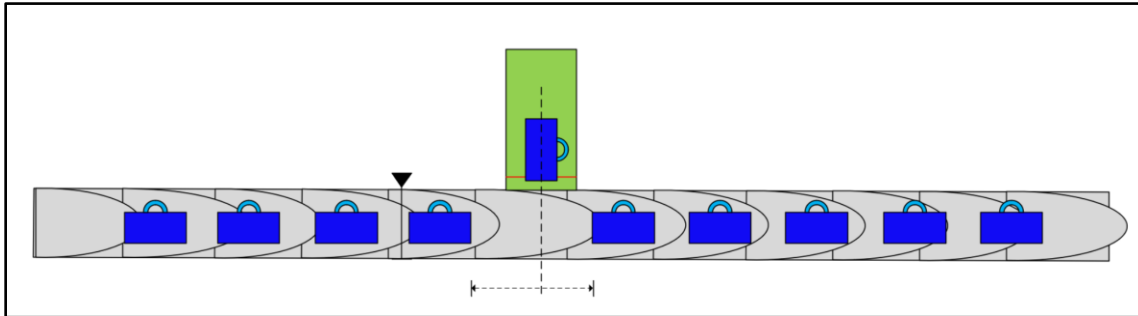


Figure 82: Degree Merge Window

When a carousel starts up, it has a slow ramp rate, taking several seconds to reach its full speed. During the acceleration period, the merge function is temporarily disabled.

22.6. Carousel Full

When a carousel has run for its full length without detecting a mergeable gap, indicating no available merge windows, a SCADA warning message is issued to notify operators that the carousel is full.

When the carousel infeed line has reached to its cascade stop capacity, the divert function to this destination is inhibited. To reactivate the diverter, bags must be removed from the make-up carousel to create merge windows. **This functionality will be disabled for this stage.**

22.7. Carousel Not Emptied

If a make-up carousel operates for a SCADA-configurable duration (initially set to 5 minutes) while continuously detecting the presence of bags, and there are no bags being merged onto the Carousel, a SCADA warning will be triggered to indicate that there are potentially unclaimed bags remaining on the carousel that have not been removed by the handling staff.

22.8. Problem Bag Destination

In the sortation area, unsortable bags are directed to the problem bag destinations. For this stage, it will be the MU.03 carousel.

Bags that will be sent to the problem carousel will be:

- 1) Mis-tracked bags after the Manual Encode entry plough AS0310P
- 2) Bags that are destined for MU06 but have failed to be diverted at Plough AS0319P.

22.9. Integrated RFID

Scope

22.9.1 The existing DTB Make-up carousel is equipped with 2-3 RFID reader antenna gantries, depending on the carousel size. Most of these are offline and have exceeded their service life. The scope of the new system involves utilising the existing locations, panels, and metal frames of these antennas. The reading hardware, controllers, and software will be updated to the latest versions. MU01 and MU03 will have the new RFID scanners installed as part of the 4-lane strategy.

Operation Procedure

22.9.2 The RFID reader features an internal model calibrated to the running speed of the carousel. Beyond the gantry, there are six configurable zones, the distances to which can be adjusted through the controller's software.

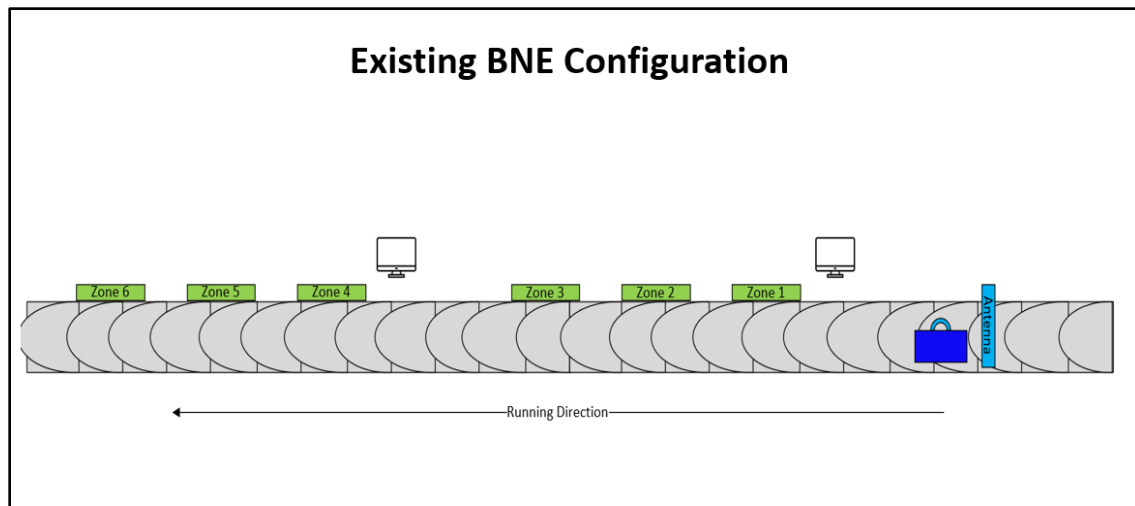


Figure 83: Make-up Carousel RFID Equipment

When a bag with a valid RFID tag passes by the antenna, a visual object is created within this model. As the bag moves through downstream zones, local displays show pertinent information, such as the flight number, destination airport, class, and whether the bag is of a connection flight. Given the notable delays in deceleration and acceleration, accurate bag tracking is only feasible when the carousel operates at a constant speed. Consequently, any cessation of carousel movement prompts the RFID controller to clear all bag data from its internal model. Tracking resumes only when the carousel restarts and attains its full operational speed.

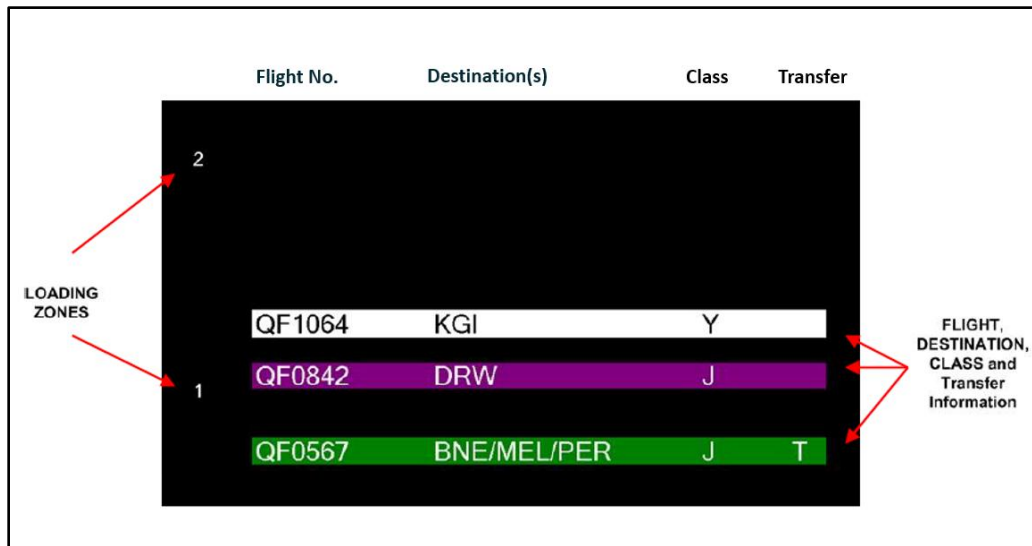


Figure 84: *Make-up Carousel RFID Screen UI*

With the details displayed, an operator can easily determine whether the bag's information aligns with the flight currently being processed.

The controller cross-references the date information stored on the tag, considering only those with the day's flight information as valid. Tags that do not meet this criterion are ignored. Error messages will display if multiple valid tags are detected in close proximity, or if a tag indicates Brisbane Airport as its destination.

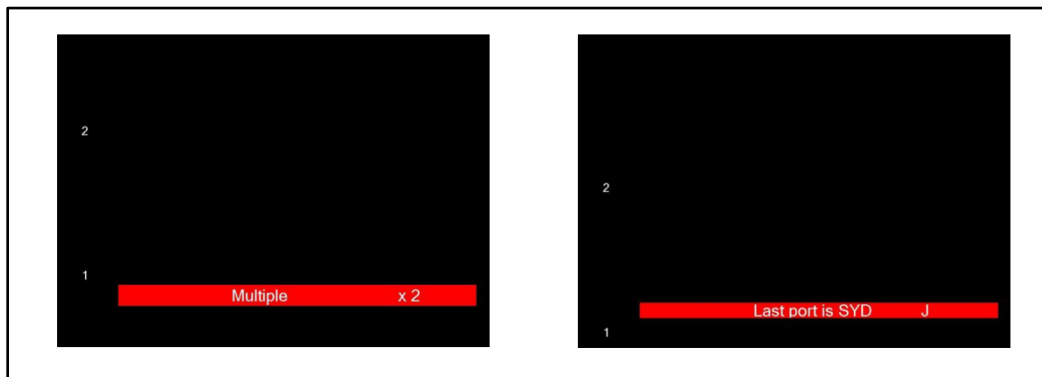


Figure 85: *RFID Error Message Examples*

22.10. Energy Conservation Mode

Energy conservation mode is available for make-up carousels. The carousels use the PEC, the speed of the carousel and the length of the carousel to determine the bag capacity. If the carousel goes a full belt rotation without any bags detected at the PEC, the carousel will go into ECM.

22.11. Motor Redundancy

Make-up carousels are designed to have at least two motors, one for each elongated side of the carousel.

Three Motor Combinations

For some Carousels, a third motor is needed as redundancy in case one drive fails, therefore, the carousel can run on two motors.

22.11.1. This means two of the three motors will be on the same side of the carousel. However, these two motors on the same side of the Carousel cannot be run at the same time.

- 1) The Carousel can only run with two Isolators ON under normal operation. It will not run with all three Isolators ON or only one Isolator ON.
- 2) Therefore, the carousel will run only with two Motors so long as both motors are not on the same side of the carousel.
 - a. Motor 1 & 2 can run together.
 - b. Motor 2 & 3 can run together.
 - c. Motor 1 & 3 cannot run together as they will be located on the same side of the carousel.
- 3) If the running feedback/Isolator/VFD ready signal is lost from either of the two operational motors:
 - a. the Carousel will immediately stop and alarm the fault to SCADA.
 - b. The operator must go to the faulted motor and turn its Isolator OFF.
 - c. The operator then goes to the standby motor and turn its Isolator ON.
 - d. The Carousel will restart after the start up warning.

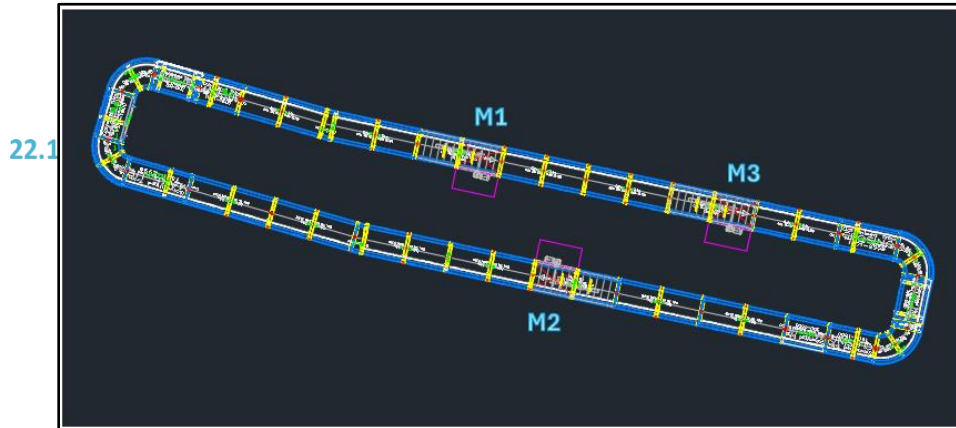
22.11.2.

Two Motor Combinations

For Carousels with only two motors, both can run together, or one motor can run the Carousel on its own.

- 1) If the running feedback/Isolator/VFD ready signal is lost from either of these two motors
 - a. The Carousel will immediately stop and alarm the fault to SCADA.
 - b. The operator must go to the faulted motor and turn its Isolator OFF.
 - c. The Carousel will restart after the pre-start warning.

MU 01



MU_01

Combinations:

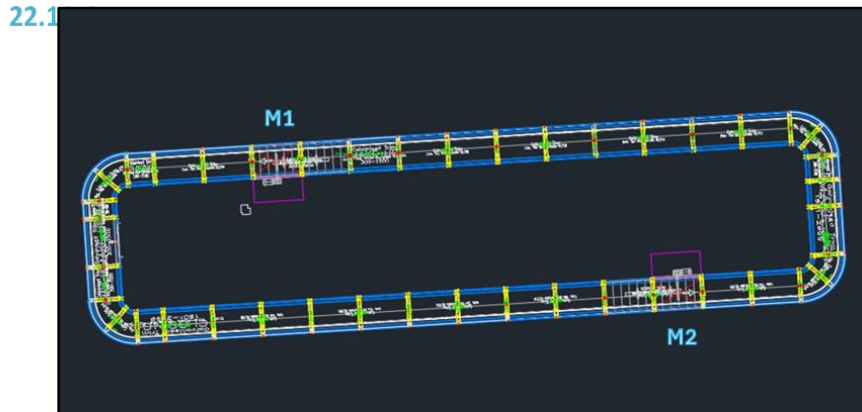
M1 & M2 – Running at the same time

M2 & M3 – Running at the same time

(Motor labels are for presentation purposes only)

Figure 86: MU_01 Motor Combinations

MU 03



MU_03

Combinations:

M1 – Running by itself

M2 – Running by itself

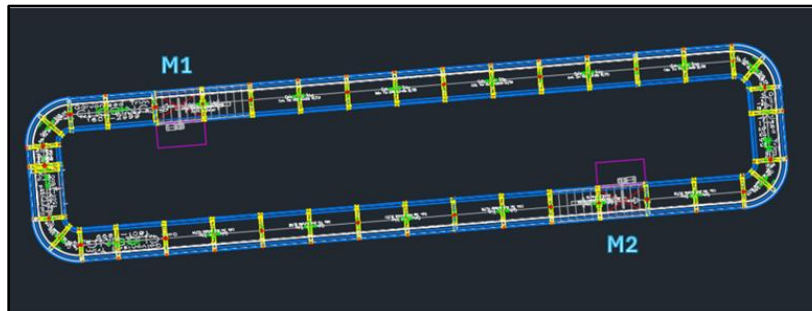
M1 & M2 – Running at the same time (Load Sharing)

(Motor labels are for presentation purposes only)

22.11.5.

Figure 87: MU_03 Motor Combinations

MU 04



MU_04

Combinations:

M1 – Running one at a time or both (Load sharing)

M2 – Running one at a time or both (Load sharing)

(Motor labels are for presentation purposes only)

Figure 88: MU_04 Motor Combinations

23. APPENDICES

23.1. Load Balancing

TC_01_28P

For the 4-lane strategy, TC_01_28P will use the load balancing algorithm outlined in the Emulation FDS.

Green line to yellow line as illustrated below.

23.1.1.

23.1.1.1. Location

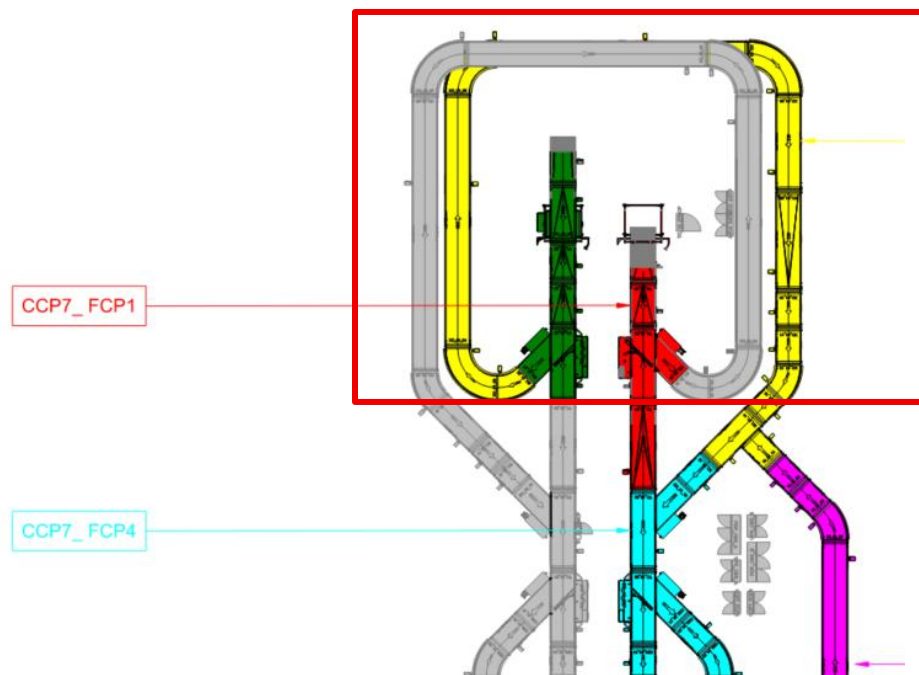


Figure 89: TC01_28P Divert

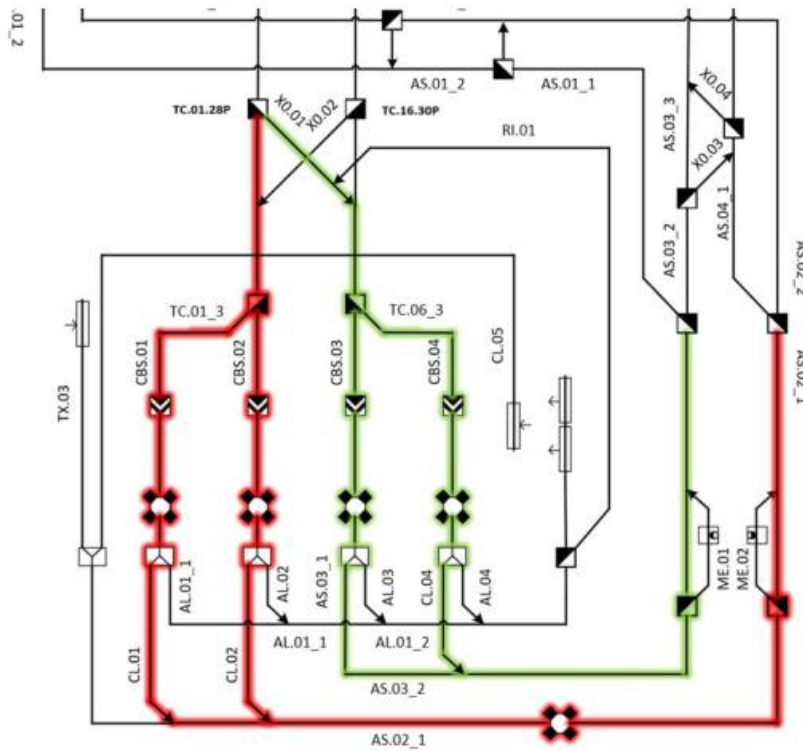


Figure 90 Primary (Red) and Secondary (Green) paths for TC01_28P

23.1.1.2. Scheme

Preferred Path	Fallback Path	Primary status	Secondary status	Bag status	Destination	Divert Decision
South Bag - Secondary	None	N/A	any	Tracked	MU01	Secondary
North Bag - Secondary	None	N/A	any	Tracked	MU03 or MU04	Secondary
South Bag - Primary	Green Line	any	N/A	Tracked	MU01	Primary
North Bag - Primary	Green Line	any	N/A	Tracked	MU03 or MU04	Primary
South Bag - Primary	Green Line	full	any	Tracked	MU01	Secondary
North Bag - Primary	Green Line	full	any	Tracked	MU03 or MU04	Secondary

TC_16_30P

For the 4-lane strategy, TC_16_30P will need to operate in divert operation red to grey line as illustrated below.

TC_16_30P will use the load balancing algorithm from the emulation FDS.

23.1.2.

23.1.2.1. Location

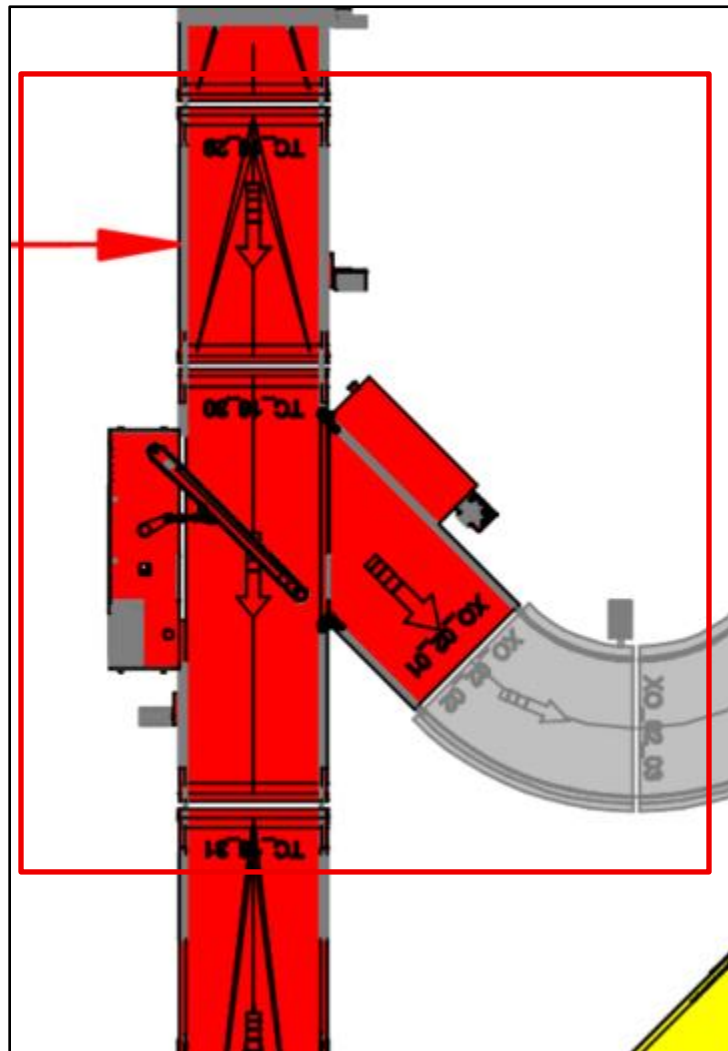


Figure 91: TC_16_30P

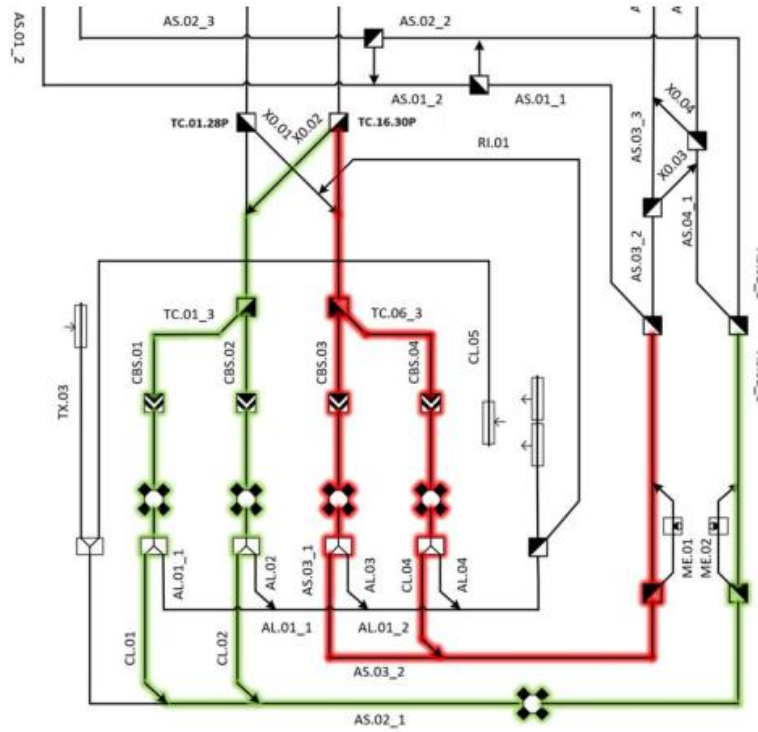


Figure 92 Primary (Red) and Secondary (Green) paths for TC16_30P

23.1.2.2. Scheme

Preferred Path	Fallback Path	Primary status	Secondary status	Bag status	Destination	Divert Decision
South Bag - Primary	Green Line	any	N/A	Tracked	MU01	Primary
North Bag- Primary	Green Line	any	N/A	Tracked	MU03 or MU04	Primary
South Bag - Primary	Green Line	full	any	Tracked	MU01	Secondary
North Bag - Primary	Green Line	full	any	Tracked	MU03 or MU04	Secondary
South Bag - Secondary	None	N/A	any	Tracked	MU01	Secondary
North Bag- Secondary	None	N/A	any	Tracked	MU03 or MU04	Secondary

TC_16_33P

The load through balancing will be applied to TC_16_33P using the load balancing algorithm detailed in Section 4.8 of the Simulation FDS.

23.1.3.1. Location

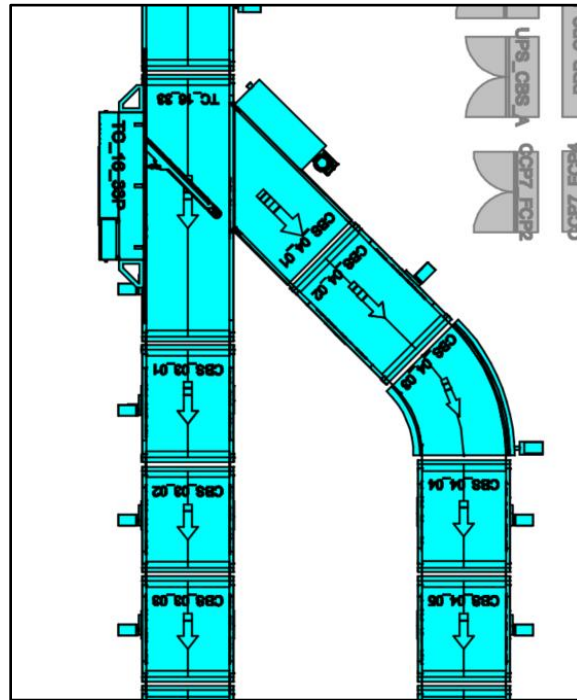


Figure 93: TC_16_33P

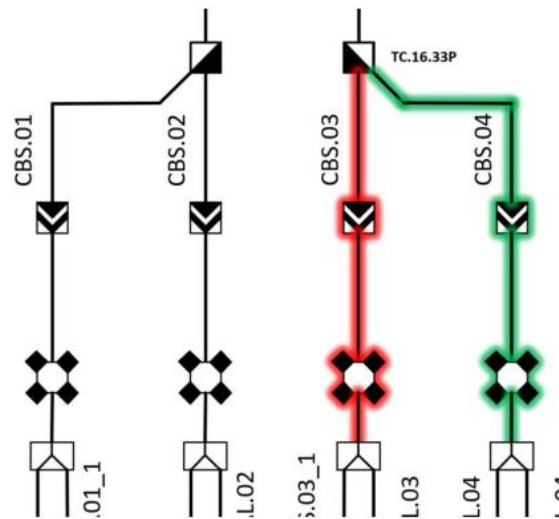


Figure 94 Primary (Red) and Secondary (Green) paths for TC16_33P

23.1.3.2. Scheme

Note:

- Preferred Path is determined by which line has the preferred EDS (See Section 19.2.2).
- Input pitch size monitoring is required.
- Weighing the “Preferred EDS” is less crucial than rectifying temporary traffic imbalances.

Scenario	Primary Path status - Full	Secondary Path status - Full	Secondary Path status - Busy	Primary Path Available	Secondary Path Available	Divert Decision
	Primary path Full threshold is 21 which including conveyors CBS0301-CBS0307	Secondary path Full threshold is 22 which including conveyors CBS0401-CBS0408	Secondary path busy threshold is 18.	Check for conveyor fault from CBS0301 to AS0305	Check for conveyor fault from CBS0401 to CL0405	
1	< full	< full	< busy	Yes	Yes	Primary (CBS03)
2	< full	>= full	>= busy	Yes	Yes	Primary (CBS03)
3	>= full	< full	< busy	Yes	Yes	Secondary (CBS04)
4	>= full	< full	>= busy	Yes	Yes	Secondary (CBS04)
5	< full	< full	< busy	No	Yes	Secondary (CBS04)
6	< full	< full	< busy	Yes	No	Primary (CBS03)
7	>= full	n/a	n/a	n/a	No	Hold Bag at upstream
8	n/a	>= full	n/a	No	n/a	Hold Bag at upstream
9	>= full	>= full	n/a	n/a	n/a	Hold Bag at upstream
10	n/a	n/a	n/a	No	No	Hold Bag at upstream

AS_03_19P

23.1.4.1. Location

23.1.4.

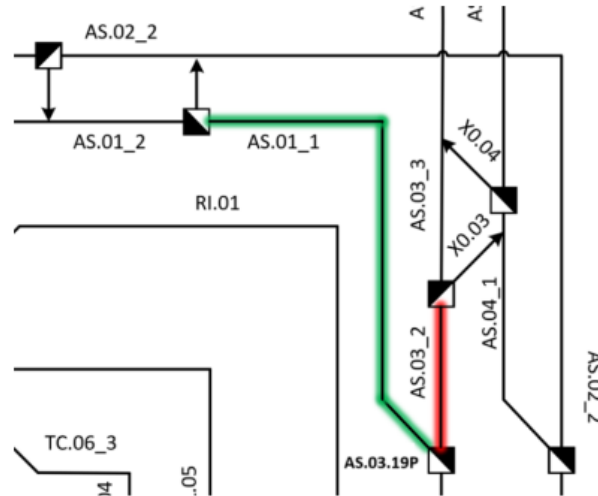


Figure 95 Primary (Red) and Secondary (Green) Paths for AS03_19P

23.1.4.2. Scheme

Note:

- No load balancing occurs at this diverter
- Primary Path is the preferred path for north bags and fallback path for south bags
- Secondary path is the preferred path for south bags and fallback path for north bags
- No alternative path for this diverter

Preferred Path	Fallback Path	Primary status	Secondary status	Divert Decision
North Bag - Primary	South Bag - Primary	any	any	Primary
South Bag - Secondary	North Bag - Secondary	any	any	Secondary

AS_02_24P

23.1.5.1. Location

23.1.5.

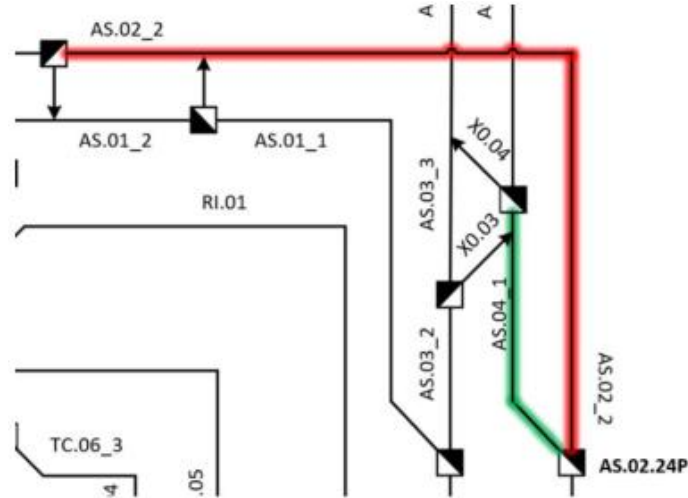


Figure 96 Primary (Red) and Secondary (Green) Paths for AS02_24P

23.1.5.2. Scheme

Note:

- No load balancing occurs at this diverter
- Primary Path is the preferred path for south bags and fallback path for south bags
- Secondary path is the preferred path for north bags and fallback path for north bags
- No alternative path for this diverter

Preferred Path	Fallback Path	Primary status	Secondary status	Divert Decision
South Bag - Primary	North Bag - Primary	any	any	Primary
North Bag - Secondary	South Bag - Secondary	any	any	Secondary

AS_03_21P

23.1.6.1. Location

23.1.6.

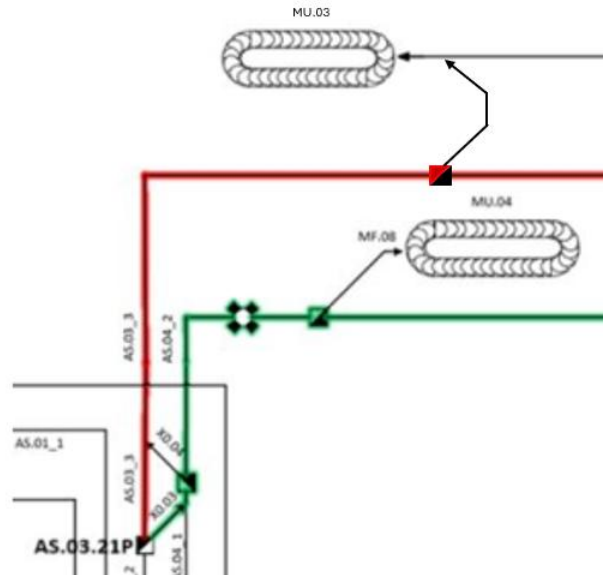


Figure 97 Primary (Red) and Secondary (Green) Paths for AS03_21P

23.1.6.2. Scheme

Note:

- Primary path is the preferred path for MU03 and mis-tracked bags
- Secondary path is the preferred path for MU04
- Bags destined for MU04 must be sent to the secondary path or will not have the option to recirculate back
- MU04 bags will be rerouted to MU03 if the secondary path is not able to receive

Preferred Path	Fallback Path	Primary status	Secondary status	Divert Decision
MU03 and Mistrack-Primary	MU04 - Primary	< full	any	Primary
MU03 and Mistrack - Primary	MU04 - Primary	>= full	< busy	Secondary
MU03 and Mistrack - Primary	MU04 - Primary	>= full	>= busy	Primary
MU04 - Secondary	None	any	< full	Secondary
MU04 - Primary	None	< full	>= full	Primary

AS_04_05P

23.1.7.1. Location

23.1.7.

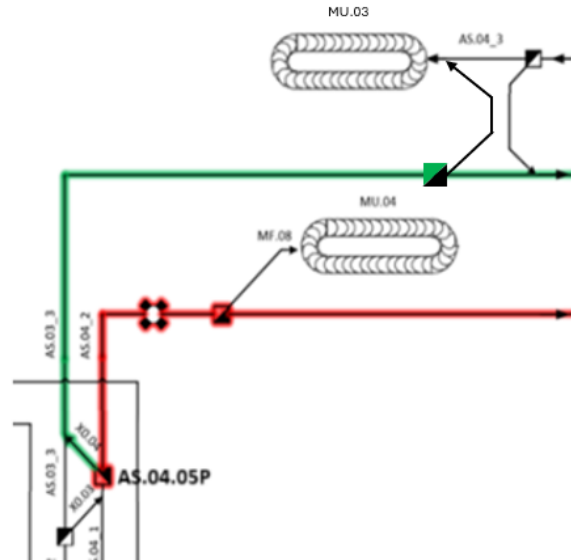


Figure 98 Primary (Red) and Secondary (Green) Paths for AS04_05P

23.1.7.2. Scheme

Note:

- Primary path is the preferred path for MU04
- Secondary path is the preferred path for MU03 and mis-tracked bags
- Bags destined for MU04 must be sent to the primary path or will not have the option to recirculate back
- MU04 bags will be rerouted to MU03 if the primary path is not able to receive

Preferred Path	Fallback Path	Primary status	Secondary status	Divert Decision
MU03 and Mistrack - Secondary	MU04 - Secondary	< full	>= busy	Primary
MU03 and Mistrack - Secondary	MU04 - Secondary	any	< busy	Secondary
MU03 and Mistrack - Secondary	MU04 - Secondary	>= full	< full	Secondary
MU03 and Mistrack - Secondary	MU04 - Secondary	>= full	>= full	Primary
MU04 - Primary	None	< full	any	Primary
MU04 - Secondary	None	>= full	< full	Secondary

23.2. ME Load Balancing

AS_03_10P

23.2.1.1. Location

23.2.1.

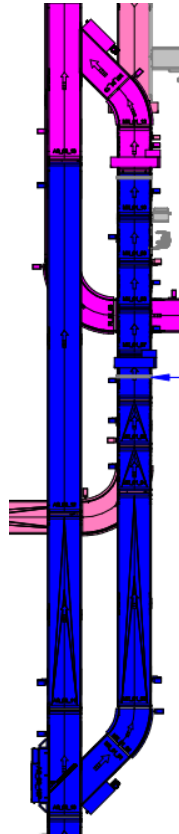


Figure 99 AS_03_10P ME01 plough location

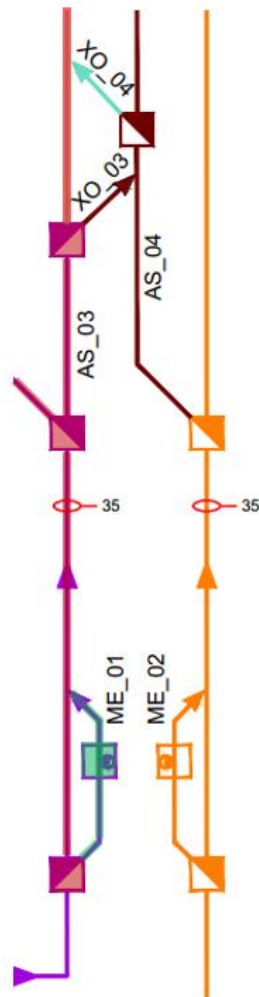


Figure 100 Primary (Light blue) and Secondary (Pink) Paths for ME01

23.2.1.2. Scheme

Bag Status	Preferred Path	Fallback Path	AS_03 status	ME status	Divert Decision
Bag – Clear status	AS03_11 onwards	ME	< full, healthy	any	Primary
Bag – Mistrack status	ME01	MU03	any,	< full, healthy	Secondary
Bag – No Destination	ME01	MU03	any	< full, healthy	Secondary
Bag – Unclear screening status	ME01	MU03	any	< full, healthy	Secondary
Bag – Mistrack status	ME01	MU03	any	>=full, healthy	Primary
Bag – No Destination	ME01	MU03	any	>=full, healthy	Primary
Bag – Unclear screening status	ME01	MU03	any	>=full, healthy	Primary
Bag – Mistrack or Unclear status	ME01	MU03	>= full or unhealthy	>= full or unhealthy	Wait at upstream

Table 47 ME01 decision matrix

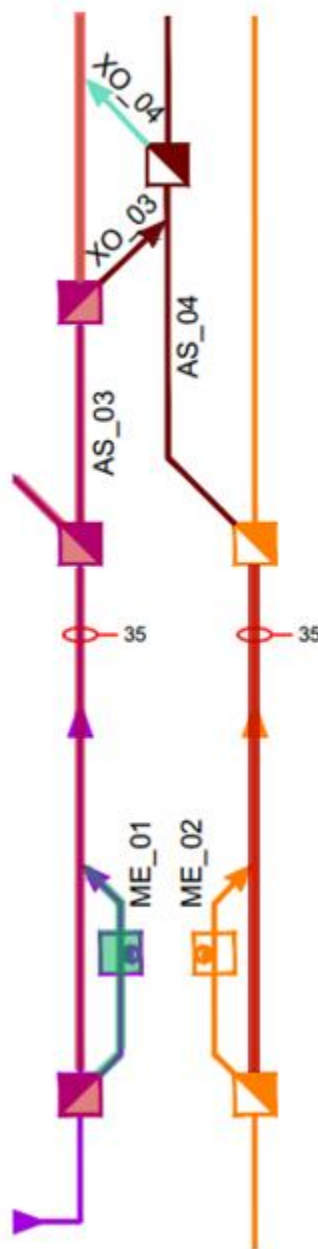


Figure 102 Primary (Yellow) and Secondary (Red Highlight) Paths for ME02

23.2.2.2. Scheme

Bag Status	Preferred Path	Fallback Path	AS_03 status	ME status	Divert Decision
Bag – Clear status	AS02_16 onwards	ME	< full, healthy	any	Primary
Bag – Mistrack status	ME01	MU03	any,	< full, healthy	Secondary
Bag – No Destination	ME01	MU03	any	< full, healthy	Secondary
Bag – Unclear screening status	ME01	MU03	any	< full, healthy	Secondary
Bag – Mistrack status	ME01	MU03	any	>=full, healthy	Primary
Bag – No Destination	ME01	MU03	any	>=full, healthy	Primary
Bag – Unclear screening status	ME01	MU03	any	>=full, healthy	Primary
Bag – Mistrack or Unclear status	ME01	MU03	>= full or unhealthy	>= full or unhealthy	Wait at upstream

Table 48 ME02 decision matrix

23.3. Roller Doors

Door types for CBS

23.3.1.

Name	Type
AL_01_20SD	Security Shutter
AL_01_26SD	Security Shutter
CBS_01_07SD	Security Shutter
CBS_01_12SD	Security Shutter
CBS_02_05SD	Security Shutter
CBS_02_10SD	Security Shutter
CBS_03_05SD	Security Shutter
CBS_03_10SD	Security Shutter
CBS_04_07SD	Security Shutter
CBS_04_12SD	Security Shutter
CL_05_01SD	Security Shutter
ME_01_06SD	Security Shutter
ME_01_11SD	Security Shutter
ME_02_07SD	Security Shutter
ME_02_12SD	Security Shutter
OS_01_02SD	Security Shutter
TC_01_05SD	Security Shutter
TC_03_01SD	Security Shutter

Table 49: Roller Doors in the 4 Lane Strategy

23.4. Metering Conveyors Setup

Conveyor Name	CCP	Bag Gap Min. Size	Bag Gap Min. Interval
TX0108	CCP1	10	10
TX0106	CCP1	10	10
TX0105	CCP1	10	10
TC0912	CCP1	10	10
TC0911	CCP1	10	10
TC0909	CCP1	10	10
TC0538	CCP1	10	10
TC0127	CCP1	28	22
TC0126	CCP1	25	22
TC0125	CCP1	10	10
TC0124	CCP1	28	10
TC0123	CCP1	10	10
TC0122	CCP1	10	10
TC0118	CCP1	20	22
TC0115	CCP1	15	22
TC0114	CCP1	15	22
TC0113	CCP1	15	22
TC0112	CCP1	15	22
TC0111	CCP1	15	22
TC0110	CCP1	15	22
TC0109	CCP1	15	22
TC0108	CCP1	15	22
TC0107	CCP1	15	22
TC0106	CCP1	15	22
TC0105	CCP1	15	22
TC0104	CCP1	15	22
TC0103	CCP1	15	22
TC0102	CCP1	15	22
TC0101	CCP1	15	22
CC0902	CCP1	10	10
TC0207	CCP2	10	10
MF0107	CCP2	10	10
MF0106	CCP2	20	10
MF0102	CCP2	10	10
AS0244	CCP2	10	10
AS0126	CCP2	20	10

Conveyor Name	CCP	Bag Gap Min. Size	Bag Gap Min. Interval
AS0125	CCP2	25	22
AS0123	CCP2	20	10
AS0122	CCP2	10	10
AS0116	CCP2	10	10
AS0402	CCP4A	10	10
AS0403	CCP4A	25	22
AS0414	CCP4A	20	10
AS0416	CCP4A	25	22
AS0417	CCP4A	22	22
MF0805	CCP4A	10	10
MF0806	CCP4A	10	10
PB0303	CCP4A	10	10
PB0304	CCP4A	10	10
XO0302	CCP4A	20	10
XO0303	CCP4A	10	10
XO0403	CCP5	20	10
XO0402	CCP5	20	10
AS0336	CCP5	20	10
AS0332	CCP5	25	22
AS0331	CCP5	10	10
AS0324	CCP5	10	10
AS0323	CCP5	15	10
AS0322	CCP5	20	10
AL0105	CCP6	10	10
AL0106	CCP6	10	10
AL0107	CCP6	10	10
AL0112	CCP6	15	12
AL0113	CCP6	15	12
AL0115	CCP6	22	22
AL0118	CCP6	10	10
AL0119	CCP6	15	10
AL0120	CCP6	10	10
AL0121	CCP6	11	10
AL0122	CCP6	11	10
AL0123	CCP6	11	10
AL0124	CCP6	11	10
AL0125	CCP6	11	10
AL0126	CCP6	16	10

<i>Conveyor Name</i>	<i>CCP</i>	<i>Bag Gap Min. Size</i>	<i>Bag Gap Min. Interval</i>
AL0205	CCP6	10	10
AL0305	CCP6	10	10
AL0405	CCP6	10	10
AS0205	CCP6	15	10
AS0206	CCP6	10	10
AS0208	CCP6	30	10
AS0214	CCP6	25	22
AS0216	CCP6	20	22
AS0217	CCP6	10	10
AS0222	CCP6	25	22
AS0224	CCP6	20	22
AS0229	CCP6	10	10
AS0231	CCP6	25	22
AS0233	CCP6	20	22
CBS0103	CCP6	15	22
CBS0104	CCP6	15	22
CBS0105	CCP6	15	22
CBS0106	CCP6	10	22
CBS0107	CCP6	10	22
CBS0108	CCP6	15	22
CBS0115	CCP6	35	25
CBS0117	CCP6	25	22
CBS0201	CCP6	10	22
CBS0202	CCP6	15	22
CBS0203	CCP6	15	22
CBS0204	CCP6	10	22
CBS0205	CCP6	10	22
CBS0206	CCP6	15	22
CBS0213	CCP6	35	25
CBS0215	CCP6	25	22
CL0105	CCP6	25	22
CL0106	CCP6	10	10
CL0205	CCP6	20	22
CL0206	CCP6	10	10
CL0511	CCP6	10	10
ME0214	CCP6	10	10
TC0129	CCP6	20	15
TC0130	CCP6	20	15

<i>Conveyor Name</i>	<i>CCP</i>	<i>Bag Gap Min. Size</i>	<i>Bag Gap Min. Interval</i>
TC0131	CCP6	15	22
TX0303	CCP6	20	10
TX0309	CCP6	10	10
XO0203	CCP6	15	18
XO0204	CCP6	22	15
XO0205	CCP6	15	22
XO0207	CCP6	15	22
XO0212	CCP6	30	24
XO0213	CCP6	20	22
XO0116	CCP7	20	22
XO0115	CCP7	30	24
XO0113	CCP7	10	10
XO0108	CCP7	15	10
XO0107	CCP7	15	10
XO0105	CCP7	15	22
XO0104	CCP7	15	22
TC1633	CCP7	15	22
TC1632	CCP7	25	22
TC1631	CCP7	25	22
TC1630	CCP7	20	22
TC1629	CCP7	20	22
TC1628	CCP7	15	22
TC1626	CCP7	25	25
TC1625	CCP7	15	22
TC1623	CCP7	15	22
TC1622	CCP7	15	22
TC1621	CCP7	15	22
TC1620	CCP7	15	22
TC1619	CCP7	15	10
TC1618	CCP7	15	10
TC1617	CCP7	20	20
TC1616	CCP7	10	10
TC0308	CCP7	10	10
TC0307	CCP7	35	10
TC0209	CCP7	10	10
TC0208	CCP7	20	10
RI0110	CCP7	10	10
RI0109	CCP7	10	15

<i>Conveyor Name</i>	<i>CCP</i>	<i>Bag Gap Min. Size</i>	<i>Bag Gap Min. Interval</i>
ME0113	CCP7	10	10
ME0109	CCP7	11	10
ME0108	CCP7	50	22
ME0107	CCP7	20	22
CL0405	CCP7	10	10
CL0404	CCP7	20	22
CBS0417	CCP7	25	10
CBS0415	CCP7	35	25
CBS0408	CCP7	15	22
CBS0407	CCP7	15	22
CBS0406	CCP7	15	22
CBS0405	CCP7	15	22
CBS0404	CCP7	15	22
CBS0403	CCP7	15	22
CBS0402	CCP7	15	22
CBS0401	CCP7	15	22
CBS0315	CCP7	25	22
CBS0313	CCP7	35	20
CBS0306	CCP7	15	25
CBS0305	CCP7	15	22
CBS0304	CCP7	15	22
CBS0303	CCP7	15	22
CBS0302	CCP7	15	22
CBS0301	CCP7	15	22
AS0320	CCP7	10	10
AS0318	CCP7	25	22
AS0317	CCP7	25	22
AS0312	CCP7	10	10
AS0308	CCP7	25	22
AS0306	CCP7	20	22
AS0305	CCP7	10	10
AS0105	CCP7	10	10
AS0104	CCP7	12	18
AS0103	CCP7	25	22
AL0404	CCP7	20	10
AL0304	CCP7	20	10

23.5. Indexing Conveyors Setup

Conveyor Name	CCP	Indexing Activation Condition(s)	Indexing Distance (0.1 sec)
MF0105	CCP2	MF0106 not ready to receive bags	8
MF0104	CCP2	MF0105 not ready to receive bags	8
MF0804	CCP4A	MF0805 not ready to receive bags	8
PB0302	CCP4A	PB0303 not ready to receive bags	8
AL0114	CCP6	AL0115 not ready to receive bags	8
ME0103	CCP7	ME0104 not ready to receive bags	8

23.6. Merge Configuration

Merge Name	CCP	Type No	Type Description
TC0538_onto_TC0109	CCP1	1	One on One
TC0912_onto_TC0105	CCP1	1	One on One
TX0108_onto_TC0113	CCP1	1	One on One
AS0233P_onto_AS0108	CCP2	1	One on One
AS0244_onto_AS0117	CCP2	1	One on One
MF0107_onto_MU01	CCP2	2	Merge Priority
MF0806_onto_MU04	CCP4A	4	Primary priority
PB0304_onto_MU03	CCP4A	4	Primary priority
XO0303_onto_AS0404	CCP4A	1	One on One
XO0403_onto_AS0324	CCP5	1	One on One
AL0205_onto_AL0106	CCP6	1	One on One
AL0305_onto_AL0107	CCP6	1	One on One
AL0405_onto_AL0108	CCP6	1	One on One
AS0105P_onto_AS0230	CCP6	1	One on One
CL0106_onto_AS0206	CCP6	1	One on One
CL0206_onto_AS0207	CCP6	1	One on One
ME0214_onto_AS0218	CCP6	5	Primary priority with Timeout and Flush
XO0213_onto_TC0130	CCP6	1	One on One
CL0405_onto_AS0306	CCP7	1	One on One
ME0113_onto_AS0313	CCP7	5	Primary priority with Timeout and Flush
RI0110_onto_XO0114	CCP7	5	Primary priority with Timeout and Flush
TC0209_onto_TC1619	CCP7	1	One on One
TC0308_onto_TC1617	CCP7	1	One on One
XO0116_onto_TC1632	CCP7	1	One on One

23.7. Bag Length Change Detection Conveyors

Conveyor Name	CBS
CBS0110	1
CBS0111	1
CBS0112	1
CBS0113	1
CBS0114	1
CBS0115	1
CBS0116	1
CBS0117	1
CBS0208	2
CBS0209	2
CBS0210	2
CBS0211	2
CBS0212	2
CBS0213	2
CBS0214	2
CBS0215	2
CBS0308	3
CBS0309	3
CBS0310	3
CBS0311	3
CBS0312	3
CBS0313	3
CBS0314	3
CBS0315	3
CBS0410	4
CBS0411	4
CBS0412	4
CBS0413	4
CBS0414	4
CBS0415	4
CBS0416	4
CBS0417	4

23.8. CCP failure & the Impact on the bag path

CCP1 Failure

23.8

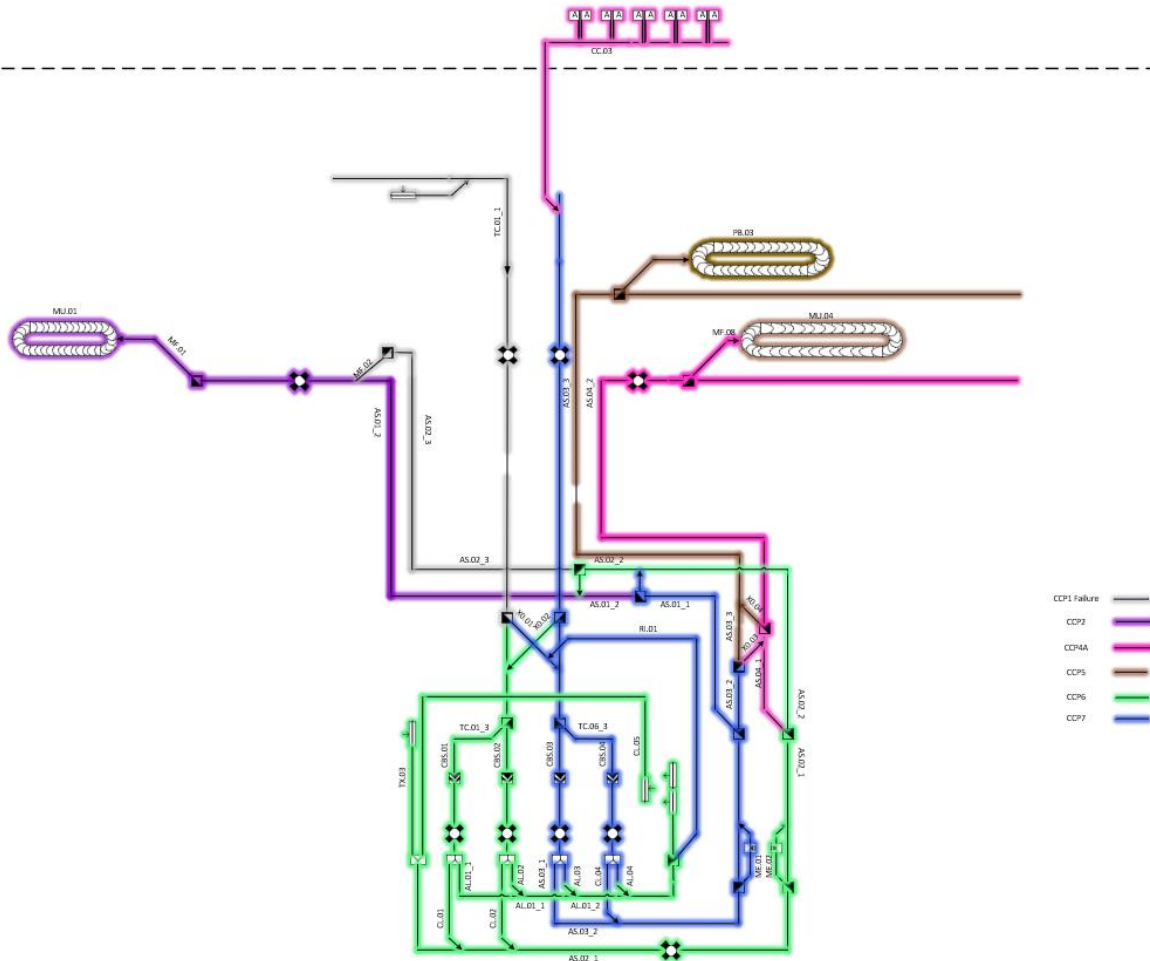


Figure 102 CCP1 Failure

CCP2 Failure

23.8

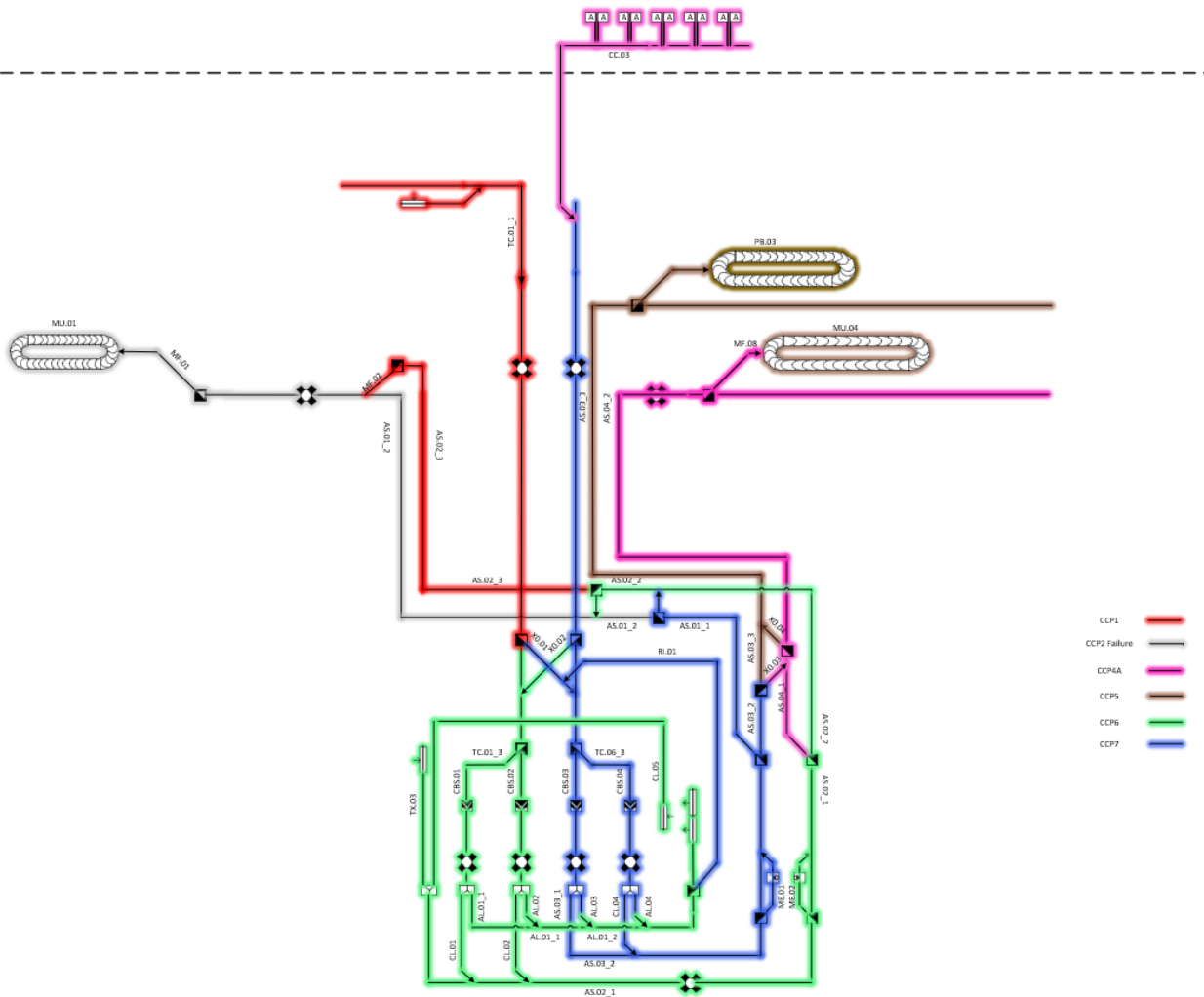


Figure 103 CCP2 Failure

CCP5 Failure

23.8

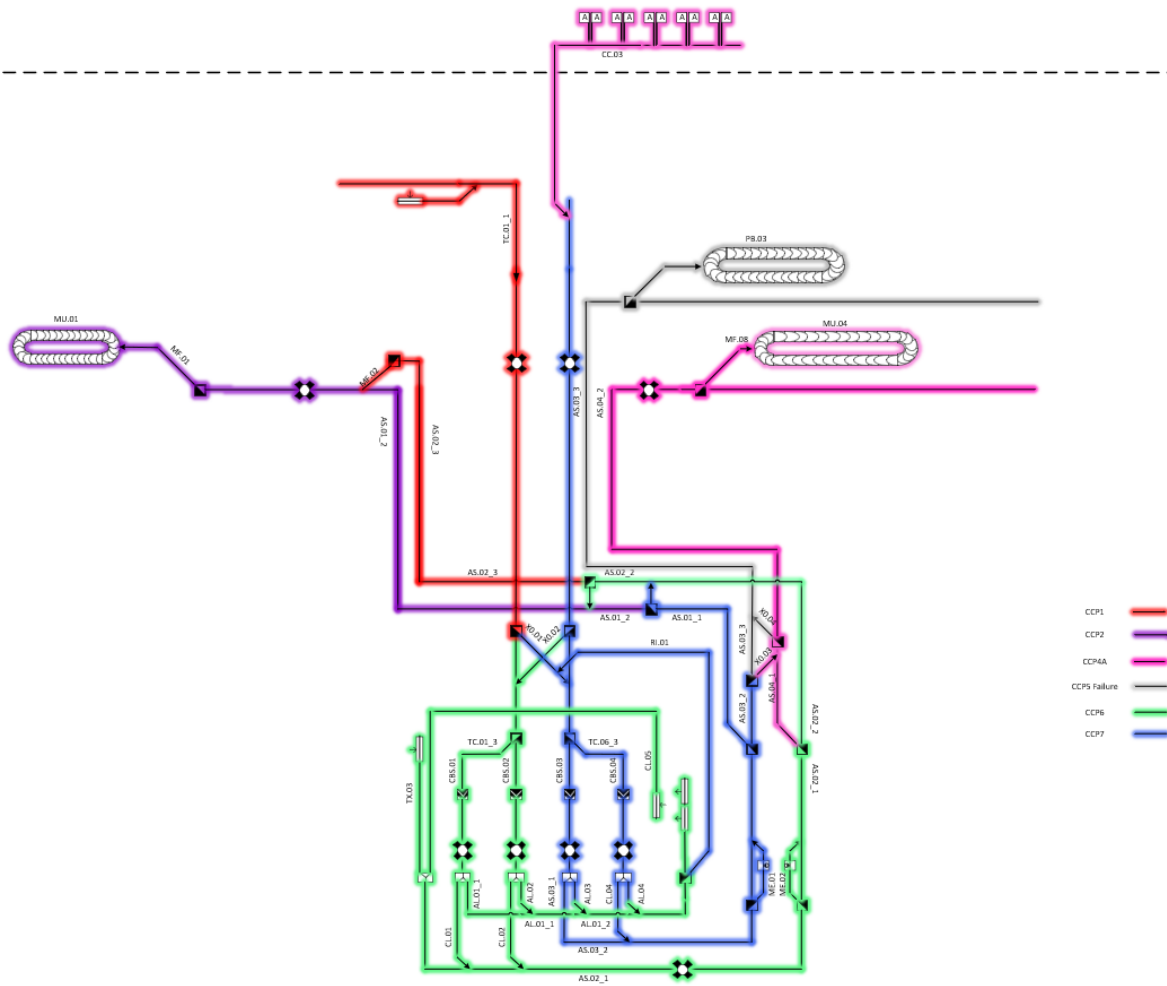


Figure 105 CCP5 Failure

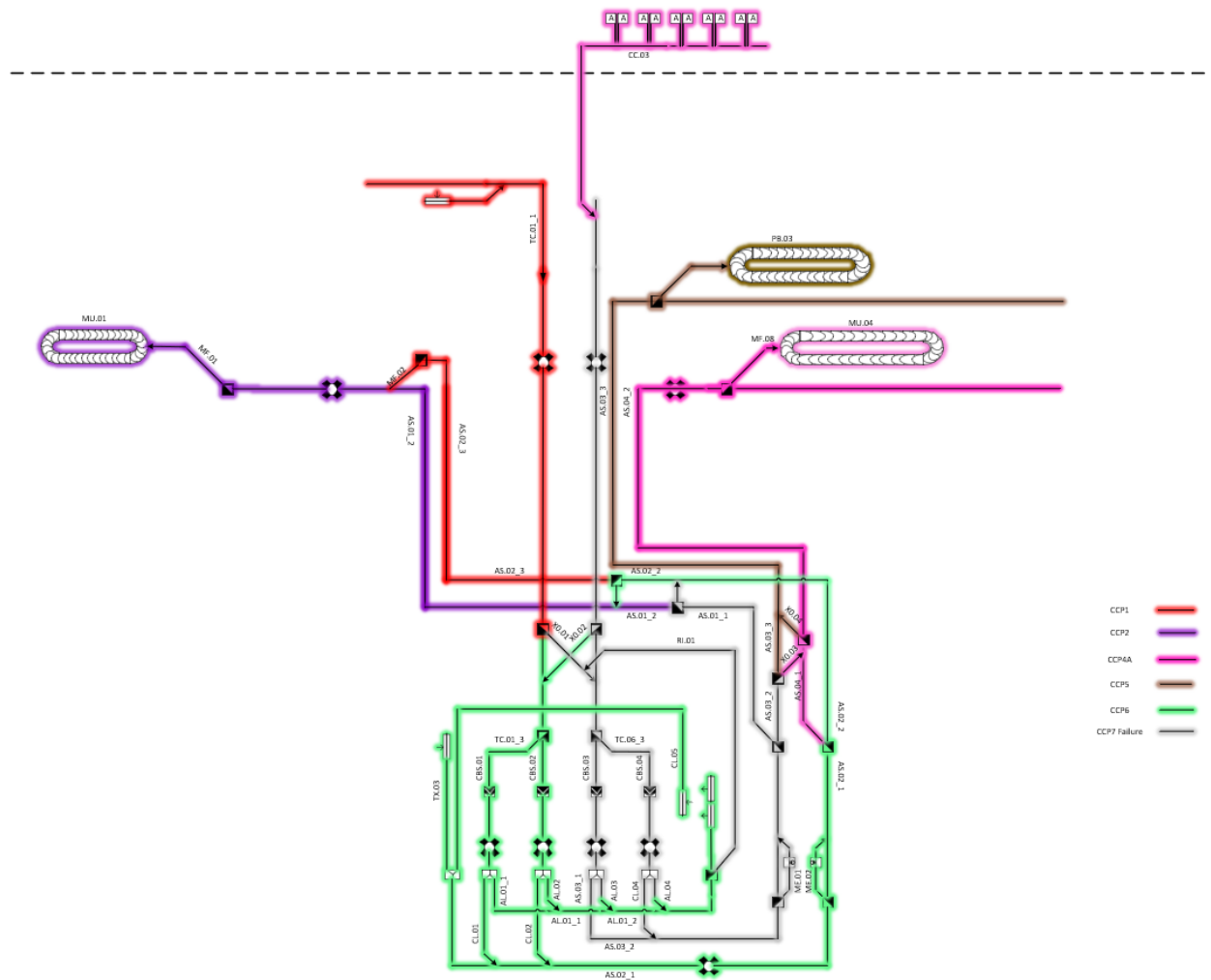
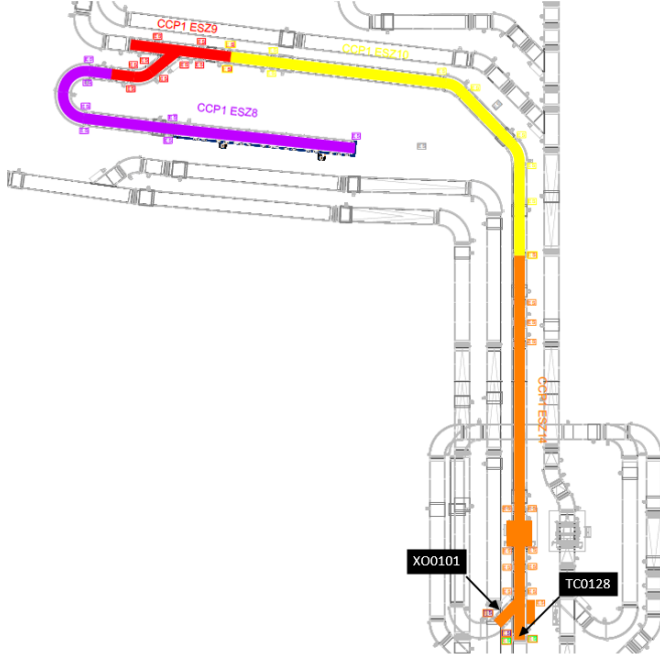
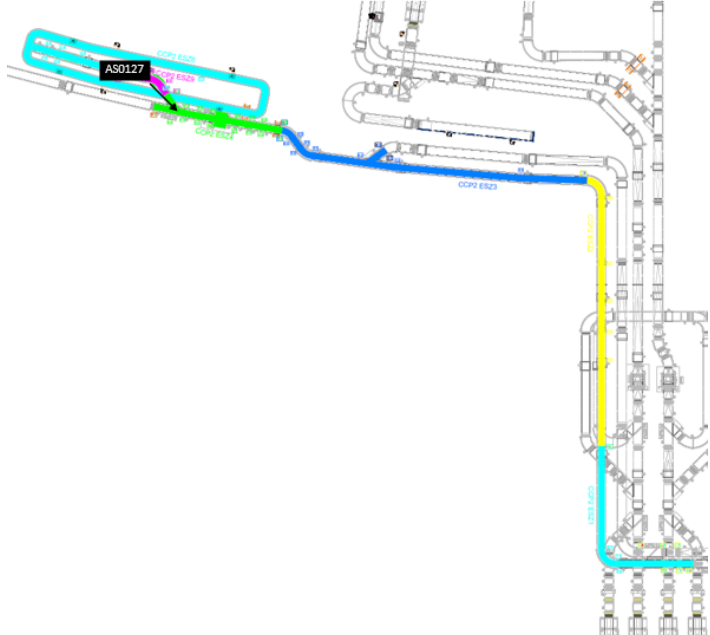


Figure 107 CCP7 Failure

23.9. “Can Start First” Conveyors

Conveyor Name	CCP
XO0101	CCP1
TC0128	CCP1
CCP1 Conveyor's location Indication	
AS0127	CCP2
CCP2 Conveyor's location Indication	
AS0405	CCP4A

Conveyor Name	CCP
AS0418	CCP4A
AS0421	CCP4A
	
MF7T04	CCP5
AS0340	CCP5
AL0116	CCP6
AL0127	CCP6
AS0216	CCP6
AS0224	CCP6
AS0233	CCP6
CBS0101	CCP6
CBS0118M	CCP6
CBS0206	CCP6
CBS0216M	CCP6
XO0201	CCP7
TC1633	CCP7
TC1630	CCP7
CBS0418M	CCP7
CBS0417	CCP7
CBS0316M	CCP7
CBS0315	CCP7

<i>Conveyor Name</i>	<i>CCP</i>
AS0321	CCP7
AS0319	CCP7
AS0310	CCP7
AS0105	CCP7
AL0404	CCP7